VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

C25 Curriculum & Course Contents

B.Tech. CSE

w.e.f. batch 2025-29

B.Tech. COMPUTER SCIENCE AND ENGINEERING

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B.TECH. PROGRAM.

MAJOR REVISION - R25 & C25

COURSE STRUCTURE – C25

Induction program

Course Code	Course Title	L	T	P	SL	C	Course category
25SA101	Orientation program (Induction Program)	0	2	0	0	1	Binary graded
Total		**0**	**2**	**0**	**0**	**1**	

Pre - semester

Course Code	Course Title	L	T	P	SL	C	Course category
25MT101	Foundations of Engineering Mathematics	0	3	2	0	1	Basic Sciences
25MT102	Engineering Mathematics Fundamentals						
25EN101	English Language and Communication	0	3	2	0	1	Humanities
25TP101	Foundation to Quantitative Aptitude and Logical Reasoning	0	5	0	0	1	Humanities
25AM101	IT Tools	0	0	2	0	1	Basic Engineering
25SS104	Stories of Indian Independence	0	2	0	0	1	Binary graded
25EN102	Universal Human Values	0	2	0	0	1	Binary graded
Sub-Total		**0**	**15**	**06**	**0**	**06**	
Total		**21**			**0**	**06**	

I Year I Semester (MPC Stream)

Course Code	Course Title	L	T	P	SL	C	Course category
25MT103	Linear Algebra	3	2	0	3	4	Basic Sciences
25MT105	Calculus and Ordinary Differential Equations						
25PY101	Engineering Physics	3	0	2	3	4	Basic Sciences
25CT102	Engineering Chemistry						
25EE101	Basic of Electrical & Electronics Engineering	2	0	2	2	3	Basic Engineering
25ME101	Engineering Drawing						
25CS101	Programming in C	2	0	4	2	4	Basic Engineering
25EN103	English Proficiency and Communication Skills (PET)	0	0	2	0	1	Humanities
25CT101	Environmental Studies	2	2	0	2	3	Basic Sciences
25MS101	Management Studies						Humanities
						1	Binary graded
						1	Binary graded
						1	Binary graded
Total						**22**	

I Year II Semester (MPC Stream)

Course Code	Course Title	L	T	P	SL	C	Course category
25MT103	Linear Algebra	3	2	0	3	4	Basic Sciences
25MT105	Calculus and Ordinary Differential Equations						
25PY101	Engineering Physics	3	0	2	3	4	Basic Sciences
25CT102	Engineering Chemistry						
25EE101	Basic of Electrical & Electronics Engineering	2	0	2	2	3	Basic Engineering
25ME101	Engineering Drawing						
25CS102	Problem Solving through Python	2	0	2	2	3	Basic Engineering
25EN104	Technical English Communication	1	0	2	1	2	Humanities
25CT101	Environmental Studies	2	2	0	2	3	Basic Sciences
25MS101	Management Studies						Humanities
25CY101	Cyber security	0	1	1	0	1	Basic Engineering
						1	Binary graded
						1	Binary graded
						1	Binary graded
						1	Binary graded
Total						**22**	

II Year I Semester

Course Code	Course Title	L	T	P	SL	C	Course category
25MT204	Probability and Statistics	3	0	2	3	4	Basic Sciences
25MT205	Discrete Mathematical Structures	2	2	0	2	3	Basic Sciences
25CS201	Data Structures	2	2	2	2	4	Basic Engineering
25CS202	Web Technology	0	2	2	0	2	Professional Core-1
25CS203	Database Management Systems	2	2	2	2	4	Professional Core-2
25CS204	Object-Oriented Programming through JAVA	2	2	2	2	4	Professional Core-3
25CS205	Digital Logic Design	2	0	2	2	3	Professional Core-4
Sub-Total		**13**	**10**	**12**	**13**	**24**	
Total		**35**			**13**	**24**	

II Year II Semester

Course Code	Course Title	L	T	P	SL	C	Course category
25MT203	Optimization Techniques	2	2	0	2	3	Basic Sciences
25CS801	Industry – Interface (I^2) Course	0	2	0	0	1	Dept. Elective-1
	Open Elective-1	2	0	2	2	3	Open Elective-1
25CS206	Computer Organization and Architecture	2	2	0	2	3	Professional Core-5
25CS207	Design and Analysis of Algorithms	2	2	2	2	4	Professional Core-6
25CS208	Operating Systems	2	0	2	2	3	Professional Core-7
25CS209	Computer Networks	3	0	2	3	4	Professional Core-8
25CS210	Design Thinking and Engineering Orientation	0	0	2	0	1	Basic Engineering
25CS211	Field Project	0	0	2	0	1	Project
Total						**23**	

III Year I Semester

Course Code	Course Title	L	T	P	SL	C	Course category
25EN301	Soft Skills Laboratory	0	0	2	0	1	Humanities
	Dept. Elective-2	2	0	2	2	3	Dept. Elective-2
25CS301	Artificial Intelligence	2	2	0	2	3	Professional Core-9
25CS302	Machine Learning	2	2	2	2	4	Professional Core-10
25CS303	Cryptography and Network Security	2	0	2	2	3	Professional Core-11
25CS304	Software Engineering	2	2	2	2	4	Professional Core-12
25CS305	Theory of Computation	2	2	0	2	3	Professional Core-13
Sub-Total						**21**	
	Honours/Minors (Add-on)	2	2	2	2	**4**	Honours/Minors
Total						**25**	

III Year II Semester

Course Code	Course Title	L	T	P	SL	C	Course category
25TP301	Quantitative Aptitude & Logical Reasoning	0	2	0	0	1	Humanities
25EN302	Professional Communication Skills	0	0	2	0	1	Humanities
	Dept. Elective-3	2	2	2	2	4	Dept. Elective-3
	Dept. Elective-4	2	2	2	2	4	Dept. Elective-4
	Open Elective-2	2	0	2	2	3	Open Elective-2
25CS306	Privacy Preserving and Intrusion Detection	2	2	0	2	3	Professional Core-14
25CS307	Parallel and Distributed Computing	2	2	0	2	3	Professional Core-15
25CS308	Inter Department Project	0	0	2	0	1	Project
Sub-Total						**20**	
	Honours/Minors (Add-on)					**4**	Honours/ Minors
Total						**24**	

IV Year I Semester

Course Code	Course Title	L	T	P	SL	C	Course category
25CS401	Computing Ethics	0	2	2	0	2	Humanities
	Dept. Elective-5	2	2	2	2	4	Dept. Elective-5
	Dept. Elective-6	2	2	2	2	4	Dept. Elective-6
25CS802	Work-in-Lieu of Course – Research paper publications/patents/presentations/ global certifications	0	4	4	0	4*	Dept. Elective-7
	Open Elective-3	2	0	2	2	3	Open Elective-3
Sub-Total						**17**	
	Honours/Minors (Add-on)	2	2	2	2	**4**	Honours/Minors
Total						**21**	

IV Year II Semester

Course Code	Course Title	L	T	P	SL	C	Course category
25CS403 / 25CS402	Project/Internship	0	2	22	0	12	Project/ Internship
Sub-Total		**0**	**2**	**22**	**0**	**12**	
	Honours/Minors (Add-on)	2	2	2	2	**4**	Honours/Minors
Total						**16**	

DEPARTMENT ELECTIVES

Course Code	Course Title	L	T	P	SL	C
25CS803	Advanced Data Structures	2	2	2	2	4
25CS804	Advanced JAVA Programming	2	2	2	2	4
25CS805	Digital Image Processing	2	2	2	2	4
25CS806	Computer Graphics	2	0	2	2	3
25CS807	Mobile-Adhoc Networks	2	2	2	2	4
25CS808	Text Mining	2	2	2	2	4
25CS809	Operating System Design	2	2	0	2	3
25CS810	Cloud Computing	2	2	2	2	4
25CS811	Deep Learning	2	2	2	2	4
25CS812	Cloud Security	2	2	0	2	3
25CS813	Software Project Management	2	2	2	2	4
25CS814	Compiler Design	2	2	2	2	4
25CS815	Numerical Algorithms	2	2	0	2	3
25CS816	Simulation and Modelling	2	2	0	2	3

ORIENTATION PROGRAM & PRE SEMESTER

B.Tech.

COURSE CONTENTS

ORIENTATION & PRE-SEMESTER

ORIENTATION PROGRAM

25SA101	-	Orientation Program (Induction)

PRE SEMESTER

25MT101	-	Foundations of Engineering Mathematics (MPC Streams)
25EN101	-	English Language and Communication
25TP101	-	Foundations of Quantitative Aptitude and Logical Reasoning
25AM101	-	IT Tools
25SS104	-	Stories of Indian Independence
25EN102	-	Universal Human Values

25SA101 ORIENTATION PROGRAM



Hours Per Semester :

L	T	P	SL	C
0	30	0	0	1

Hours Per Week :

L	T	P	SL	C
0	2	0	0	1

PREREQUISITE KNOWLEDGE: Nil

COURSE DESCRIPTION AND OBJECTIVES:

This course is intended to help the students in getting adjusted with the new environment, to develop bondage with other students and faculty along with learning the institutional policies. The purpose of the orientation course is to create awareness on various issues that are essential for any individual. To familiarize the fresh entrants with the new environment and system in engineering education.To make the students a responsible citizens.To promote smooth transition from +2 level education to graduate program among students.

MODULE-1

0L+12T+0P+0SL = 12 Hours

UNIT-1

SOCIALIZING:

Interaction with faculty members, Deans, peers, student bodies and alumni.

UNIT-2

ASSOCIATING:

Eco-friendly practice and sustainable living, teaching sessions to school children of adopted villages, Interaction with farmers and identifying the problems, Field Visits.

PRACTICES:

- Stress assessment and management.
- Sustainable living.
- Iconic visit.

MODULE-2

0L+18T+0P+0SL = 18 Hours

UNIT-1

GOVERNING:

Brief about regulations and assessment pattern, Interaction with the functionaries i.e. Chancellor, Vice Chancellor and Registrar.

UNIT-2

EXPERIENCING:

Physical activity, Creative arts, Literary activities.

UNIT-3

MEDITATION:

Customizing Daily routines and Habits, Practicing Meditation.

SKILLS:

- ✓ *Understanding of organizational knowledge.*
- ✓ *Compliance Awareness.*
- ✓ *Team Integration.*
- ✓ *Adaptability and Confidence.*

PRACTICES:

- Yoga and meditation.
- Sports and Games.
- Any of the one creative art.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Become familiar with the institute ethics and culture.	Apply	7,8,9,10,11
2	Cope up with stress management through meditation.	Analyze	7,8,9,10,11
3	Socially conscious towards the society and environment.	Apply	7,8,9,10,11
4	Build bonds with peers and faculty members.	Apply	7,8,9,10,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGs) :



25MT101 FOUNDATIONS OF ENGINEERING MATHEMATICS

Hours Per Week :

L	T	P	SL	C
0	3	2	0	1

PREREQUISITE KNOWLEDGE: Basics of set theory, Algebra.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims to provide students with fundamental mathematical concepts essential for engineering applications. It covers topics such as set theory, functions, matrices, determinants, calculus, and vectors. Students will develop problem-solving skills and analytical reasoning to apply mathematical principles effectively in various engineering domains.

MODULE-1

0L+06T+04P+0SL=10 Hours

UNIT-1

SETS AND SET OPERATIONS:

Definition and representations of sets, Cardinality, Operations: union, intersection, complement, symmetric difference, Cartesian product, Venn diagrams (basic) and De Morgan's law.

UNIT-2

FUNCTIONS AND EQUATIONS:

Functions and their properties, Domain, Range, Composition of two functions, Types of functions, Inverse of a function.

Polynomials, Factorization, Quadratic equations, Inequalities (linear).

PRACTICES:

- Solve set operation problems using Venn diagrams.
- Compute the domain and range of given functions.
- Verify De Morgan's law with examples.
- Factorize quadratic equations and determine their roots.
- Identify and classify different types of functions.

MODULE-2

0L+12T+08P+0SL=20 Hours

UNIT-1

MATRICES AND DETERMINANTS:

Types of matrices, Basic matrix operations: addition, multiplication, transpose (up to 3×3). Determinants and their properties, Inverse of a matrix (if it exists) and its computation (up to 3×3).

UNIT-2

CALCULUS:

Limits and continuity: simple numerical approach, Basic differentiation rules (product, quotient, chain rules), Derivatives of elementary functions (polynomials, exponentials, logarithms, trigonometric functions), Basic integration techniques (substitution, by parts), Definite and indefinite integrals, Geometric interpretation of integrals (area under curves).

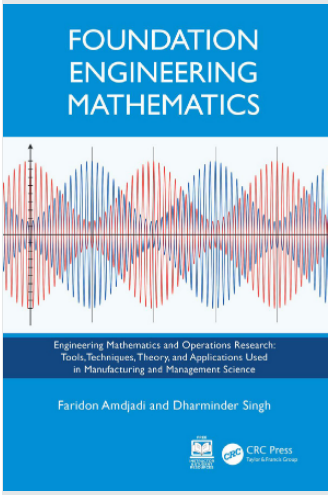


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SKILLS:

- ✓ *Mathematical problem-solving.*
- ✓ *Application of algebra, calculus and vectors in engineering.*
- ✓ *Logical reasoning and analytical thinking.*
- ✓ *Graphical and numerical representation of data.*

UNIT-3

VECTORS AND GEOMETRY:

Basic vector operations (addition, scalar multiplication, dot product, cross product, triple product), Geometric interpretation of vectors in 2D and 3D.

PRACTICES:

- Compute determinants and inverse of given matrices.
- Compute matrix operations.
- Solve differentiation problems using product and quotient rules.
- Evaluate definite and indefinite integrals using appropriate techniques.
- Perform vector operations and verify their geometric interpretations.
- Apply matrices to solve systems of linear equations.
- Matrix Multiplication Using the ***Urdhva-Tiryagbhyam Vedic Sutra.*** Compare results with standard methods and reflect on computational efficiency.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Solve set theory and function-related problems effectively.	Apply	1, 3
2	Utilize matrices and determinants for engineering applications.	Apply	1, 5
3	Analyze and solve differential and integral calculus problems.	Analyze	2, 6
4	Use vector analysis to interpret real-world problems.	Apply	1, 3
5	Implement mathematical problem-solving skills in engineering.	Apply	2, 10

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGs) AND INDIAN KNOWLEDGE SYSTEM (IKS) :



TEXT BOOKS:

1. B.S. Grewal, "Higher Engineering Mathematics", Khanna Publishers, 44th Edition, 2017.
2. Erwin Kreyszig, "Advanced Engineering Mathematics", John Wiley & Sons, 10th Edition, 2011.

REFERENCE BOOKS:

1. R.K. Jain, S.R.K. Iyengar, "Advanced Engineering Mathematics", Narosa Publishing, 5th Edition, 2016.
2. H.K. Dass, "Advanced Engineering Mathematics", S. Chand Publishing, 28th Edition, 2018.
3. M.D. Greenberg, "Advanced Engineering Mathematics", Pearson Education, 2nd Edition, 1998.

25EN101 ENGLISH LANGUAGE AND COMMUNICATION

Hours Per Week :

L	T	P	SL	C
0	3	2	0	1

PREREQUISITE KNOWLEDGE: Basic proficiency in English (A1-A2 level), with the ability to understand and participate in simple conversations and write short texts.

COURSE DESCRIPTION AND OBJECTIVES:

This course is designed to strengthen intermediate language skills, focusing on improved fluency, vocabulary expansion, grammatical accuracy, and effective communication in academic, social, and professional settings. The course prepares learners for B1 level proficiency, enabling them to handle a wider range of communication tasks.

MODULE-1

0L + 6T + 4P + 0SL = 10 Hours

UNIT-1

ENHANCING EVERYDAY COMMUNICATION:

- **Reading:** Interpreting short articles, messages, and dialogues with implied meanings.
- **Writing:** Completing guided writing tasks such as formal requests and informal responses.
- **Listening:** Understanding longer conversations and identifying tone and purpose.
- **Speaking:** Describing routines, experiences, and making comparisons using various tenses.

UNIT-2

SOCIAL INTERACTIONS AND OPINIONS:

- **Reading:** Understanding text structures and making inferences.
- **Writing:** Correcting and rewriting passages with grammatical errors.
- **Listening:** Extracting specific and general information from monologues.
- **Speaking:** Expressing personal opinions with justifications in short discussions.

PRACTICES MAPPED WITH IKS:

- Reading emails, blogs, and notices with detailed comprehension: ***Script styles from palm leaf manuscripts.***
- Listening to authentic audio clips and identifying perspectives: ***Oral transmission techniques.***
- Engaging in pair and group discussions on daily topics: ***Community storytelling/council culture.***
- Writing short reports and summaries: ***Panchayat records and administrative logs.***

MODULE-2

0L+12T+08P+0SL=20 Hours

UNIT-1

EXPRESSING THOUGHTS AND OPINIONS:

- **Reading:** Analyzing articles and opinion pieces.
- **Writing:** Writing coherent paragraphs expressing opinions.
- **Listening:** Listening to debates and identifying arguments.
- Speaking: Participating in structured group discussions on likes/dislikes and choices.

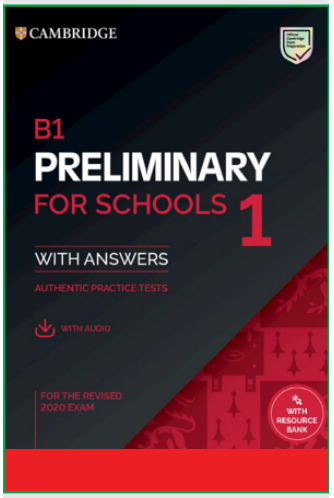


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SKILLS:

- ✓ *Receptive Skills – Reading and Listening.*
- ✓ *Productive Skills – Writing and Speaking.*
- ✓ *Public Speaking –Presentation.*

UNIT-2

GOALS, ASPIRATIONS, AND PERSONAL DEVELOPMENT:

- **Reading:** Understanding and summarizing goal-setting articles.
- **Writing:** Writing short essays on career plans.
- **Listening:** Comprehending longer audio clips related to personal stories and achievements.
- **Speaking:** Delivering short talks on aspirations and future plans.

UNIT-3

RELATIONSHIPS AND IDENTITY:

- **Reading:** Understanding and analyzing email correspondence.
- **Writing:** Writing formal and informal emails and messages.
- **Listening:** Listening for relationship dynamics and speaker attitude.
- **Speaking:** Giving presentations on personal relationships and milestones.

PRACTICES MAPPED WITH IKS:

- Reading narratives and informative texts: ***Panchatantra, Jataka tales.***
- Listening to podcasts and interviews: ***Oral tradition in storytelling, folk wisdom.***
- Writing structured essays and reflections: ***Spiritual reflection in Indian philosophy.***
- Role-playing interviews and discussions: ***Courtroom and sabha simulations.***

COURSE OUTCOMES:

Upon successful completion of this course, students will be able to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Interpret intermediate-level texts, including articles, blogs, and personal communications.	Apply	6,7,8,9,11
2	Write structured content such as essays, formal emails, and descriptive paragraphs.	Apply	6,7,8,9,11
3	Analyze speaker's intent, tone, and point of view in spoken texts.	Analyze	6,7,8,9,11
4	Evaluate and present arguments with clarity and coherence.	Evaluate	6,7,8,9,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGs) AND INDIAN KNOWLEDGE SYSTEM (IKS) :



TEXT BOOKS:

1. B1 Preliminary for Schools – Student's Book by Cambridge University Press, 2020.

   Aligned with B1-B2 level learners, this textbook provides comprehensive practice in all four skills using real-world contexts.

REFERENCE BOOKS:

1. B1 Preliminary for Schools Trainer (Six Practice Tests) – Cambridge University Press, 2020.

   Includes exam-style tasks, strategies, and explanations to build confidence and fluency.

2. English Grammar in Use (Intermediate) by Raymond Murphy – Cambridge University Press.

   Self-study grammar book with clear explanations and practice exercises, suitable for B1–B2 learners.
3. Write & Improve – Cambridge Write & Improve online tool.

   An interactive writing platform providing real-time feedback to enhance writing skills.
4. BBC Learning English – Intermediate Section – BBC Learning English.

   Free listening, vocabulary, and grammar materials tailored for intermediate learners.
5. Cambridge English Online Dictionary and Thesaurus – https://dictionary.cambridge.org

   Useful for building vocabulary and understanding word usage with examples.

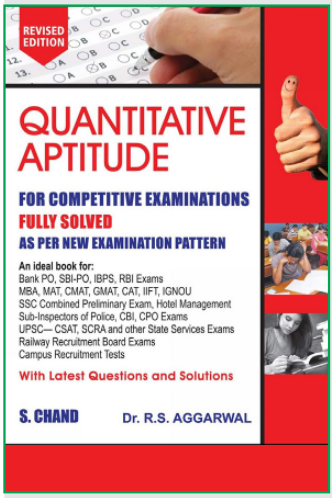


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25TP101 FOUNDATIONS OF QUANTITATIVE APTITUDE AND LOGICAL REASONING

Hours Per Week :

L	T	P	SL	C
0	5	0	0	1

PREREQUISITE KNOWLEDGE: Basic Mathematics Proficiency.

COURSE DESCRIPTION AND OBJECTIVES:

The "Foundation to Intermediate Quantitative Aptitude & Logical Reasoning" course is designed to develop both foundational and advanced skills in arithmetic, data interpretation, puzzles, and logical reasoning. By incorporating hands-on problem-solving exercises, strategic thinking, and time management techniques, the course enhances critical thinking and systematic problem-solving abilities. It equips students with the tools to tackle complex challenges efficiently, improve speed and accuracy, and prepare effectively for competitive exams, placements, and professional tasks.

MODULE-1

0L+10T+0P+0SL=10 Hours

UNIT-1

QUANTITATIVE APTITUDE 1:

Understanding Numbers and Their Types, Finding Common Multiples and Divisors, Percentages and Their Applications of Ratios and Proportions.

UNIT-2

QUANTITATIVE APTITUDE 2:

Calculating Profit, Loss, and Discount, Finding Averages and Solving Mixture Problems, Calculating Simple and Compound Interest, Time and Work Problems.

PRACTICES:

- Each concept would be taught in detail in the class.
- Followed by 10 problems solved in the class.
- Timer based questions solving during class room practice.
- Students would have to solve 10 additional problems as homework assignment in each concept.

MODULE-2

0L+20T+0P+0SL=20 Hours

UNIT-1

QUANTITATIVE APTITUDE 3:

Problems on Time and Distance, Average Speed, Relative Speed and its application on Trains Ages related with Ratios, Introduction to Permutations and Combinations, Probability Problems.

UNIT-2

LOGICAL REASONING 1:

Finding Patterns in Number Series, Identifying Patterns in Letter Sequences, Letter and Number Analogies, Identifying the Odd One Out with multiple categories, Decoding different coded patterns, Syllogism Problems with Venn diagrams.

UNIT-3

LOGICAL REASONING 2:

Puzzles involving Seating related problems, Blood Relations with Family Trees, Solving Problems on Directions, Order and Ranking Problems with multiple criteria, Solving Seating Arrangement Problems, Problems involving Days, Dates in Calendars and Finding different angles in Clocks.

PRACTICES:

- Each concept would be taught in detail in the class.
- Followed by 10 problems solved in the class.
- Timer based questions solving during class room practice.
- Students would have to solve 10 additional problems as homework assignment in each concept.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Develop foundational skills in arithmetic, algebra, and equations for effective problem-solving.	Apply	1, 5
2	Enhance data interpretation accuracy for graphs and charts.	Analyze	1, 4, 5
3	Build confidence and expertise for aptitude tests and professional exams.	Evaluate	2, 5
4	Master time-efficient problem-solving techniques and systematic approaches to tackle complex challenges.	Create	1, 3, 5
5	Strengthen logical reasoning to solve puzzles, sequences, and arrangements efficiently.	Analyze	2, 6, 11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGs):



TEXT BOOKS:

1. 1. R. S. Aggarwal - Quantitative Aptitude for Competitive Examinations - S.Chand Publications - Edition-2023.
2. R. S. Aggarwal - A Modern Approach to Logical Reasoning - S.Chand Publications - Edition-2022.

REFERENCE BOOKS:

1. Trishna Knowledge Systems- Quantitative Aptitude for Competitive Examinations - Pearson Publication - First Edition-2013.
2. R. S. Aggarwal - A Modern Approach to Verbal &Non-Verbal Reasoning - S.Chand Publication - Revised Edition-2018.
3. B.S. Sijwali & S. Sijwali - A New Approach to Reasoning - Arihant Publication - Edition-2018.

SKILLS:

- ✓ *Develop the ability to analyze problems logically and derive effective solutions.*
- ✓ *Master calculations and numerical reasoning for tackling arithmetic and algebraic challenges.*
- ✓ *Gain proficiency in interpreting and analyzing data through charts, graphs, and tables.*
- ✓ *Enhance skills in pattern recognition, critical thinking, and solving puzzles.*
- ✓ *Learn strategies to solve problems quickly and efficiently within time constraints.*

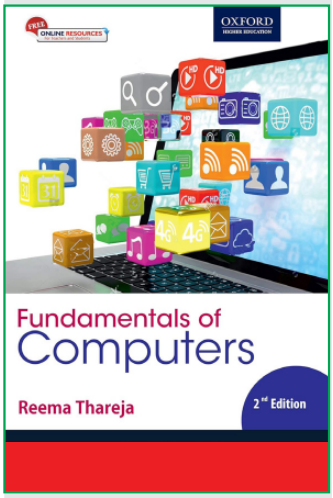


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25AM101 IT TOOLS

Hours Per Week :

L	T	P	SL	C
0	0	5	0	1

PREREQUISITE KNOWLEDGE: Nil

COURSE DESCRIPTION AND OBJECTIVES:

- Equip students with practical knowledge and skills in using essential IT tools such as:
  - Microsoft Word, Excel, and PowerPoint.
  - LaTeX for professional document preparation.
- Enable students to apply these tools in academic, personal, and professional.

MODULE-1

0L+0T+10P+0SL=10 Hours

UNIT-1

WORD:

- Create a table of student marks with subjects, apply shading, change border styles, and align text.
- Create a form letter for a notice and use Mail Merge to send it to a list of student names and roll numbers.
- Design a basic resume with sections for personal info, education, skills, and interests. Use bullet points, alignment, and simple borders.
- Create a 2-column newsletter layout with headers, images, text boxes, and WordArt.
- Create a content index at the top of the page. Add hyperlinks or bookmarks to navigate to each section.
- Create a brochure for a 2-day academic conference with schedule, keynote speakers, and contact info using creative layout and styles.

UNIT-2

EXCEL:

- Format a student mark sheet with at least 5 subjects and calculate total and average.
- Create a payroll system in Excel for 5 employees given Basic salary, 40% for D.A;12% for HRA; 10% for TA; 10% for CCA; 12% for PF; IT 10%. Calculate the gross and net salary.
- Use SUM, AVERAGE, MIN, MAX functions on sales data for five products across 6 months.
- Use an IF function to assign "Pass" or "Fail" based on marks (>35).
- Create a bar chart comparing quarterly sales across different regions.

MODULE-2

0L+0T+18P+0SL=18 Hours

UNIT - 1

ADVANCED EXCEL:

- Highlight all cells in a sales column that are above ₹50,000 using conditional formatting.
- Sort employee records by department and filter to show only those with experience > 5 years.
- Use a Pivot Table to summarize sales by product and region.
- Create an employee lookup sheet where entering an Employee ID returns name, department, and salary.
- Build a monthly attendance sheet for a class, calculate attendance percentage, and highlight students below 75%.
- Develop an EMI calculator that takes principal, interest rate, and tenure to calculate monthly payments.

UNIT-2

POWERPOINT:

- Create a 5-slide presentation introducing yourself—name, interests, hobbies, and goals. Apply different slide layouts and basic transitions.
- Create a 3-slide presentation on your favourite place. Include images, shapes, and text boxes, and use picture styles.
- Create a presentation showing sales data. Insert a table and a chart (bar/pie) to visualize the data.
- Use Slide Master to create a consistent design for all slides (e.g., university (VFSTR) logo in header/footer and common font styling).
- Create a presentation containing an embedded video or audio clip, and set it to play automatically.

UNIT-3

LATEX:

- Create a LaTeX document with a title, author, and date. Include two paragraphs and a section heading.
- Demonstrate text formatting by creating a paragraph using bold, italic, underline, and monospace fonts.
- Create a bulleted list of at least 3 hobbies and a numbered list of your top 3 goals.
- Write the quadratic formula and a few basic math expressions such as fractions, exponents, and square roots.
- Insert an image of your choice (e.g., example-image.jpg) into a document and add a caption below it.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Create word documents, presentations and spread sheets by applying various tools	Apply	2,5
2	Usage of latex tool for report writing	Apply	5,6

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGs) :



TEXT BOOKS:

1. Fundamentals of Computers by Reema Thareja, Oxford University Press 2nd edition 2019, India.
2. Lamport, Leslie (1994). LaTeX: A Document Preparation System, User's Guide and Reference Manual (2nd ed.). Pearson Education.
3. LaTeX: A Document Preparation System by Leslie Lamport.

REFERENCE BOOKS:

1. John Walkenbach, Herb Tyson, Michael R.Groh and FaitheWempen, "Microsoft Office 2010 Bible", Wiley.

SKILLS:

- ✓ *Formatting, document design, resume & letter writing, mail merge.*
- ✓ *Data entry, formulas, payroll systems, pivot tables.*
- ✓ *Slide creation, use of multimedia and animations.*
- ✓ *Posters, brochures, magazine cover creation.*
- ✓ *Basic webpage creation using hyperlinks and styles.*
- ✓ *Using Excel for- mulas and data analysis.*
- ✓ *Document design and presentation delivery.*
- ✓ *LaTeX for academic and professional documents.*

25SS104 STORIES OF INDIAN INDEPENDENCE

Hours Per Week :

L	T	P	SL	C
0	2	0	0	1

PREREQUISITE KNOWLEDGE: Basic understanding of Indian history at the high school level.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces students to the inspiring narratives of India's struggle for independence. It focuses not only on well-known leaders but also on forgotten heroes, movements, and lesser-known revolts that collectively shaped India's freedom story. The course aims to instill national pride, civic consciousness, and leadership qualities in students while strengthening their appreciation for democratic values and social responsibility.

MODULE-1

0L+12T+04P+0SL=12 Hours

UNIT-1

EARLY RESISTANCES AND UNSUNG HEROES:

Revolts Before 1857: Vellore Mutiny, Paika Rebellion, and Sanyasi-Fakir Rebellion; Revolt of 1857: Causes, Course, and Consequences; Tribal and Peasant Uprisings: Santhal Rebellion and Munda Ulgulan; Lesser-Known Freedom Fighters: Rani Gaidinliu; Veerapandiya Kattabomman, and Birsa Munda.

UNIT - 2

PATHBREAKING MOVEMENTS AND LEADERS:

Formation of Indian National Congress and Early Nationalism; Swadeshi and Boycott Movements (1905); Role of Revolutionary Activities: Anushilan Samiti and Hindustan Socialist Republican Association; Transnational and Pan-Indian Movements: Ghadar Movement and Khilafat Movement (Ali Brothers).

PRACTICES:

- Research and present a profile of a lesser-known freedom fighter featured in the unit.
- Prepare a comparative chart on the Revolt of 1857 and the Ghadar Movement.
- Participate in a debate or panel discussion on the role of religion and revolution in India's freedom struggle.
- Create a digital timeline covering key events from early revolts to transnational movements (1800–1920).

MODULE-2

0L+18T+0P+0SL=18 Hours

UNIT-1

MASS MOVEMENTS AND PEOPLE'S PARTICIPATION:

Non-Cooperation Movement (1920–22); Civil Disobedience Movement and Salt Satyagraha; Quit India Movement (1942): "Do or Die" Spirit; Role of Women and Students in the Freedom Struggle.

UNIT - 2

VISIONARIES AND THEIR LEGACIES:

Mahatma Gandhi: Philosophy of Satyagraha and Ahimsa; Subhas Chandra Bose and INA; Bhagat Singh and Revolutionary Nationalism; Dr. B.R. Ambedkar and the Struggle for Social Justice.

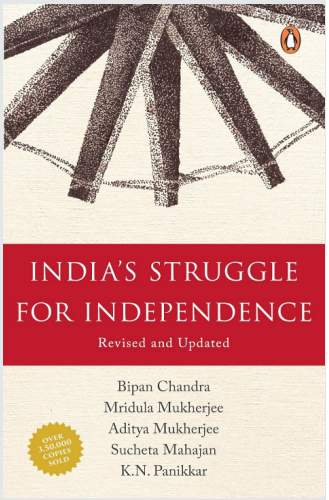


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SKILLS:

- ✓ *Leadership and teamwork by understanding collective struggles.*
- ✓ *Analytical skills through study of strategies and methods used in the freedom struggle.*
- ✓ *Communication skills via storytelling, debates, and presentations.*
- ✓ *Ethical thinking by appreciating sacrifice, social responsibility, and patriotism.*

UNIT - 3

TRANSITION TO INDEPENDENCE:

The Cabinet Mission Plan and Negotiations; Partition and Independence: Causes and Consequences; Integration of Princely States: Sardar Patel's Role; Building a New Nation: Early Challenges.

PRACTICES:

- Group discussions on major movements.
- Role-play / dramatizations: Dandi March, INA Trials, Quit India Movement.
- Documentary review assignments: Freedom struggle films and documentaries.
- Quiz competitions on key events and leaders.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply knowledge of early revolts and lesser-known freedom fighters to understand regional contributions to India's independence.	Apply	6, 11
2	Analyze the development of nationalist ideologies and the role of revolutionary, religious, and transnational movements.	Analyze	6, 8, 9
3	Evaluate the contributions of women, students, and marginalized communities in shaping the mass movements of the freedom struggle.	Evaluate	6, 7, 11
4	Apply lessons of patriotism, leadership, and social responsibility to contemporary societal roles and ethical decision-making.	Apply	6, 7, 11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGs) AND INDIAN KNOWLEDGE SYSTEM (IKS) :



TEXT BOOKS:

1. Bipin Chandra, India's Struggle for Independence, Penguin, 2016.
2. Sekhar Bandyopadhyay, From Plassey to Partition and After: A History of Modern India, Orient Blackswan, 2014 (Reprint Edition).

REFERENCE BOOKS:

1. Ramachandra Guha, Gandhi: The Years That Changed the World, Penguin, 2018.
2. Sugata Bose, His Majesty's Opponent: Subhas Chandra Bose and India's Struggle against Empire, Harvard University Press, 2011.
3. S. Gopal, Jawaharlal Nehru: A Biography, Harvard University Press.

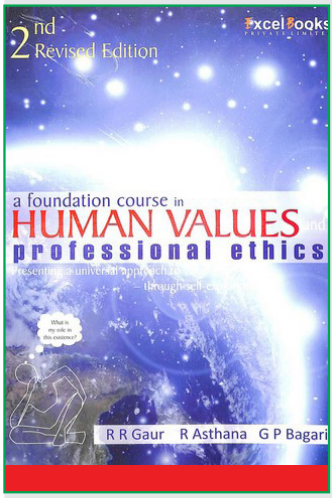


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25EN102 UNIVERSAL HUMAN VALUES

Hours Per Week :

L	T	P	SL	C
0	2	0	0	1

PREREQUISITE KNOWLEDGE: Nil

PREREQUISITE KNOWLEDGE:

UHV (Universal Human Values) doesn't necessarily require specific prerequisite knowledge. It's designed to be accessible for students with various backgrounds. However, there are some foundational concepts like 'Basic understanding of ethics, Openness to different perspectives, Ability for critical thinking' might be helpful.

COURSE DESCRIPTION AND OBJECTIVE:

This course helps the students to appreciate the essential complementarily between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings It also helps to develop students' holistic perspective based on self-exploration about themselves (human being) and their family. It further touches issues very briefly related to their role in the society and the nature.

MODULE-1

0L+16T+0P+2SL=16 Hours

UNIT-1

INTRODUCTION TO VALUE EDUCATION:

Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) Understanding Value Education, ***(concept of Dharmasastra)*** self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human aspirations. Happiness and Prosperity – Current Scenario.

UNIT-2

HARMONY IN THE HUMAN BEING:

Understanding Human being as the Co-existence of the Self and the Body -Distinguishing between the Needs of the Self and the Body- The Body as an Instrument of the Self- Understanding Harmony in the Self- Harmony of the Self with the Body ***(Concept of Sanyam)*** - Programme to ensure self-regulation and Health.

PRACTICES:

- Sharing about Oneself.
- Talk on Everyday life situation.
- Reflect on what she/he would do in this situation.
- Exploring Natural Acceptance.

MODULE-2

0L+18T+0P+0SL=18 Hours

UNIT-1

HARMONY IN THE FAMILY AND SOCIETY:

Harmony in the Family – the Basic Unit of Human Interaction-'Trust' – the Foundational Value in Relationship, 'Respect and Excellence' – as the Right Evaluation, Other Feelings, Justice in Human-to- Human Relationship, Understanding Harmony in the Society ***(Concept of Nyaya)***.

UNIT-2

HARMONY IN THE NATURE/EXISTENCE:

Harmony in the Nature- Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature- Realizing Existence as Co-existence at All Levels- The Holistic Perception of Harmony in Existence ***(Concept of Pancha Mahabhuta).***

UNIT-3

IMPLICATIONS OF THE HOLISTIC UNDERSTANDING – A LOOK AT PROFESSIONAL ETHICS:

Natural Acceptance of Human Values- Definitiveness of (Ethical) Human Conduct- Competence in Professional Ethics-: Holistic Technologies, Production Systems and Management Models- Typical Case Studies- Strategies for Transition towards Value-based Life and Profession (Concept of Universal Human Values).

PRACTICES:

- Consider the potential consequences of each option (returning vs. keeping).
- Case studies: Analyze real-world examples.
- Group discussions: Sharing perspectives and learn from others' experiences.
- Role-play scenarios to navigate conflict resolution and promote healthy interaction.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply the essentials of human values and skills, self-exploration, happiness and prosperity.	Apply	7,8,9,10
2	Apply to self-regarding the coexistence of the "I" with the body.	Apply	7,8,9,10
3	Identify and evaluate the role of harmony in family, society and universal order.	Evaluate	7,8,9,10
4	Analyze the holistic perception of harmony at all levels of existence.	Analyze	7,8,9,10
5	Develop appropriate technologies and management patterns to create harmony in professional and personal life.	Create	7,8,9,10

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGs) AND INDIAN KNOWLEDGE SYSTEM (IKS) :



TEXT BOOK:

1. R R Gaur, R Sangal, G P Bagaria, "Human Values and Professional Ethics", Excel Books, New Delhi, 2010.

REFERENCE BOOKS:

1. A.N. Tripathi, "Human Values", New Age Intl. Publishers, New Delhi, 2004.
2. Gergen, K. J. "Relational Being: Beyond Self and Community", United Kingdom: Oxford University, Press, 2009.
3. A Nagraj, "Jeevan Vidya ek Parichay", Divya Path Sansthan, Amarkantak", 1998.
4. E.F. Schumacher, "Small is Beautiful: a study of economics as if people mattered, Blond & Briggs, Britain" 1973.
5. R R Gaur, R Sangal, G P Bagaria, "A Foundation Course in Human Values and Professional Ethics", 2009.

I

YEAR

B.Tech.

COURSE CONTENTS
(MPC STREAM)

25MT103	-	Linear Algebra
25MT105	-	Calculus and Ordinary Differential Equations
25PY101	-	Engineering Physics
25CT102	-	Engineering Chemistry
25EE101	-	Basics of Electrical and Electronics Engineering
25ME101	-	Engineering Drawing
25CS101	-	Programming in C
25EN103	-	English Proficiency and Communication Skills
25CT101	-	Environmental Studies
25MS101	-	Management Studies
25CS102	-	Problem Solving through Python
25EN104	-	Technical English Communication
25CY101	-	Cyber Security
25SS101	-	Constitution of India
25SA103	-	Physical fitness
25SA104	-	Life Skills
25SA102	-	Self-understanding and Gender Sensitization
25SS102	-	Indian Knowledge Systems
25SS103	-	Indian Culture and Heritage
25SS105	-	Gerontology
25ME102	-	Do it yourself

25MT103 LINEAR ALGEBRA

Hours Per Week :

L	T	P	SL	C
3	2	0	3	4

PREREQUISITE KNOWLEDGE: Basics of Geometry and Algebra.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces students to the core concepts of linear algebra, including matrix operations, linear systems, vector spaces and inner product spaces. The objective is to equip students with the necessary mathematical tools to model, analyze and solve real-world problems across various engineering and scientific fields.

MODULE-1

UNIT-1 **18L+12T+0P+18SL= 48 Hours**

MATRICES:

Elementary row and column operations, Elementary matrices, Similar Matrices, Echelon form, Row reduced echelon form, Rank of a matrix, Inverse of a matrix by Gauss-Jordan method, LU decomposition.

UNIT-2

SYSTEMS OF LINEAR EQUATIONS:

Systems of linear equations, Matrix representation, Consistency using rank of the coefficient matrix, Solutions: Gaussian elimination method, Gauss-Jordan method, Do-Little method.

PRACTICES:

- Perform elementary row/column operations and find the rank of a matrix.
- Compute the inverse of a matrix using the Gauss-Jordan method and LU decomposition.
- Check consistency of a system of linear equations.
- Solve homogeneous and non-homogeneous linear systems using Gauss elimination, Gauss-Jordan and Do-Little methods.
- Solve linear systems using the ***Vyaktāvyakta-Nyāya*** (Substitution) and compare with Gaussian elimination. Analyze consistency and computational efficiency.

MODULE-2

UNIT-1 **27L+18T+0P+27SL=72 Hours**

EIGENVALUES, EIGENVECTORS AND DIAGONALIZATION:

Characteristic Equation, Eigenvalues, Eigenvectors and their Properties (without proofs), Cayley-Hamilton Theorem (without proof), Diagonalization of a Matrix (only for diagonalizable matrices).

Applications: Inverse of a matrix by Cayley-Hamilton Theorem, Power of a diagonalizable square matrix.

UNIT-2

REAL VECTOR SPACE:

Real vector space, Subspace, Linear dependence and independence, Linear span, Bases and dimension.

Applications: Row space and column space of a matrix, Determining rank of a matrix using row space and column space, Row and column spaces of similar matrices.

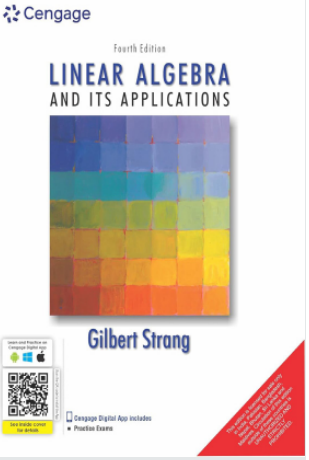


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SKILLS:

- ✓ *Understanding abstract algebraic structures.*
- ✓ *Solving linear systems using various numerical techniques.*
- ✓ *Evaluating eigenvalues, eigenvectors and matrix transformations.*
- ✓ *Applying linear algebra in engineering applications.*
- ✓ *Using rthogonalization and decomposition in advanced algorithms*

UNIT-3

REAL INNER PRODUCT SPACE:

Real inner product space, Norm of a vector, Orthogonal set, Orthonormal set, Cauchy-Schwarz inequality, Gram-Schmidt orthogonalization process.

Applications: QR decomposition and singular value decomposition (SVD).

PRACTICES:

- Find eigenvalues and eigenvectors of a given matrix and verify Cayley-Hamilton theorem.
- Determine the diagonalization of a matrix and compute its power using diagonalization.
- Identify vector subspaces, bases and dimensions.
- Determine rank of a matrix using row and column spaces.
- Apply Gram-Schmidt process to find an orthonormal basis of a vector space.
- Perform QR decomposition.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1.	Determine the rank of matrices using row operations and solve systems of linear equations to assess their consistency.	Apply	1, 2
2.	Compute eigenvalues, eigenvectors and diagonalization for advanced matrix computations.	Evaluate	1, 2, 5, 9
3.	Analyse and apply inner product space concepts for orthogonalization and decomposition techniques.	Analyze	1, 3, 5
4.	Compare the row and column spaces of similar matrices and evaluate their structural properties for advanced computations.	Apply	1, 2, 9, 10
5.	Utilize linear algebra concepts in real-world applications such as optimization, control systems and artificial intelligence.	Analyze	1, 2, 4, 6, 7, 11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXT BOOKS:

1. Gilbert Strang, "Linear Algebra and Its Applications", Cengage Learning, 5th Edition, 2016.
2. Kanti B. Datta, Varma, P. Sekhar, "Calculus, Linear Algebra and Ordinary Differential Equations", Cengage Learning, 1st Edition, 2024.

REFERENCE BOOKS:

1. Erwin Kreyszig, "Advanced Engineering Mathematics", Wiley, 10th Edition, 2011.
2. Sheldon Axler, "Linear Algebra Done Right", Springer, 3rd Edition, 2015.
3. George B. Thomas & Ross L. Finney, "Calculus and Analytic Geometry", Addison-Wesley, 9th Edition, 1995.

25MT105 CALCULUS AND ORDINARY DIFFERENTIAL EQUATIONS

Hours Per Week :

L	T	P	SL	C
3	2	0	3	4

PREREQUISITE KNOWLEDGE: Basics of Algebra, Functions, Differentiation and Integration.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims to develop a fundamental understanding of calculus and differential equations, focusing on their applications in engineering and science. Students will explore single-variable and multi-variable calculus, integral calculus and various techniques for solving ordinary differential equations (ODEs). The objective is to equip students with analytical skills and problem-solving techniques that are essential for real-world applications.

MODULE–1

UNIT-1 **18L+12T+0P+18SL=48 Hours**

DIFFERENTIAL CALCULUS (FUNCTIONS OF ONE VARIABLE):

Mean value theorems and applications, Taylor's and Maclaurin's series expansion with remainders, Indeterminate forms, Concavity and convexity of a curve, Points of inflection, Curve sketching.

UNIT-2

INTEGRAL CALCULUS:

Integrations as the limit of a sum, Fundamental theorem of integral calculus, Mean value theorems, Reduction formulae, Rectification, Improper integrals, Tests of convergence (Only statements).

PRACTICES:

- Verify and interpret the Mean Value Theorem.
- Expand functions using Taylor & Maclaurin series and analyse errors.
- Analyse concavity and classify points of inflection.
- Apply the fundamental theorem of calculus to compute definite integrals.

MODULE–2

UNIT-1 **27L+18T+0P+27SL=72 Hours**

MULTIPLE INTEGRALS:

Double and triple integrals, Change of order of integration, Change of variables - Jacobians of Transformations.

Applications: Computations of areas and volumes, Center of mass and centroid, Moment of inertia.

UNIT-2

FIRST ORDER ORDINARY DIFFERENTIAL EQUATIONS:

Introduction, Order and degree, Formation of ODE from a given one-parameter family of curves, Solutions: Variable separable method, homogeneous, equation reducible to variable separable type, linear and Bernoulli's equations.

Applications: Newton's law of cooling, Natural growth and decay.

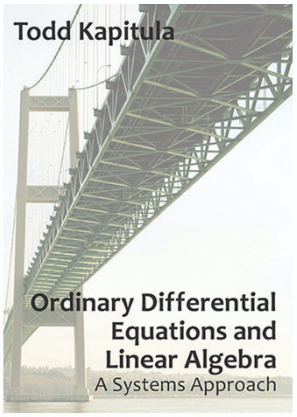


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SKILLS:

- ✓ *Understanding function behaviour using calculus principles.*
- ✓ *Evaluating integrals and applying reduction formulas.*
- ✓ *Solving multiple integrals with variable changes and transformations.*
- ✓ *Mastering techniques for solving ODEs and applying them in scientific contexts.*

UNIT-3

SECOND ORDER ORDINARY DIFFERENTIAL EQUATIONS:

Homogeneous and non-homogeneous equations, Complementary functions, Particular integrals, Solution with constant equations: method of undetermined coefficients, variation of parameters.

PRACTICES:

- Solve simple first-order ODEs using separation of variables.
- Evaluate the area under a curve using definite integrals.
- Solve a real-world problem by computing the volume of a solid using triple integrals with appropriate limits.
- Compare and contrast the use of Cartesian vs. polar coordinates in evaluating a given double integral.
- Apply ODEs to Newton's Law of Cooling and natural growth/decay problems.
- Solve circular motion problems using ***Āryabhaṭa's Bhūgola-Nyāya*** (Rotating Earth Principle) and compare with modern calculus. Understand angular speed and how fast things change.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply calculus concepts in problem-solving	Apply	1, 2, 5
2	Analyze the behavior of functions and integrals	Analyze	1, 2, 6
3	Apply multiple integrals and transformations	Apply	1, 4, 5
4	Evaluate solutions of ODEs	Evaluate	1, 4, 10
5	Create solutions for real-world applications	Create	1, 11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXT BOOKS:

1. Erwin Kreyszig, "Advanced Engineering Mathematics", Wiley, 10th Edition, 2011.
2. Kanti B. Datta, Varma, P. Sekhar, "Calculus, Linear Algebra and Ordinary Differential Equations", Cengage Learning, 1st Edition, 2024.

REFERENCE BOOKS:

1. B.S. Grewal, "Higher Engineering Mathematics", Khanna Publishers, Latest Edition.
2. S.L. Ross, "Differential Equations for Engineers", New Age International Publishers, Latest Edition.
3. James Stewart, "Calculus", Cengage Learning, 8th Edition, 2015.

25PY101 ENGINEERING PHYSICS

Hours Per Week :

L	T	P	SL	C
3	0	2	3	4

PREREQUISITE KNOWLEDGE: Basics of semiconductors and Quantum mechanics.

COURSE DESCRIPTION AND OBJECTIVES:

This course ensures commensurable understanding of semiconductors and their potential applications. It enunciates the electron dynamics in solids through the conceptual grasp of principles of quantum mechanics. This embarks perspective outlook on optoelectronic devices and optical fibres in the backdrop of semiconductor physics.

MODULE-1

UNIT-1 **18L+0T+12P+18SL = 48 Hours**

METALS AND SEMICONDUCTORS:

Introduction to metals, expression for electrical conductivity, Introduction to semiconductors-intrinsic, n-type and p-type semiconductors. Electrical conductivity of semiconductors for intrinsic and extrinsic. Hall effect- applications. ***(Concept of Panchabhuta-Five Elements)***

UNIT-2

Introduction to Quantum Mechanics, Dual nature of radiation, de Broglie's concept of matter waves, Uncertainty principle, Schrödinger's time-independent wave equation, Particle confined in a one-dimensional infinite potential well, quantum dots; finite potential well- Quantum Tunnelling: Scanning tunnelling microscope and tunnelling diode (qualitative in view of quantum tunnelling).
Kanada's atomic theory (paramanu)

PRACTICES:

- Determination of Energy Band gap of p-n junction diode.
- Hall Effect - Determination of Hall coefficient.
- Photoelectric Effect-Determination of Planck's constant.
- To study the variations of thermo emf of a copper-constantan thermo-couple with temperature (Seebeck effect).
- Demonstration of tunnelling effect in tunnel diode.

MODULE-2

UNIT-1 **27L+0T+18P+27SL = 72 Hours**

Quantum free electron theory, Fermi-Dirac distribution, electronic specific heat of solids, Density of states (qualitative), success and failures of quantum free electron theory of solids. E-K diagram- classification of materials based on bands in solids, Fermi level in semiconductors- intrinsic and extrinsic.

UNIT-2

p-n junction diode- forward and reverse bias conditions; Solar cell-construction-working-characteristics-applications. Direct and indirect bandgap semiconductors-LED-construction-working-characteristics-applications. Photodiode -construction-working-characteristics- applications. Tunnelling diode.

UNIT-3

Introduction to lasers- difference between normal light and laser light: characteristics of laser light; stimulated absorption, spontaneous and stimulated emission; population inversion, pumping; optical resonator-lasing mechanism; Diode laser- construction-working and applications- Optical fibre communications.

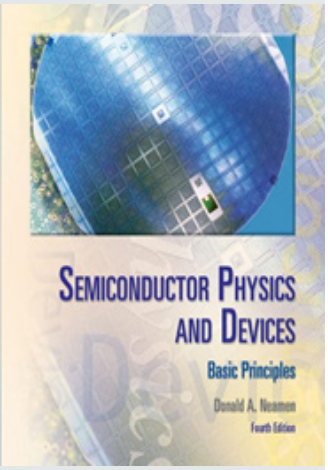


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SKILLS:

- ✓ *Apply the quantum laws to understand the electron dynamics of solids.*
- ✓ *Identification of semiconductors for optoelectronic devices.*
- ✓ *Realizing the importance of laser for industrial and commercial applications.*

PRACTICES:

- Study the characteristics of diode.
- Determination of efficiency and Fill factor of a solar cell.
- Determine the efficiency and fill factor of solar cells in Parallel and series combinations.
- Study the V-I characteristics of LED.
- Optical fibre – Determination of Numerical aperture – Acceptance angle.
- Determination of attenuation in Optical fiber.
- Laser - Determination of wavelength.
- Determination of the slit width from Fraunhofer diffraction pattern using LASER beam.
- Demonstration of monochromatic nature of laser light comparing with ordinary light by the principle of dispersion by using Prism.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Comprehend the nature of metals and semiconductors under different conditions.	Apply	1,2,3,4,9,10,11
2	Apply the principles of quantum mechanics to unravel the latest technical developments.	Apply	1,2,3,4,9,10
3	Categorize the solids based on band theory.	Analyse	1,2,3,4,11
4	Appraise the significance of P-n junction diodes in opto-electronic devices.	Evaluate	1,2,3,4,9,10,11
5	Recognize the versatility of Lasers to realize their applications in latest technical developments.	Apply	1,2,3,4,5,9,10,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



ACTIVITIES:

- P Type and N Type-Hall Effect.
- Study the temperature dependence of resistance for the given material.
- solar Cell-Series and Parallel.

TEXT BOOKS:

1. M. N. Avadhanulu, “Engineering Physics”, S. Chand publications 2010.
2. Donald A. Neamen, “Semiconductor Physics and Devices: Basic principle”, 4th edition, McGraw-Hill, New York, 2012.

REFERENCE BOOKS:

1. D. Halliday, R. Resnick and J. Walker, “Fundamentals of Physics”, 6th edition, John Wiley and Sons, New York, 2001.
2. M. N. Avadhanulu, “Engineering Physics”, S. Chand publications 2010.
3. Charles Kittel, “Introduction to solid state physics”, 7th edition, Wiley, Delhi, 2007.
4. David J. Griffiths, “Introduction to Electrodynamics”, 3rd edition, Prentice Hall of India, New Delhi, 2012.
5. Ashcroft and Mermin, “Solid State Physics”, International student edition, 2008.

25CT102 ENGINEERING CHEMISTRY

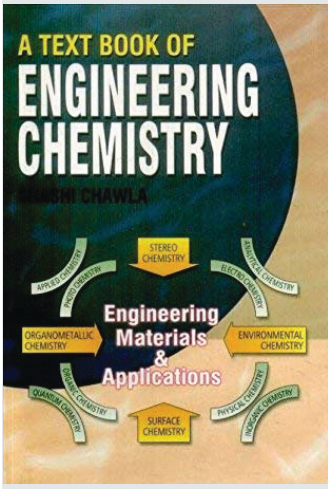


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Hours Per Week :

L	T	P	SL	C
3	0	2	3	4

PREREQUISITE KNOWLEDGE: Concept of bonding, chemical reactions and electrochemical cell.

COURSE DESCRIPTION AND OBJECTIVES:

The course aims to cover the importance of chemistry and its applications in engineering disciplines particularly focusing on developing new engineering materials (such as polymers, nanomaterials etc.) and understanding their property for scientific and engineering applications. In addition, the students are also expected to acquire knowledge on electrochemistry and construction of batteries and fuel cells.

MODULE-1

UNIT-1 **18L+0T+12P+18SL = 48 Hours**

CHEMICAL BONDING & STRUCTURE:

Introduction to Chemistry - Chemical bonding, VSEPR and VBT theory; Bond fissions, Introduction to reactive intermediates (carbanion, carbocation and free radical).

UNIT -2

INTRODUCTION TO MATERIALS AND ELECTROCHEMISTRY:

Introduction to materials, Type of materials: Polymers-Introduction and Classification, Nanomaterials-Metal and Carbon nanomaterials, Semiconductors – Introduction. Redox reactions, Electrode potential, EMF of an electrochemical cell.

PRACTICES:

- Determination of Fe (II) by Dichrometry method.
- Determination of strength of Weak acid by pH-method.
- Determination of strength of acid by conductometry.
- Synthesis of Phenol-Formaldehyde resin.
- Synthesis of Urea-Formaldehyde resin.

MODULE-2

UNIT-1 **27L+0T+18P+27SL = 72 Hours**

POLYMERS:

Types of Polymerizations, Preparation, properties, mechanism (free-radical) and applications of PE, Nylon-6,6; Conducting polymers: Introduction, Classification and Mechanism of conduction in Poly-acetylene, Applications. Introduction to Biodegradable polymers.

UNIT-2

ELECTROCHEMICAL ENERGY SYSTEMS:

Introduction - ***Agastya Samhit***ā (Agastya battery), Classification of batteries (primary and secondary). Construction, working and applications of Lead-acid and Li-ion batteries. Fuel Cells-Classification, Construction, working and applications of H_2O_2.

UNIT-3

MATERIALS CHEMISTRY:

Engineering Materials: Refractories - classification, refractoriness RUL, Chemical & Thermal stability; Lubricants – classification, viscosity, viscosity index, flash and fire points, cloud and pour points.

SKILLS:

- ✓ *Synthesize various polymers.*
- ✓ *Synthesize nanomaterials.*
- ✓ *Identify the properties of different industrially relevant engineering materials.*
- ✓ *Understand the different components of an electrochemical cell.*
- ✓ *Design electrochemical cell such as battery.*
- ✓ *Identify the types of energy conversion/storage systems.*

Carbon based Nanomaterials: Carbon nanotubes and graphene's –Synthesis, properties and applications.

Electronic memory devices: Introduction, organic semiconductors for memory devices: p-type semiconductor-pentacene, n-type-semiconductor perfluoropentacene; Introduction to organic memory devices.

PRACTICES:

- Preparation of Nylon-6,6.
- Determination of Strength of an acid in Pb-Acid battery.
- Potentiometer - determination of redox potentials and EMFs.
- Determination of viscosity of oil (Biodiesel, castor oil and coconut oil).
- Purity of lubricant- Saponification and neutralization number.
- Synthesis of magnetic Iron oxide nanoparticles.

COURSEOUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Understand chemical bonding and predicts the molecular geometry and reactivity using bonding VSEPR and VB theories.	Apply	1, 2, 3, 4, 5,8,10,11
2	Apply concepts of materials and electrochemistry to analyse the properties of polymers, nanomaterials, semiconductors, and redox systems.	Apply	1, 2, 3, 4, 5, 8, 9, 10, 11
3	Evaluate polymerization methods and mechanisms to assess the properties and applications of PE, Nylon-6,6, conducting, and biodegradable polymers.	Evaluate	1, 2, 3, 4, 5, 6, 7, 10
4	Analyze the classification, construction, and working principles of electrochemical and fuel cells, and examine their applications in energy storage and conversion.	Analyze	1, 2, 3, 4, 5, 7, 9, 10, 11
5	Apply the concepts of refractories, lubricants, carbon-based nanomaterials, and organic memory devices, and evaluate their properties and applications in engineering.	Apply	1, 2, 3, 4, 5, 6, 7, 9, 10

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXT BOOKS:

1. S. Chawala, "A Textbook of Engineering Chemistry Engineering Materials and Applications", Dhanpat Rai Publications, 3rd Edition, 2015.
2. K. S. Maheswaramma and M. Chugh, "Engineering Chemistry", Pearson, 1st Edition, 2015.

REFERENCE BOOKS:

1. V.R. Gowariker, N. V. Viswanathan, Jayadev, Sreedhar, "Polymer Science", New age Int. Publishers, 4th Edition, 2021
2. Kirby W. Beard, "Linden's Handbook of Batteries", Fifth Edition, McGraw-Hill, 2019.
3. MaliniS, K.S. Anantha Raju, "Chemistry of Engineering materials", CBS publishers Pvt. Ltd., 2022
4. T. Pradeep, "Nano: The Essentials; Understanding of Nano Science and Technology" Tata McGraw-Hill, New Delhi, 2012.
5. Takatoshi Tsujimura, "OLED Display Fundamentals and Applications", Wiley–Blackwell, 2012.

25EE101 BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING

Hours Per Week :

L	T	P	SL	C
2	0	2	2	3

PREREQUISITE KNOWLEDGE: Basics of Physics.

COURSE DESCRIPTION AND OBJECTIVES:

This course covers fundamental concepts of electrical and electronic circuits, semiconductor devices, and electrical machines. Topics include circuit analysis, AC and DC circuits, magnetic circuits, rectifiers, voltage regulators, and electrical machines like transformers and motors. Hands-on experiments reinforce theoretical concepts for real-world engineering applications.

MODULE-1

UNIT-1 **12L+0T+12P+12SL = 36 Hours**

FUNDAMENTALS OF ELECTRIC AND MAGNETIC CIRCUITS:

Circuit Elements and Laws- Types of circuit components and Sources, ohms law, KVL and KCL, series Parallel circuits, Comparison of DC and AC sources, Characteristics of AC waveform, Response of Resistor, Inductor and Capacitor for AC voltage.

Magnetic circuits- Ohm's law in Magnetic circuit (Magneto motive force, Reluctance and Flux), Analogy of Electrical and Magnetic circuits, Faraday's Law of Electromagnetism, Self and mutual inductances.

UNIT-2

SEMICONDUCTOR DEVICES:

Classification of semiconductors, Formation of PN Junction, Avalanche and Zener breakdown of diode, PN Junction Diode, Zener Diode and BJT-operation and their V-I characteristics, Piecewise linear model of a Diode.

PRACTICES:

- Verification of Ohm's law.
- Verification of Kirchhoff's current law.
- Verification of Kirchhoff's voltage law.
- Determination of R.M.S. Values of sinusoidal waveform.
- Obtain the V-I characteristics of Diode.
- Obtain the V-I characteristics of Zener diode.

MODULE-2

UNIT- 1 **18L+0T+18P+18SL = 54 Hours**

ANALYSIS OF ELECTRICAL CIRCUITS:

Mesh and nodal analysis of resistive circuits with DC source. Response of single- phase ac circuits with RL, RC, RLC loads, Resonant circuits. Introduction to three phase system, Relation between phase and line quantities in star and delta connected systems.

UNIT- 2

ELECTRICAL MACHINES:

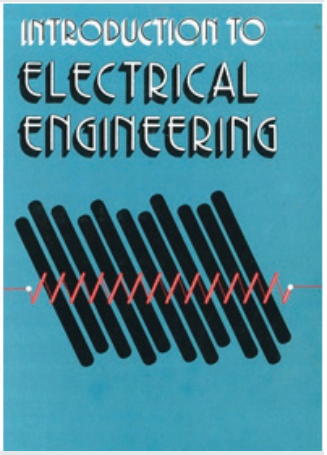


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SKILLS:

- ✓ *Distinguish between linear and nonlinear elements by looking at VI characteristics.*
- ✓ *Develop a simple loop generator.*
- ✓ *Design a voltage regulator using Zener diode.*
- ✓ *Design a half wave rectifier using PN junction diode.*
- ✓ *Design a full wave rectifier using PN junction diodes.*

DC MACHINES: Construction of DC Machine, DC Generator- working principle, EMF equation, DC Motor-Working principle and Torque equation.

AC MACHINES: Transformer- Construction, Operating Principle, and EMF Equation.

Three Phase Induction motor- Construction, Working Principle, and Torque vs. Slip Characteristic.

UNIT-3

ANALYSIS OF ELECTRONIC CIRCUITS:

Half Wave Rectifier- Operation, Efficiency, Ripple Factor, and Peak Inverse Voltage (PIV); Full Wave Rectifiers- Types, Operation, Efficiency, Ripple Factor, and PIV; Zener Diode as a Voltage Regulator.

PRACTICES:

- Transformation ratio of a single phase transformer at different loads.
- Measurement of Energy in single phase resistive load circuit.
- Measurement of Power in single phase resistive load circuit
- Determination of impedance in complex AC circuits.
- Verification of line and phase quantities in a balanced three phase system

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Analyze the resistive circuits with independent sources and find its solution.	Analyze	1,2,6,9
2	Solve the AC (single and three phase) and DC circuits using different methods.	Apply	1,2,9
3	Apply the concepts of electromagnetism for its applications.	Apply	1,2,3,9
4	Examine the different electrical equipment.	Evaluate	1,2,9
5	Acquire the knowledge of semiconductor devices to create circuits.	Create	1,2,3,9

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXTBOOK:

1. M Naidu and S. Kamakshiaih, “Introduction to Electrical Engineering”, Tata McGraw-Hill Edition, 2017.

REFERENCE BOOKS:

1. A.K. Thereja and B.L.Thereja, “Electrical Technology”, Vol.–II, S. Chand & Co., Publications, 2020.
2. V. K. Mehta, “Principles of Electrical Engineering and Electronics”, S. Chand & Co., Publications, New Delhi, 2019.
3. D.P. Kothari, “Basic Electrical and Electronics Engineering”, TMH, New Delhi, 2017.

25ME101 ENGINEERING DRAWING

Hours Per Week :

L	T	P	SL	C
2	0	2	2	3

PREREQUISITE KNOWLEDGE: Basics of Geometry.

COURSE DESCRIPTION AND OBJECTIVES:
This course provides a foundational understanding of engineering drawing, focusing on basic projections, dimensioning, and reading technical drawings relevant across various engineering disciplines. The objective is to equip learners with essential skills to visualize, interpret, and communicate design ideas effectively using standard drawing practices.

MODULE-1

UNIT-1 **12L+0T+12P+12SL = 36 Hours**

GEOMETRICAL CONSTRUCTIONS AND CONICAL CURVES:
Introduction: Lettering and Dimensioning, Types of Lines.
Geometrical Constructions: Regular polygons using angle, ARC and general methods. Conical Curves: Construction of ellipse, parabola and hyperbola by using eccentricity method.
Śulba Sūtras (Vedic Geometry)

UNIT-2

PROJECTION OF POINTS AND STRAIGHT LINES:
Introduction: Reference planes, importance of reference lines and planes; Principles of projection.
Projection of Points: Projection of points located in any of the four quadrants.
Projection of Straight Lines: Projection of straight lines - parallel to both the reference planes, parallel to the one plane and inclined to other plane, parallel to one plane and perpendicular to other plane.

PRACTICES:

- A road roller covers ‘x’ kilometres distance in one day & the roller completed ‘N’ revolutions in a minute. By using the above information, inscribe a regular hexagon and pentagon in a circle of diameter, D. Where ‘D’ is considered as the diameter of the road roller. Also evaluate the length of each side of the regular hexagon and pentagon.
- A cow is tethered in the middle of a field which is in the form of circular shape and suppose that cow grazes ‘x’ square meters in a day. By using the above information, inscribe regular hexagon and pentagon in a circle of diameter ‘D’.
- Two oranges on a tree are respectively x m and y m above the ground, and p m and q m from a r m thick wall, but on the opposite sides of it. The distance between the oranges, measured along the ground and parallel to the wall is ‘L’ m. (a) Determine the real distance between the oranges and the angle of inclination of the line joining the oranges with the ground and also with the wall.
- A room measures ‘x’ m long, ‘y’ m wide and ‘z’ m height. Assume (x>y>z). An electric bulb hangs in the centre of the ceiling and ‘p’ m below it. A thin straight wire connects the bulb to a switch kept in one of the corners of the room and ‘r’ m above the floor. Draw the projections.
- On lighting a rocket cracker, it gets projected in a parabolic path and reaches a maximum height of ‘x’ m when it is ‘y’ m away from the point of projection. Finally, it reaches the ground ‘z’ m away from the starting point. Find the angle of projection.
- Two pegs fixed on a wall are ‘x’ meters apart. The distance between the pegs measured parallel to the floor is ‘y’ meters. If one peg is ‘z’ meters above the floor. Consider (x>y>z) & assume suitable scales. Design the height of the second peg and the inclination of the line joining the two pegs with the floor. Develop the projections.

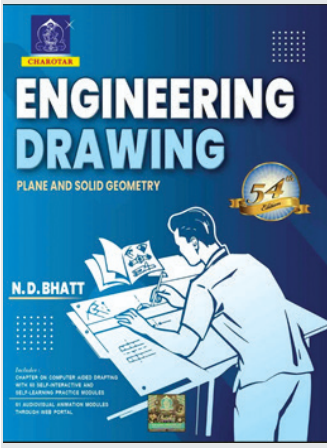


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SKILLS:

- ✓ *Convert iso-metric views of objects into orthographic views and vice versa.*
- ✓ *Visualize the shape of the 3D components.*
- ✓ *Create pictorial views by using AutoCAD.*
- ✓ *Understand projections by visualization.*

MODULE-2

UNIT-1 **18L+0T+18P+18SL = 54 hours**

PROJECTIONS OF PLANES:
Regular planes perpendicular to one reference plane and inclined to the other reference plane.

UNIT-2

PROJECTIONS OF SOLIDS:
Axis parallel to one reference plane and inclined to other plane - prisms, pyramids.

UNIT-3

CONVERSION OF ISOMETRIC PROJECTIONS TO ORTHOGRAPHIC PROJECTIONS:
Conversion of isometric views of geometrical shapes, and simple orthographic views.

PRACTICES:

- A plate having shape of an isosceles triangle has base 'x' mm long and altitude 'y' mm. It is so placed that in the front view it is seen as an equilateral triangle of 'x' mm sides and one side inclined at an angle 'θ' degrees to the reference line. Develop its front view and top view.
- Prepare the top-view (plan) of a class-room/lab, home (Drawing Room/Bedroom/ Study Room, Kitchen) along with suitable dimensions.
- A lighting point in a ceiling design is 0 mm above the HP and 50 mm in front of the VP (i.e., lies exactly on HP). Draw views and mark the location. Explain what it means physically.
- A sensor mounted on a panel lies 25 mm below the panel surface (HP) and 60 mm in front of the operator (VP). Draw the reference line and mark views. Justify point placement.
- You are given an isometric view of an L-shaped bracket with holes and steps. Draw the orthographic views (Front, Top, and Right Side views).

ACTIVITIES:

- Developing the following solids with the help of cardboard/ thick paper, cube, cuboid prisms & pyramids (triangular, square, pentagonal, and hexagonal) right circular cylinder and cone.
- Design a Bolt and nut assembly for the given dimensions.
- Design a Connecting rod of IC engine for the given dimensions.
- Design a I.C Engine Piston with piston rings for the given dimensions.

COURSE OUTCOMES:
Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Creating and interpreting technical drawings.	Apply	1,3,5
2	Apply the drawing skills in representing various geometrical features.	Apply	1,3,5
3	Develop orthographic projections and isometric views of various objects using AutoCAD.	Analyse	1,3,5
4	Sketch simple objects and their pictorial views using Auto-CAD.	Analyse	1,3,5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOK:

1. N D Bhatt, "Engineering Drawing", 54th edition, Charotar Publication, 2023.

REFERENCE BOOKS:

1. S. N. Lal, "Engineering Drawing with an Introduction to Auto CAD", 1st edition, Cengage, 2017.
2. K L Narayana, "Engineering drawing", 2nd edition, SciTech Publications, 2014.

25CS101 PROGRAMMING IN C

Hours Per Week :

L	T	P	SL	C
2	0	4	2	4

PREREQUISITE KNOWLEDGE: Fundamentals of Problem Solving.

COURSE DESCRIPTION AND OBJECTIVES:

This course is aimed to impart knowledge on basic concepts of C programming language and problem solving through programming. It covers basic structure of C program, data types, operators, decision making statements, loops, functions, strings, pointers, structures and unions. At the end of this course, students will be able to design, implement, test and debug complex problems using features of C.

MODULE-1

UNIT-1 **12L+0T+24P +12SL=48 Hours**

INTRODUCTION TO ALGORITHMS AND PROGRAMMING LANGUAGES:

Introduction to Algorithms: Introduction to computational thinking – Problem solving with algorithms and flowcharts - Introduction to C: Structure of a C program - pre-processor statement) inline comments, variable declaration statements, executable statements; C Tokens - C character set, identifiers and keywords, type qualifiers, type modifiers, variables, constants, punctuations and operators.

Data Types and Operators: Basic data types; Formatted I/O; Reading and writing characters; Operators - assignment, arithmetic, relational, logical, bitwise, ternary, address, indirection, sizeof, dot, arrow, parentheses operators; Expressions - operator precedence, associative rules.

Control Statements: Introduction to category of control statements; Conditional branching statements - if, if– else, nested-if, if – else ladder, switch case;

Iterative statements - for, while, do - while, nested loops; Jump statements - break, jump, goto and continue.

Algorithmic logic in Panini's Ashtadhyayi; problem-solving in Indian logic ***(Nyaya Shastra)***, Structured numerical systems – Vedic mathematics and early binary concepts

UNIT-2

FUNCTIONS & POINTERS:

User-defined functions: Function declaration - definition, header of a function, body of a function, function invocation; Call by value, Call by address. Recursion; Library Functions; Storage classes; Scope of a variable.

Pointers: Declaration, Initialization, Multiple indirection, Pointer arithmetic.

Recursive hymns ***(Rigveda),*** modular constructs in Shastra classification, Concepts of abstraction and references in Indian epistemology ***(Pramana).***

MODULE-2

UNIT-1 **18L+0T+36P +18SL=72 Hours**

ARRAYS:

Arrays: Introduction; Types of arrays; Single dimensional array - declaration, initialization, usage, reading, writing, accessing, memory representation, operations; Multidimensional arrays. Passing arrays to functions; Relationship between arrays and pointers, scaling up - array of arrays, array of pointers, pointer to a pointer and pointer to an array; Dynamic memory allocation functions – Creating arrays dynamically and performing operations on them.

Representation of multidimensional data in ancient Indian astronomy and calendar systems Granularity and scalability in system design found in traditional architecture (e.g., ***Vastu***).



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SKILLS:

- ✓ *Analysis of the problem to be solved.*
- ✓ *Select static or dynamic data structures for a given problem and manipulation of data items.*
- ✓ *Develop C programs that are understandable, debuggable, maintainable and more likely to work correctly in the first attempt.*

UNIT-2

STRINGS:

Strings: Character array, reading strings from the standard input device, displaying strings on the standard output device, Importance of terminating a string, Standard string library functions.

Preservation and transmission of Sanskrit and Tamil texts using structured phonetic systems.

UNIT-3

STRUCTURES, UNIONS:

Structures: Defining a structure, declaring structure variable, Operations on structures, Pointers to structure - declaring pointer to a structure, accessing structure members using pointer; Array of structures, Nested structures, passing structures to functions - passing each member of a structure as a separate argument, passing structure variable by value, passing structure variable by reference/ address; Typedef and structures.

Unions: Defining a union - declaring union variable, operations on union.

Classification of knowledge in Indian systems – Charaka Samhita, Manuscripts, Palm-leaf layouts, Group representation in indigenous knowledge frameworks – e.g., Ayurveda Tridosha classification.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Explain algorithmic problem-solving, flowchart design, and fundamental programming constructs in C.	Apply	1, 2
2	Analyze and implement various control structures (conditional, iterative, and jump statements) for effective decision-making in programs.	Analyze	2, 3, 4
3	Implement modular programming by designing user-defined functions and utilize recursion effectively for problem-solving. Apply array concepts, including single and multi-dimensional arrays, dynamic memory allocation, and pointer-array relationships in programming.	Apply	1, 2, 3, 4, 5
4	Utilize string handling techniques and standard string library functions to solve text-processing problems effectively.	Evaluate	1, 2, 3, 5
5	Design, implement, and analyze structured data types using structures and unions for efficient handling of complex data.	Create	1, 2, 3, 5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOKS:

1. Behrouz A. Forouzan, Richard F.Gilberg, "Programming for Problem Solving", 1st edition, Cengage publications, 2019.
2. Ajay Mittal, "Programming in C - A Practical Approach", 1st edition, Pearson Education, India, 2010.

REFERENCE BOOKS:

1. 1. R.G. Dromey, "How to Solve it by Computer", Prentice-Hall International Series in Computer Science, USA.
2. Reema Thareja, "Computer Fundamentals and Programming in C", 1st edition, Oxford University Press, India, 2013.
3. Herbert Schildt, "C: The Complete Reference", 4th edition, Tata McGraw-Hill, 2017.
4. Byron S Gottfried, "Programming with C", 4th edition, Tata McGraw-Hill, 2018.

25EN103 ENGLISH PROFICIENCY AND COMMUNICATION SKILLS

Hours Per Week :

L	T	P	SL	C
0	0	2	0	1

PREREQUISITE KNOWLEDGE: Basics of grammar, Read and understand for global context, Cultural sensitivity and Basic writing skills.

COURSE DESCRIPTION AND OBJECTIVES:
English Proficiency and Communication Skills seeks to develop the students' abilities in listening, speaking, reading, writing and grammar. The course will provide students an exposure on a wide range of language use in everyday situations. It will make the students to equip with functional English and make them use it confidently in their professional and social contexts. Finally, students will strengthen their listening, speaking, reading, writing skills in English.

MODULE-1

UNIT-1 **0L+0T+12P +0SL=12 Hours**

MY LIFE AND HOME – MAKING CHOICES – HAVING FUN:

Listening: Understanding short monologues or dialogues and choose the correct visual.

Speaking: Self-introduction, agreeing, disagreeing, making decision, cultural matters in a limited way.

Reading: Understanding main message, factual information global meaning, specific information and paraphrasing.

Writing: Developing hints-based mail, Writing short messages/paragraphs.

Vocabulary: Discerning use of right word suiting the context, B1 Preliminary word list.

Grammar: Frequency Adverbs, State Verbs and Prepositions. In Indian philosophies like ***Vedanta, Sankhya,*** and ***Yoga,*** the concept of cyclical time is fundamental. the ***Atman*** (soul) is eternal and always present. In Indian agriculture, astrology, and spirituality, the regular appearance of seasons or celestial events often translates to frequency. (Life on Earth, governed by cycles of karma and rebirth, reflects actions that happen at varying intervals).

UNIT-2

ON HOLIDAY – DIFFERENT FEELINGS – THAT'S ENTERTAINMENT:

Listening: Understand straightforward instructions or public announcements.

Speaking: Describing people, things and places in a photograph, making suggestions, discussing plans.

Reading: Longer text for detailed comprehension, gist and inference.

Writing: Developing notes and responding to penfriends or 'e-pals'.

Vocabulary/Grammar – Comparatives and Superlatives, Gradable and non-gradable adjectives, Cloze tests.

PRACTICES:

- Developing hints based mail.
- Writing short messages.
- Writing paragraphs.
- Expressing opinions and cultural matters.

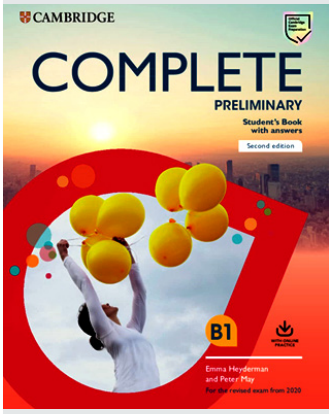


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SKILLS:

- ✓ *Use of appropriate grammar and vocabulary with syntactic patterns in short texts.*
- ✓ *Read and extract the main message, global meaning, specific information, detailed comprehension, understanding of attitude, opinion and writer purpose and inference.*
- ✓ *Listen to understand key information, specific information, gist and detailed meaning and to interpret meaning.*
- ✓ *Understand questions and make appropriate responses and talk freely on everyday topic.*

- Understanding short monologues.
- Understanding straightforward instructions and public announcements.
- Describing people, things and places in a photograph.

MODULE-2

UNIT-1 **0L+0T+18P +0SL=18 Hours**

GETTING AROUND – INFLUENCES:

Listening: Discussion activities and listening to understand the gist of each short dialogue.

Speaking: Snap Talks, Make and respond to suggestions, discuss alternatives and negotiate agreement.

Reading: Reading for understanding coherence of the text and drawing inferences.

Writing: Reading an announcement from a magazine or website for preparing an article.

Vocabulary / Grammar: Punctuation, Prepositions, (Mount Meru, in Hindu, Buddhist, and Jain cosmology, is considered the center of the universe, and many deities are believed to reside on its peaks). Phrasal Verbs, B1 Preliminary word list.

UNIT-2

STAY FIT AND HEALTHY – LOOKS AMAZING!

Listening: An interview for a detailed understanding of meaning and to identify attitudes and opinions.

Speaking: Discuss likes, dislikes, experiences, opinions, habits, etc.

Reading: Content, Communicative Achievement, Organisation and Language.

Writing: Developing a story with clear links to the given opening sentence.

Vocabulary / Grammar: Modals, Conditionals, Indian philosophy, particularly in Hinduism, Buddhism, and Jainism, the law of karma is central to understanding the relationship between actions and their consequences. Verb forms (Time and Tense).

UNIT-3

THE NATURAL WORLD – EXPRESS YOURSELF!

Listening: Listening for specific purpose, listening for identifying the situation.

Speaking: Discussing sports, discussing everyday objects in photos, talking about habits and routines.

Reading: Reading to identifying text purpose, Identifying opinion and attitude.

Writing: Reported speech, using range of tenses.

Vocabulary / Grammar: Collocations, Adjective Prefixes and Suffixes.

PRACTICES:

- Listening to understand the gist of each short dialogue.
- Listening to an interview for a detailed understanding of meaning and to identify attitudes and opinions.
- Preparing an article.
- Discuss for alternatives and negotiate agreement.
- Discussion on likes, dislikes, experiences, opinions, habits, etc.

ACTIVITIES:

- Snap talks.
- Making and responding suggestions.
- Discussion on likes, dislikes, experiences, opinions, habits, etc.
- Description of people, things and places in a photograph.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	apply to read and grasp content on a range of topics/texts related to their everyday life like notifications, advertisements, travel brochures, news reports, articles.	Apply	7, 8, 9, 10, 11
2	apply suitable strategies to achieve comprehension, like listening for main points and checking comprehension using contextual clues etc.	Apply	7, 8, 9, 10, 11
3	demonstrate vocabulary beyond that of the familiar subjects.	Analyze	7, 8, 9, 10, 11
4	show sufficient control of English grammar and sentence variety to coherently organize information at sentence and discourse levels.	Evaluate	7, 8, 9, 10, 11
5	use functional English to communicate and interact effectively in everyday situations.	Create	7, 8, 9, 10, 11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOK:

1. Emma Heyderman and Peter May, “Complete Preliminary”, Student’s Book with Answers, 2nd edition, Cambridge University Press, 2019.

REFERENCE BOOKS:

1. Annette Capel and Rosemary Nixon, “Introduction to PET”, Oxford University Press, 2009.
2. Adrian Doff and Craig Thaine, “Empower Pre intermediate”, Cambridge University Press, 2015.
3. Louise Hashemi and Barbara Thomas, “Objective PET”, Cambridge University Press, 2010.

25CT101 ENVIRONMENTAL STUDIES

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: General awareness regarding environmental problems and importance of environmental protection.

COURSE DESCRIPTION AND OBJECTIVES:
It is a multidisciplinary subject where we deal with different aspects using a holistic approach. It is evolving to be the education for sustainable and ethical development both at a local and global level. It helps to prepare the next generation and to plan appropriate strategies for addressing environmental issues. It identifies and create solutions that conserve to manage ecosystem and biodiversity and also helps to eliminate pollutants, toxicants to preserve air, water and soil quality. Environmental education recognize impacts of global issues, enhances the public awareness and helps to take decisions towards environmentally responsible actions.

MODULE-1

UNIT-1 **12L+12T+0P+12SL=36 Hours**

SUSTAINABLE DEVELOPMENT, NATURAL RESOURCES AND ECOSYSTEMS:
Sustainable Development - Concept and Sustainable Development Goals. Natural Resources- Forest, water, energy, mineral and land resources. Concept of Ecology and Ecosystems, Ecological succession, Population Ecology.

UNIT-2

BIODIVERSITY AND ECOSYSTEM CONSERVATION:
Biodiversity concept, Biodiversity hot spots, Biodiversity indices (Simpson index and Shannon index) Threats to biodiversity, Biodiversity conservation (in-situ and ex-situ). ***Traditional Knowledge*** in biodiversity conservation, Ecosystem protection and its maintenance.

PRACTICES:
- Wealth from waste, (development of value-added products from waste materials).
- Biogas production.
- Herbarium sheet preparation.
- Models: wind mills, solar cell, bioenergy, Vertical garden, green building models.
- Preparation of Compost.
- Installation of artificial nests for local bird population.
- On campus plantation programme.
- Disposal of laboratory waste- Individual practice and report writing, lab safety awareness

MODULE-2

UNIT-1 **18L+18T+0P+18SL =54 Hours**

ENVIRONMENTAL ISSUES AND WASTE MANAGEMENT:
Emerging local and global environmental issues, Waste Management-Municipal, Industrial. Biomedical, hazardous and radioactive waste management, Waste-water treatment.

UNIT-2

ENVIRONMENTAL HEALTH AND SAFETY:
Occupational health and hygiene, Toxicity studies (Heavy metal toxicity, Tobacco, Silicosis, Xenobiotics and Carcinogenicity) and its remedy.

UNIT-3

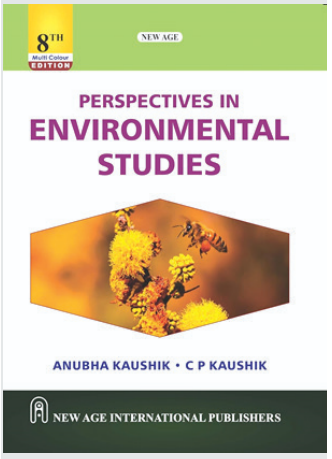


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ADVANCED TECHNIQUES IN POLLUTION CONTROL:

Pollution control devices -Catalytic converters, Electrostatic precipitators, Scrubbers, Cyclonic separators, Gravitational settling chambers, Carbon capture technology, Water pollution technologies- membrane filtration, bioremediation, Bio manipulation technology.

PRACTICES AND FIELD PROJECT:

- Comparative analysis of water quality metrics to determine whether the water is suitable for irrigation and drinking purposes.
- Survey of flora and fauna in your own remote areas for assessment of local Biodiversity.
- Comparative study of traditional versus modern waste collection systems.
- Study on the effects of household items and detergents on the quality of the water.
- Detailed study of any environmental problem/situation, Identification of factors pertaining to that problem with conclusion.
- Study of agricultural runoff in crop lands of Angalakuduru and Sekuru.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Ability to apply knowledge, research skills and field techniques to protect environment.	Apply	2, 6, 9, 11
2	Examine the impacts of ecosystems, biodiversity and apply conservation strategies to curb the issues for sustainability	Apply	2, 6,9, 11
3	Survey on various waste management approaches and develop innovative solutions to solve the issues in the future	Evaluate	2, 3, 6,11
4	Create and implement useful skills to solve occupational health and environmental issues, ensuring safer work places and healthier communities	Apply	2,3, 6, 11
5	Analyse, design and implementation of sophisticated control system to manage industrial emissions and manage environmental pollutants	Analyse	1,2 3, 6,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOKS:

1. Anubha Kaushik and C. P. Kaushik, “Perspectives in Environmental Studies”, 8th edition, New Age International Publishers, 2024.
2. M. Anji Reddy, “Textbook of Environmental Science and Technology” B. S. Publications, 2023.

REFERENCE BOOKS:

1. B. Joseph, “Environmental Studies Simplified”, 3rd edition, Mc Graw Hill Education (India), 2024.
2. John Pichtel, “Waste Management Practices: Municipal, Hazardous, and Industrial”, 2nd Edition, Taylor and Francis Books India Pvt. Ltd, 2024.
3. Pankaj Gupta,” Environmental Health and Occupational Safety”. 1st Edition, Routledge Taylor and Francis group, 2024.
4. C. S. Rao, “Environmental Pollution Control Engineering”, New Age International Publishers, 2021
5. P.D. Sharma, “Ecology and Environment”,13th Edition, Rastogi publication, 2022.
6. Anil Kumar De, “Environmental Chemistry”, 10th Edition, New Age International Private Limited, 2022.

25MS101 MANAGEMENT STUDIES

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Management Fundamentals, Operations Management, Business Environment.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims to provide foundational knowledge in the science of management by examining classical and modern management theories and their applications in business operations, HR, and marketing. The course equips students with tools to analyse markets, customers, and internal organizational processes, thereby enabling them to develop effective managerial strategies.

MODULE-1

UNIT-1 **12L+12T+0P+12SL=36 Hours**

INTRODUCTION TO MANAGEMENT:

Concepts of Management and Organization – Nature, Importance and Functions – Evolution of Management Thought: Taylor's Scientific Management, Fayol's Principles, Administrative Theory, Mayo's Experiments. Theories of Motivation: Maslow and Herzberg. Leadership Styles. Corporate Social Responsibility from Indian Perspectives.

UNIT-2

OPERATIONS MANAGEMENT:

Plant Location Principles, Plant Layout Types, Productivity Measures, Project Management – Planning, Scheduling and Controlling, Inventory Control Techniques – EOQ, ABC Analysis, Stores and Purchase Management.

PRACTICES

- Design a management strategy for a Start-up using planning, organizing, leading, controlling.
- Conduct a leadership style analysis comparing crisis vs. stability periods.
- Prepare a CSR initiative aligned with Indian ethics and practices.
- Create a simplified Gantt chart for a semester project schedule.
- Compute EOQ for a sample product with demand, ordering, and holding cost data.
- Simulate inventory classification using ABC analysis for a retail store.
- Demonstrate statistical process control using chart data sets.
- Evaluate functions of management through a case study of a real-life business.

MODULE – II

UNIT-1 **18L+18T+0P+18SL=54 Hours**

HUMAN RESOURCE MANAGEMENT:

Concepts and Functions of HRM – Manpower Planning – Recruitment & Selection – Training & Development – Wage & Salary Administration – Promotion, Transfer and Separation – Performance Appraisal – Grievance Handling – Welfare Measures – Job Evaluation and Merit Rating, ***Ancient Gurukul systems,*** mentorship models, values-based leadership.

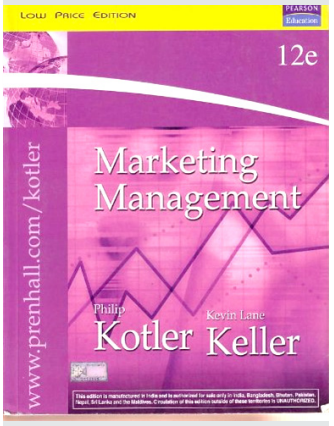


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SKILLS:

- ✓ *Strategic Thinking and Managerial Decision-Making.*
- ✓ *Human Resource Planning and Administration*
- ✓ *Application of Operations and Inventory Techniques.*
- ✓ *Marketing Analysis and Digital Campaign Planning.*
- ✓ *Ethical and Sustainable Business Practices.*

UNIT-2

MARKETING MANAGEMENT:

Evolution and Functions – Selling vs. Marketing – 4Ps & 7Ps of Marketing – Product Mix & Life Cycle – Price Mix & Pricing Methods – Place Mix & Channels – Promotion Mix Tools – Packaging & Process in Marketing.

UNIT-3

EMERGING TRENDS IN BUSINESS MANAGEMENT:

Digital Transformation in Business – Role of AI in business, Remote & Hybrid Work Models, Digital Marketing Innovations, FinTech & Digital Payments, E-Commerce & Social Commerce.

PRACTICES:

- Draft a job description and specification for a chosen designation.
- Prepare an HR training calendar for a small business.
- Design a grievance handling mechanism with a workflow.
- Conduct a mock interview and document evaluation criteria.
- Analyze the marketing mix for a consumer product or service.
- Conduct a class debate on: "Is Digital Marketing replacing Traditional Marketing?"
- Analyze an Indian e-commerce platform's business model (e.g., Flipkart, Nykaa).
- Compare and contrast HR practices across two industry sectors.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Analyze core management principles and classical theories	Analyze	1,2
2	Apply operations management techniques	Apply	1,2,3
3	Evaluate human resource management practices	Evaluate	1,2
4	Evaluate the markets, customers and competition	Evaluate	1,2,3
5	Critically assess emerging trends such as digital transformation	Apply	1,2,3

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOKS:

1. Kotler Philip & Keller Kevin Lane, Marketing Management, 12th edition, PHI, 2005.
2. Koontz & Weihrich, Essentials of Management, 6th edition, TMH, 2005.
3. Aryasri, Management Science, 4th edition, TMH, 2004.

REFERENCE BOOKS:

1. Stoner, Freeman, Gilbert, Management, 6th edition, Pearson Education, 2018.
2. Thomas N. Duening & John M. Ivancevich, Management: Principles and Guidelines, Biztantra, 2003.

25CS102 PROBLEM SOLVING THROUGH PYTHON

Hours Per Week :

L	T	P	SL	C
2	0	2	2	3



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PREREQUISITE KNOWLEDGE: Prior knowledge of any programming language and object-oriented concepts is helpful but not mandatory.

COURSE DESCRIPTION AND OBJECTIVES:
This course offers sufficient knowledge required to understand the fundamental concepts of Python programming language. This course enables students to choose appropriate data structures (lists, dictionaries, tuples, sets, strings) for the given problem. In addition, the students will be able to create reliable, modular and reusable applications using Object- Oriented Programming approaches. At the end they will get an idea of how to access database using python programming, develop web applications, and using web Services using python Programming.

MODULE-1

UNIT-1 **12L+0T+12P+12SL=36 Hours**

INTRODUCTION:
Introduction to python, Variables, Assignment, Keywords, Built-in functions, Indentation, Comments, Basic data types - integers, float, complex, Boolean, strings; Python program development, running python using REPL shell, Python scripts.
Operators and Expressions: Operators- arithmetic operators, comparison (relational) operators, assignment operators, logical operators, bitwise operators, membership operators, identity operators; Expressions and order of evaluations.
Control Structures: Conditional control structures - if, else if, else; Loop control structures - for, while, for... else, while…else, nested loops, break, continue, pass.
Structured rule-based logic as in ***Panini's grammar;*** algorithmic thinking in ***Nyaya philosophy.***

UNIT-2

PYTHON DATA STRUCTURES AND FUNCTIONS:
Data Structures: Lists, Tuples, Sets, Strings, Dictionary - creation, accessing, basic operators and methods; List comprehensions.
Functions: Defining functions, calling functions, Passing arguments - keyword arguments, default arguments, positional arguments, variable-length arguments; Types of functions- anonymous functions, fruitful function, recursive functions; Scope of the variables- global and local variables, Development of recursive and non-recursive functions.
Recursive and modular thinking in Vedic systems; structured data modeling akin to Ayurveda and traditional knowledge.

MODULE-2

UNIT-1 **18L+0T+18P+18SL=54 Hours**

MODULES, PACKAGES AND OOPS:
Creating modules, Import Statement, From...Import Statement, Name Spacing, Creating user defined modules.
Standard Modules: sys, math, date, time, os, random and itertools modules.
Packages: Package Initialization, Importing * From a Package, Sub packages.
Modular organization of knowledge in Indian philosophical schools; abstraction in ancient systems like ***Arthashastra.***

UNIT 2

OBJECT ORIENTED PROGRAMMING:

Classes and Objects, Attributes and Methods, Constructors and Destructors- (init and del methods), Encapsulation - bundling of data and methods within a class, promotes data integrity and prevents unauthorized access, use of access modifiers to control access to attributes and methods; Inheritance, Method Overriding, Polymorphism, Abstraction, Class and Instance Variables - Differentiate between class variables (shared among all instances) and instance variables (unique to each instance), Access Modifiers-access modifiers and their role in controlling access to class members.

Operator Overloading: customization of operators for user- defined classes- define methods such as add, sub, mul, etc.

Traditional agricultural methods (Krishi-Parashara), Ayurveda-based healthcare practices.

UNIT-3

ERRORS AND EXCEPTIONS AND FILES:

Introduction to Exceptions, Handling Exception, Try Except Else and Finally Block, Raising Exceptions. File Processing: Reading and writing files -creating a new file, writing to a file, reading text files, opening and closing files, reading, writing, tell (), seek (), rename ().

Risk assessment in ancient Indian governance (Arthashastra); traditional knowledge documentation and preservation.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Analyze various features of programming language and their application in problem solving in computer programming to write, compile, and debug programs in python language.	Analyze	1,2
2	Make use of different data types to design programs involving decisions, loops, and functions.	Apply	1,2,5
3	Analyze the usage of different data structures for practical and contemporary applications for a given problem.	Analyze	1, 2, 3, 5
4	Choose appropriate programming paradigms, interrupt and handle exceptions using files to propose solution through reusable modules	Apply	1, 2, 3,5
5	Develop applications for a range of problems using object-oriented programming techniques.	Apply	1, 2,3,4,5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOKS:

1. Kenneth A. Lambert, “The Fundamentals of Python: First Programs”, Cengage Learning, 2011.
2. Mark Lutz, “Learning Python”, 5th edition, Orielly Publishers, 2013.
3. Reema Thareja, ‘Python Programming using problem solving approach”, oxford university press,2017.

REFERENCE BOOKS:

1. Introduction to Computation and Programming Using Python. John V. Guttag, The MIT Press.
2. James Payne, “Beginning Python using Python 2.6 and Python 3”, Wrox publishing.
3. Paul Gries, “Practical Programming: An Introduction to Computer Science using Python 3”, The Pragmatic Bookshelf, 2nd edition, 4 Oct. 2013.
4. Allen B. Downey, “Think Python”, 1st edition, Orielly publishing.

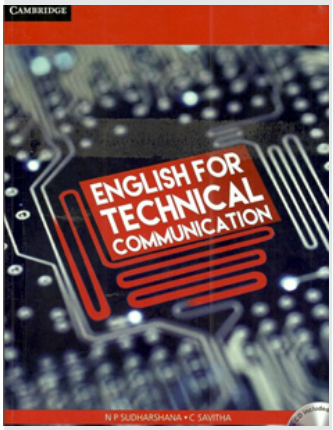


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25EN104 TECHNICAL ENGLISH COMMUNICATION

Hours Per Week :

L	T	P	SL	C
1	0	2	1	2

PREREQUISITE KNOWLEDGE: Basic sentence formation, understanding contextual meanings, basic writing skills and moderate fluency in English.

COURSE DESCRIPTION AND OBJECTIVES:
In this course students will read, analyze, and interpret material from technical and general fields, and practice reading, writing, listening and speaking skills to gain exposure and functional English on a variety of contemporary topics. The overall course objective is to provide English for Specific Purposes(ESP) instruction to enhance students' reading, writing, listening and speaking skills through a practice in the language. It will aim to build students' confidence and motivation through exposure to academic skills like Note making/taking, Paraphrasing, Summarizing, Report Writing, Making Presentations etc., so as to generate interest in the language from an ESP perspective. Finally, students are expected through the course to gain key strategies and expression for communicating with professionals and non- specialists.

MODULE-1

UNIT-1 **6L+0T+12P+6SL= 24Hours**

EFFECTIVE COMMUNICATION: EXPLORING CHANNELS AND METHODS:

Sub-themes: Types of communication, non-verbal communication, communication strategies and Barriers of communication, digital communication tools.

*IKS- Traditional Communication Methods in Ancient India and the ***Gurukul system*** emphasized face-to-face communication, clarity in speech, and respectful dialogue.

Vocabulary Building: Roots, Prefixes and Suffixes-commonly used 50.

Grammar: Articles and Prepositions.

Reading: Skimming and scanning for information.

Writing: Paragraph Writing, Summarizing.

Listening: Note Making, Note Taking.

Speaking: Introducing One- self.

UNIT- 2

A NEW ERA OF HEALTH CONSCIOUSNESS: IN PURSUIT OF BETTER HEALTH:

Sub-themes: Importance of physical and mental well-being, benefits of healthy lifestyle choices, current health trends.

*IKS- "Traditional Indian Approaches to Health and Well-being"

Vocabulary Building: Homophones and Homonyms (commonly Miss spelt Words).

Grammar: Tenses.

Reading: Reading for specific details – Reasons.

Writing: Process Description.

Listening: Listening for main ideas and supporting details on health topics.

Speaking: Object Description (JAM).

PRACTICES:

- Note making and Note Taking.
- Summarizing.
- Paragraph Writing.
- Reading Comprehension
- Error Correction and Restructuring.
- Vocabulary building.
- Process Description.
- Tenses.

MODULE-2

UNIT-1 **9L+0T+18P+9SL= 36Hours**

DISRUPTIVE TECHNOLOGIES: ETHICAL IMPLICATIONS AND FUTURE PROSPECTS:

Sub-themes: Introduction to artificial intelligence, machine learning, Augmented and Virtual Realities, AI in different sectors, Colonizing the Space and Cloning, ethical implications of AI, future prospects.

Vocabulary Building: Idioms and Phrases

Grammar: Subject-Verb Agreement

Reading: Reading for Tone, intention and Irony for Global understanding

Writing: Paraphrasing, Letter Writing

Listening: Listening for main ideas

Speaking: JAM

UNIT-2

ENTREPRENEURSHIP AND START-UP CULTURE: EMERGING TRENDS:

Sub-themes: Startup ecosystem, emerging industries, the role of technology in entrepreneurship, global market trends.

Vocabulary Building: Words Confused

Grammar: Active and Passive Voice

Reading: Reading for specific details

Writing: E-mail Etiquette, Data Interpretation, Report Writing

Listening: Note Making from a Long Lecture

Speaking: Group Discussion

UNIT-3

ETHICAL LEADERSHIP: INTEGRATING SOCIAL RESPONSIBILITY AND CORE VALUES:

Sub-themes: Corporate social responsibility (CSR), ethical business practices, environmental responsibility.

Principles of Dharma and Leadership in Ancient Indian Thought

Vocabulary Building: Homo Phones-Spellings (commonly Miss Spelt Words).

Grammar: Synthesis – use of coordinating and subordinating Conjunctions.

Reading: Reading for tone, intention and irony.

Writing: Report Writing, Reviews and Social Media Messaging-Etiquette.

Listening: Listening for main ideas and supporting details on health topics.

SKILLS:

- ✓ *Apply different sub-skills like skimming, scanning, reading for information, reading for inference etc. to understand different kinds of text.*
- ✓ *Apply different sub skills like top down, bottom up approaches to listening.*
- ✓ *Use functional vocabulary relevant to engineering and technology to express ideas lucidly.*
- ✓ *Use appropriate sentence structure, cohesive devices to construct simple text in regular correspondence like e-mails and letters.*

Speaking: Team Presentations

PRACTICES:

- E-mail writing.
- Letter writing.
- Paraphrasing
- Report writing.
- Process Description.
- Data Interpretation.
- Messaging in Social media.
- Writing Reviews.
- Group Discussion.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply a variety of strategies to interpret and comprehend spoken texts/discourse using contextual clues.	Apply	6, 7, 8, 9, 10, 11
2	Apply appropriate reading strategies to interpret content / material related to engineering and technology domain.	Apply	6 7, 8, 9, 10, 11
3	Participate in discussions and make short presentations on general and technical topics.	Analyze	6 7, 8, 9, 10, 11
4	Possess an ability to write clearly on topics related to technology and workplace communication.	Evaluate	6 7, 8, 9, 10, 11
5	Choose functional language, grammar structures, cohesive devices and skills of organization to express clearly in speaking.	Create	6, 7, 8, 9, 10, 11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOK:

1. Tabitha Chekuri & Ravi Perimulla, INGENUITY: A Course Book on Technical English Communication, Cengage Learning, 2025.

REFERENCE BOOKS:

1. Balasubramanian T, “A Text book of Phonetics for Indian Students”, Orient Longman, New Delhi, 1989.
2. Krishnaswamy, N and Sriraman, T, “Current English for Colleges”, Trinity publications, 2016.
3. Ashraf Rizvi M, “Effective Technical Communication”, 2nd Edition, McGraw Hill Education, 2017.

25CY101 CYBER SECURITY

Hours Per Week :

L	T	P	SL	C
0	1	1	0	1

PREREQUISITE KNOWLEDGE: Basic familiarity with computers, internet usage, mobile apps, and fundamental digital communication concepts.

COURSE DESCRIPTION AND OBJECTIVES:
This course offers a broad overview of cybersecurity, designed for students from all engineering disciplines. It introduces the basic concepts of security, threats, and preventive practices in the digital world. Students will learn cyber hygiene, common attack vectors, legal and ethical considerations, and safe digital practices applicable to personal and professional environments.

MODULE-1

UNIT-1 **0L+6T+6P+0SL = 12 Hours**

INTRODUCTION TO CYBERSECURITY:

Cybersecurity: Definition and Goals, Scope and Importance in Modern Society, Threat landscape: Statistics and recent real-life cyber-attacks, Basics of Information Security: Confidentiality, Integrity, and Availability (CIA Triad).

UNIT-2

TYPES OF CYBER ATTACKS:

Malware (Viruses, Worms, Trojans, Ransomware), Phishing, Spoofing, Denial of Service (DoS/DDoS), Man-in-the-middle (MitM), SQL Injection, Zero-day attacks, Social engineering attacks.

MODULE-2

UNIT-1 **0L+9T+9P+0SL = 18 Hours**

CYBER SAFETY AND BEST PRACTICES:

Password protection and Multi-Factor Authentication, Safe browsing habits and email hygiene, Mobile device security, Secure use of social media platforms, Safe use of public Wi-Fi.

UNIT-2

CYBER LAWS AND ETHICS:

Overview of Indian IT Act 2000 & amendments, Digital signature, cyber-crime laws, Data Privacy & GDPR, Cyber ethics & responsible digital behaviour.

UNIT-3

THREATS, ATTACKS, AND VULNERABILITIES:

Online Threat Vectors: Web, Email, USB, Social Media, Introduction to Vulnerabilities and Risk Concepts, Case Studies: Real-world Cyber Attacks.



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SKILLS:

- ✓ *Awareness of common cyber threats and online risks.*
- ✓ *Ability to identify phishing, fraud messages, and unsafe links.*
- ✓ *Practice of strong password habits and secure digital behaviour.*
- ✓ *Understanding of real-world cybercrime scenarios.*

PRACTICES:

- Working of phishing emails.
- Secure password creation.
- Browser settings and privacy tools.
- Use of antivirus/firewalls/Two-Factor Authentication (2FA).
- Introduction to tools like Wireshark, Burp Suite demo.
- Cybercrime reporting portal walk-through.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Understand and apply the fundamentals of cybersecurity and cyber threats.	Apply	1,2,6,8
2	Identify and explain common cyber threats and personal protection techniques.	Apply	1,2,3,4,6
3	Demonstrate safe browsing, secure communication practices, and recognize scam patterns.	Apply	2,3,4,6,8
4	Appreciate the role of cybersecurity across different fields and in daily personal use.	Apply	1, 2, 3,5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXTBOOK:

1. EC-Council, “Cybersecurity Essentials”, Cengage Learning, 2021.

REFERENCE BOOKS:

1. Nina Godbole & Sunit Belpure, “Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives”, Wiley India.
2. James Graham, “Introduction to Cyber-Warfare: A Multidisciplinary Approach”, Syngress.
3. Peter W. Singer & Allan Friedman, “Cybersecurity and Cyberwar: What Everyone Needs to Know”, Oxford University Press.

25SS101 CONSTITUTION OF INDIA

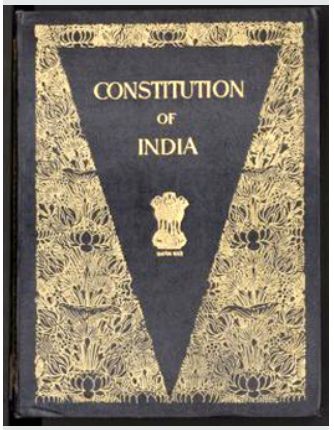


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Hours Per Week :

L	T	P	SL	C
0	2	0	0	1

PREREQUISITE KNOWLEDGE: Nil

COURSE DESCRIPTION AND OBJECTIVES:

This foundational course introduces students to the basic philosophy and functioning of the Indian Constitution. It explores constitutionalism, fundamental rights, duties, directive principles, and the organization of the Indian state. Emphasis is placed on participatory democracy, local self-governance, and ethical foundations using insights from Indian Knowledge Systems (IKS) and alignment with Sustainable Development Goals (SDGs).

MODULE-1

UNIT-1 **0L+12T+0P+0SL=12 Hours**

Historical Background to the Indian Constitution- Meaning of Constitution, Constitutional Law and Constitutionalism- Historical evolution: colonial legacy to the Constituent Assembly- Salient features and values of the Indian Constitution, Rajadharma, Sabha–Samiti traditions of governance, Peace, Justice, and Strong Institutions.

UNIT-2

Fundamental Rights, Duties, Directive Principles, and Amendments-- Right to Equality, Freedom, and Life-- Directive Principles of State Policy- Fundamental Duties: relevance and legal status- Amendment procedure and Basic Structure Doctrine, Dharma & Nyaya principles in ***Manusmriti & Arthashastra,*** Reduced Inequality; Peace, Justice and Strong Institutions.

PRACTICES:

- Skit or role-play on framing the Preamble using Sabha–Samiti models.
- Group discussion: Comparison of Fundamental Duties with Dharma-based duties.
- Poster making: Timeline of Indian Constitutional development.

MODULE - 2

UNIT-1 **0L+18T+0P+0SL=18 Hours**

Structure and Form of Government- Federalism: Centre-State Power Distribution-Parliamentary Form of Government- Emergency Provisions: Sabha advisory traditions; checks and balances.

UNIT-2

Local Self-Government- 73rd and 74th Constitutional Amendments- Panchayats and Municipalities- Participatory democracy and grassroots governance: Gram Sabha, Panchayat Raj in Vedic and Smriti texts: Sustainable Cities and Communities.

UNIT-3

Constitution and Contemporary Citizenship- Importance of the Constitution in everyday life- Digital citizenship and democratic awareness- Rights of marginalized and vulnerable groups- Civic duties, RTI, electoral literacy: Duty-based ethics; community participation traditions- Gender Equality.

SKILLS:

- ✓ *Understand foundational constitutional principles and values.*
- ✓ *Identify and apply rights and duties in every-day civic life.*
- ✓ *Analyze government structure and local governance mechanisms.*

PRACTICES:

- Mock Gram Sabha or Ward Committee meeting.
- RTI Application writing workshop.
- Constitution Quiz based on rights and duties.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Implement the core principles and features of the Indian Constitution in real- life civic and democratic situations.	Apply	6
2	Dissect the significance of Fundamental Rights, Duties, and Directive Principles in promoting constitutional morality.	Analyze	6,7
3	Differentiate the roles of constitutional institutions and intergovernmental relations in sustaining India's federal framework.	Analyze	6,7
4	Evaluate the performance of the Indian Constitution in upholding democratic values and social justice.	Evaluate	6,7

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOK:

1. B.R. Ambedkar – The Constitution of India, Educreation Publishing, 2020.

REFERENCE BOOKS:

1. P.M. Bakshi – The Constitution of India, Universal Law Publishing, 15th Ed.
2. Subhash Kashyap – Our Constitution, National Book Trust, 2011.
3. Arun Thiruvengadam – The Constitution of India: A Contextual Analysis, Hart, 2017.
4. Constituent Assembly Debates – Lok Sabha Secretariat.
5. Arthashastra (Kautilya) – Translations on governance.
6. Manusmriti (Select chapters on duties and justice).

25SA103 PHYSICAL FITNESS

Hours Per Week :

L	T	P	SL	C
0	0	2	0	1

PREREQUISITE KNOWLEDGE: Nil

COURSE DESCRIPTION AND OBJECTIVES:

This course offers students a structured approach to improve their physical fitness and mental wellness using evidence-based exercise science and traditional yogic wisdom. It emphasizes developing strength, endurance, flexibility, and emotional balance, and encourages students to incorporate lifelong wellness habits.

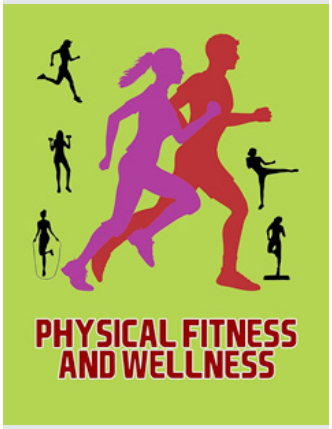


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MODULE-1

UNIT-1 **0L+0T+12P+0SL = 12 Hours**

FUNDAMENTALS OF PHYSICAL FITNESS:

Health-related and skill-related fitness components, Fitness assessment techniques and health screening, Posture and gait analysis.

UNIT-2

ASSOCIATING:

Endurance training methods – HIIT, continuous running, basic circuit training, Strength training using bodyweight and resistance, Importance of core stability and functional fitness, Injury prevention and recovery basics.

PRACTICES:

- Warm-up and mobility exercises, Stretching exercises.
- Running ABC Exercises.
- Speed Training - Short Sprints (30 Meter Run), Shuttle Runs.
- 400 Meters run, 100 Meters run.
- Bodyweight strength circuit (Sit- Ups and Push-Ups).
- Endurance (running/jogging/skipping).
- Static & dynamic stretching.
- Postural correction activities.

MODULE-2

UNIT-1 **0L+0T+18P+0SL = 18 Hours**

FLEXIBILITY, STRENGTH & COORDINATION:

Static and dynamic stretching, Balance drills and neuromuscular coordination, Plyometric and agility ladder exercises, Integration of yoga practices for cool-down.

UNIT-2

NUTRITION, RECOVERY & FITNESS HABITS:

Basics of sports and yogic nutrition, Macronutrients, hydration, meal planning, Role of sleep and active recovery, Fitness myths and self-monitoring.

SKILLS:

- ✓ *Improved strength, endurance, and flexibility.*
- ✓ *Understanding of fitness principles.*
- ✓ *Discipline and goal-setting.*
- ✓ *Teamwork and communication.*

UNIT-3

YOGA PRACTICES FOR WELLBEING:

Asanas (10–15) for flexibility, balance, and spine health, Pranayama: Anulom-Vilom, Bhramari, Kapalabhati, Meditation: Breath focus, Om chanting, Yoga Nidra and relaxation techniques.

PRACTICES:

- Guided Sukshma Vyayam and Asana sequences.
- Meditation.
- Yoga Nidra.
- Nutrition logging.
- Breath-focused mindfulness activities.
- Group Reflections.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Describe the physiological effects of exercise on the body systems.	Apply	1	7,8,9,10,11
2	Apply the FITT (Frequency, Intensity, Time, Type) principle to personal fitness plans.	Apply	1	7,8,9,10,11
3	Assess personal fitness levels through various fitness tests.	Analyze	2	7,8,9,10,11
4	Evaluate progress toward personal fitness goals and make adjustments as needed.	Evaluate	2	7,8,9,10,11
5	Construct a lifestyle plan that integrates regular physical activity for long-term health and fitness.	Create	2	7,8,9,10,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



25SA104 LIFE SKILLS

Hours Per Week :

L	T	P	SL	C
0	0	2	0	1

PREREQUISITE KNOWLEDGE: Nil

COURSE DESCRIPTION AND OBJECTIVES:

This course offers orientation in life skills. Its objective is to acquaint students with the social and inter-personal skills that will enable them to cope with the constantly changing environment. Developing general self-awareness & self-expression.

KUCHIPUDI

MODULE-1

0L+0T+12P+0SL = 12 Hours

Origin of dance according to Natya Shastra, 1-10 Asamyukta hastaalu, 11-20 Asamyukta hastaalu.

MODULE-2

0L+0T+18P+0SL = 18 Hours

21-28 Asamyukta hastaalu, Chaturvidha Abhinayaalu,1-12 Samyukta hastaalu, 13-24 Samyukta hastaalu, Pancha jaatulu, Rasa theory, Classical dance forms of India.

WESTERN DANCE

MODULE-1

0L+0T+12P+0SL = 12 Hours

Basic of Freestyle dancing, Understanding the Rhythm.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Adapting the basic steps to different music's, Assignments – using same choreography routines to different tracks.

CARNATIC MUSIC

MODULE-1

0L+0T+12P+0SL = 12 Hours

Basics of carnatic music, Sarali swarali 3 speeds, Akaara sadhana, Janta swaralu introduction, Importamce of Raag, Taal and about different instruments.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Bhadrachala ramadasu keertanalu, Composing a song, Sing along with karaoke, Popular writers compositions from different languages, Students should do project about music and a famous person related to music.

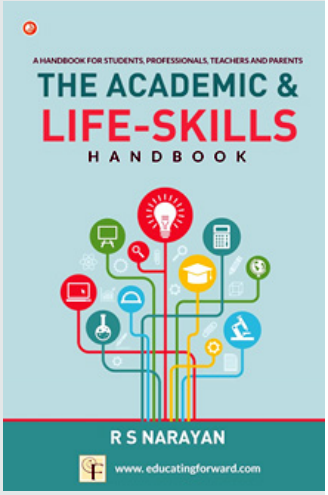


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SKILLS:

- ✓ *Bouncing back from setbacks and adversity.*
- ✓ *Thinking outside the box to generate new ideas.*
- ✓ *Discipline and goal-setting.*
- ✓ *Teamwork and communication.*

PUBLIC SPEAKING

MODULE-1

0L+0T+12P+0SL = 12 Hours

Introduction to Public Speaking, Informative Speaking, Listening Effectively.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Organizing and Outlining, Delivering Your Speech, Introductions and Conclusions, Critical Thinking and Reasoning.

PAINTING

MODULE-1

0L+0T+12P+0SL = 12 Hours

Free hand outlines drawing on the basis of creativity, originality and individuality.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Memory drawings on various topics like games, festivals, animal study and daily life of society, Still Life with light and shade, Nature study like Landscape, waterfall and Forests etc.

YOGA

MODULE-1

0L+0T+12P+0SL = 12 Hours

Introduction to Yoga, Introduction to Yogasana Sports, Brief History of Yoga, Types, schools of Yoga, Sun salutations Variations.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Yoga for Stress, Introduction to Astanga Yoga, Introduction to classical texts of Yoga, Yogic Diet, Yoga Nidra, Introduction to Meditation, Yoga for Sports, Injury Management with Yoga, Introduction to Yoga therapy, Yoga for weight loss, Diabetes.

COOKERY

MODULE-1

0L+0T+12P+0SL = 12 Hours

Preparation of Bread Halwa, Flavoured Milk, Veggie soups, Herbal Tea.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Fritters, Rasagulla, Onion Samosa, Soya Nuggets, Dhokla, Cutlet.

VIDEO EDITING

MODULE-1

0L+0T+12P+0SL = 12 Hours

Necessity & Principles of Editing, Editing for the genre, Film Appreciation, Digital Filmmaking.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Post Production Tools, Film Editing Styles, Audio Editing and Production, Enhancing Video with Color Grading, Advanced editing techniques, Motion Graphics & Video Editing Portfolio Development.

GERMAN

MODULE-1

0L+0T+12P+0SL = 12 Hours

Alphabet, numbers, greetings, geschwister.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Familie,wo wohnen sie, nachbharn von familie wiegel, inline skaten, wetter, fernsehen, Hobbies.

FRENCH

MODULE-1

0L+0T+12P+0SL = 12 Hours

Pour saluer, Les jours de la semaine, Les mois de l' année.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Les nombres, Décrire l'heure, Les verbes être,avoir et aller, Les nationalités, Les professions, Les fruits et les legumes, Les adjectifs pour décrire une person.

PHOTOGRAPHY

MODULE-1

0L+0T+12P+0SL = 12 Hours

Introduction to Photography, Photography Evolution, Different types of Nature lights.

MODULE-2

0L+-0T+18P+0SL = 18 Hours

DSLR Parts and Types of DSLR's, Anatomy of Camera and Lenses, Types of Lenses, Photo framing, Image resolution, Exposure triangle.

APSSDC ELECTRICAL HOME

MODULE-1

0L+0T+12P+0SL = 12 Hours

Understanding and testing of live switch boards, DB distribution boards, Staircase wiring, complete wiring of 1 BHK, 2 BHK.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Erection and maintenance of fan & tube light, Understanding the protection devices,20 Minutes timer, LDR with timer control, Impulse relay.

APSSDC ELECTRONICS HOME

MODULE-1

0L+0T+12P+0SL = 12 Hours

Understanding of series and parallel connection of bulbs.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Working of ceiling fan, tubelight, mixer grinder, iron box.

BLOCK PRINTING

MODULE-1

0L+0T+12P+0SL = 12 Hours

Introduction to Kalamkari Painting, Tools and Materials for Kalamkari.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Basic Techniques and Sketching for Kalamkari, Dyeing Techniques for Kalamkari Introduction to Block Printing, Block Printing Techniques and Final Projects.

HANDCRAFTS

MODULE-1

0L+0T+12P+0SL = 12 Hours

Plastic bottle with Shilpkar work, Paper craft for room decoration, Wall hanging with ice cream sticks.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Table stand with cotton buds, 3D outline with bottle decoration, Thermocol work.

SELF DEFENSE BY TAEKWONDO

MODULE-1

0L+0T+12P+0SL = 12 Hours

Self Defense by Taekwondo- Side Wrist Grip Escape and Cross Wrist Grip escape, Side Wrist Grip - Arm Bar, Wrist Lock, Shoulder Lock.

MODULE-2

0L+0T+18P+0SL = 18 Hours

Double Wrist Grip Escape, Upper Double Grip Escape, Major Outer Reaping Throw, Major Hip Throw, One Arm Shoulder Throw.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Demonstrate proficiency in a variety of techniques	Apply	1	7,8,9,10,11
2	Identify, demonstrate and understand the structures and characteristics	Apply	1	7,8,9,10,11
3	Cultivate concentration, self-control, and emotional regulation.	Analyze	2	7,8,9,10,11
4	Demonstrate increased confidence and the ability to respond assertively and calmly in threatening situations.	Evaluate	2	7,8,9,10,11
5	Understand and apply sustainable and ethical practices.	Create	2	7,8,9,10,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



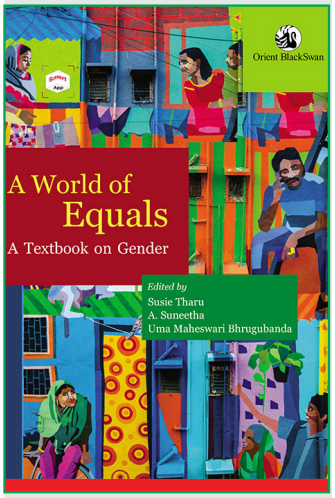


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25SA102 SELF-UNDERSTANDING AND GENDER SENSITIZATION

Hours Per Week

L	T	P	SL	C
0	0	2	0	1

PREREQUISITE KNOWLEDGE: Nil

COURSE DESCRIPTION AND OBJECTIVES:

To comprehend the dynamics of gender roles and relations within social and cultural contexts, and to evaluate legal and behavioral frameworks that promote gender equality and personal growth.

MODULE-1

0L+0T+12P+0SL=12 Hours

UNIT-1

UNDERSTANDING GENDER:

Concept of Gender: Distinction between Gender and Sex - Socialization and Gender Roles - Gender and Culture: Stereotypes and Social Conditioning - Gender-Based Violence: Forms, Impact and Prevention. ***Shakti philosophy, contributions of women in Vedas.***

Activity-Based Learning: Group discussions on gender roles in media - Role plays on gender stereotyping - Case study analysis of real-life incidents of gender-based violence

UNIT-2

GENDER LAWS AND RIGHTS:

Gender and Labour: Equal Pay, Workplace Harassment, Maternity Benefits - Legal and Constitutional Provisions for Gender Equality in India: Article 14, 15, 16, 39, and 42, POSH Act, Domestic Violence Act, Dowry Prohibition Act - Women's Rights and Human Rights Framework. ***Dharma-based equality, Sarvodaya philosophy.***

Activity-Based Learning: Legal case study presentations -Debate on gender justice and law implementation - Quiz on gender-related legal provisions

PRACTICES:

- Conduct a survey on gender roles and relations in the banking and IT sectors to assess employee attitudes and resistance to policy changes.
- Undertake a study to identify the impact of cultural perceptions on gender-based violence and its effects on workplace productivity.
- Analyze the characteristics and components of gender laws to evaluate their effectiveness in promoting labor rights and reducing gender discrimination.
- Perform a study on the determinants of gender roles and culture by reviewing media content and social norms to understand their influence on attitudes.
- Conduct a comparative analysis of constitutional and legal perspectives on gender equality to assess their impact on labor practices and gender sensitivity in various industries.

MODULE-2

0L+0T+18P+0SL=18 Hours

UNIT-1

SELF –EMPOWERMENT:

Self-awareness and Self-confidence -Building Personal Identity -Values and Ethics for Personal Growth - Goal Setting and Motivation.

Activity-Based Learning: Self-reflection journals - Vision board creation - Group sharing sessions

UNIT-2

PERSONALITY DEVELOPMENT SKILLS:

Types of Personality: Introvert, Extrovert, Ambivert - Good Manners and Professional Etiquette - Developing a Positive Attitude.

Activity-Based Learning: Role plays on etiquette, Personality-type assessment and group analysis, Positivity circles

UNIT-3

LIFE MANAGEMENT SKILLS:

Decision Making and Problem Solving - Interpersonal Communication Skills - Time and Stress Management - Conflict Resolution and Leadership.

Activity-Based Learning: Situational games for decision making, Time log and stress diary activity, Group project on conflict resolution strategies

PRACTICES:

- Survey on different personality types and their impact on workplace performance and team dynamics.
- Undertake a study on the effectiveness of etiquette training programs in improving professional interactions and workplace relationships.
- Analyze strategies for developing a positive attitude and their impact on employee engagement and job satisfaction.
- Conduct a study on time management practices and their influence on productivity and work-life balance.
- Perform a study on conflict management techniques and their effectiveness in resolving workplace disputes and improving team cohesion.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Analyze the influence of gender roles and cultural norms on individual behavior and social structures.	Analyze	6
2	Apply legal perspectives and gender laws to assess their impact on gender equality and labor rights.	Apply	6
3	Analyze various personality types and develop Strategies to apply good manners and etiquette in diverse settings.	Analyze	6
4	Apply decision-making and time management skills to enhance personal and professional effectiveness.	Apply	6
5	Analyze conflict management and leadership skills to effectively handle interpersonal and organizational Challenges.	Analyze	6

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGs) AND INDIAN KNOWLEDGE SYSTEM (IKS) :



REFERENCE BOOKS:

1. A World of Equals: A Textbook on Gender, Susie Tharu, Uma Maheswari Bhrugubanda (2022), Orient Blackswan Pvt. Limited.
2. Chitra Chellam (2022), Personality Development, Sara Book Publication.

SKILLS:

- ✓ *Self Esteem Skills.*
- ✓ *Decision Making Skills.*
- ✓ *Problem Solving Skills.*
- ✓ *Team Building Skills.*
- ✓ *Interpersonal Skills.*
- ✓ *Leadership Skills.*

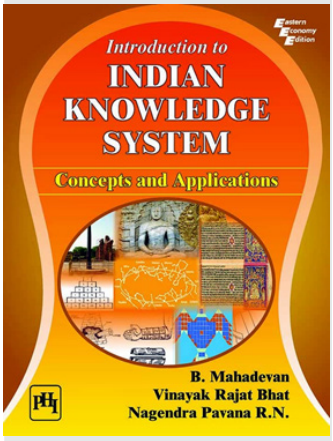


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25SS102 INDIAN KNOWLEDGE SYSTEMS

Hours Per Week :

L	T	P	SL	C
0	2	0	0	1

PREREQUISITE KNOWLEDGE: High School – level knowledge on Indian history and society.

COURSE DESCRIPTION AND OBJECTIVES:
The course introduces students to India's rich and diverse knowledge traditions and heritage. The profound insights, timeless wisdom, and diverse perspectives, originating from the ancient past have played a major role in shaping the country's cultural identity. The richness of India's intellectual heritage is vindicated by the existence of innumerable texts and thinkers in the country. By exploring India's knowledge traditions, the course aims to help students learn how the country's culture developed across the ages and to enable them look beyond conventional methods in finding solutions to scientific and technological issues.

MODULE-1

UNIT-1 **0L+12T+0P+0SL = 12 Hours**

INTRODUCTION TO INDIAN KNOWLEDGE SYSTEMS:
Definition, Scope, and Significance of IKS; Various Fields of Study in IKS (Science, Medicine, Arts, Linguistics, Philosophy, etc.); Key Texts and Sources of information for IKS: Vedic, Buddhist, Jain, and Other Traditions.

UNIT-2

PHILOSOPHICAL AND SCIENTIFIC FOUNDATIONS OF IKS:
Overview of Major Schools of Thought (Nyaya, Samkhya, Vedanta, etc.); Early Scientific and Technological Contributions (Mathematics, Astronomy, Medicine); Centers of Learning: Nalanda, Takshashila, and Others.

PRACTICES:

- Mapping IKS Disciplines – Create a chart or mind map showcasing various fields of study in IKS, including Science, Medicine, Arts, and Philosophy.
- Textual Exploration – Select and summarize insights from classical IKS texts (Vedas, Upanishads, Buddhist & Jain texts).
- IKS in Daily News – Collect newspaper articles on IKS-related discoveries (archaeological findings, indigenous technologies, art forms) and discuss their relevance.
- Field Visit/Virtual Tour – Explore ancient centers of learning like Nalanda and Takshashila (via museum visits, documentaries).
- Poster Display of Philosophers – Create visual profiles of prominent ancient philosophers (e.g., Charaka, Sushruta, Panini, Aryabhata).

MODULE-2

UNIT-1 **0L+18T+0P+0SL = 18 Hours**

DISCIPLINE-SPECIFIC INDIAN KNOWLEDGE SYSTEMS:
Role of IKS within the discipline (Science, Technology, or Law); Key Concepts and Applications within the discipline; Demonstration of IKS via real-life examples (Traditional Art, Architecture, and Indigenous Engineering Practices).

UNIT-2

RELEVANCE OF IKS IN THE MODERN WORLD:
Contributions of IKS to Global Knowledge Systems; Revival and Integration of IKS with Contemporary Science and Technology; Challenges and Opportunities in Preserving IKS.

UNIT-3

IKS AND SUSTAINABLE PRACTICES:
Indigenous Knowledge in Agriculture, Water Management, and Medicine; Environmental Ethics in Ancient Indian Traditions; Case Studies on Sustainable Development Using IKS.

PRACTICES:

- Discipline-Specific IKS Study – Research and present how IKS applies to law, governance, science, architecture, or technology.
- Exploring Indigenous Techniques – Engage in practical demonstrations of traditional knowledge such as Yoga, Meditation, Martial Arts (Karra Saamu), or Body Art.
- IKS Documentation Project – Collect and document indigenous knowledge traditions, such as traditional medicine, water harvesting, or agriculture.
- Awareness on Intellectual Property Risks – Use the internet to study indigenous intellectual properties that are at risk of international appropriation.
- Group Discussion on Challenges to IKS – Debate the modern relevance and challenges in preserving IKS, and propose solutions for its integration into science and technology.

COURSE OUTCOMES:
Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply basic concepts of Indian Knowledge Systems to recognize their relevance in contemporary science, engineering, and society.	Apply	6, 11
2	Analyze selected indigenous practices in areas such as health, environment, water management, or architecture to understand their principles.	Analyze	6, 8, 9, 11
3	Evaluate the ethical and ecological values embedded in IKS and their relevance to sustainability and social responsibility.	Evaluate	6, 7
4	Develop an informed understanding of Indian Knowledge Systems by applying them in reflections, discussions, or illustrative examples.	Apply	6, 8, 9, 11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):





TEXTBOOKS:

1. B. Mahadevan, Vinayak Rajat Bhat, Nagendra Pavana R. N., Introduction to Indian Knowledge System: Concepts and Applications, PHI Learning Private Limited, Delhi, 2023.
2. Mukul Chandra Bora, Foundations of Bharatiya Knowledge System, Khanna Book Publishing, 2023.

REFERENCE BOOKS:

1. Dhirendranath Banerjee and Sanjit Kumar Sadhukan (ed.), Ancient Indian Scientific Thought and Modern Theories: An Overview, National Mission for Manuscripts and D. K. Printworld, 2019.
2. Shaik Taufiq Khalil, Indian Knowledge System: Arthshastra by Kautilya, Notion Press, 2023.
3. Chauhan Bhag Chand, IKS: The Knowledge System of Bharata, Garuda Prakashan, 2023.

SKILLS:

- ✓ *Ability to analyze and evaluate Indian Knowledge System (IKS) practices in the context of modern challenges.*
- ✓ *Problem-solving skills through the application of IKS principles to contemporary issues.*
- ✓ *Strong research skills in gathering and analyzing information related to IKS.*
- ✓ *Collaborative skills for preserving and promoting IKS.*
- ✓ *Enthusiasm for learning and practicing traditional skills such as weaving, pottery, and herbal medicine.*



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25SS103 INDIAN CULTURE AND HERITAGE

Hours Per Week :

L	T	P	SL	C
0	2	0	0	1

PREREQUISITE KNOWLEDGE: Basic understanding of Indian history and culture at the high school level.

COURSE DESCRIPTION AND OBJECTIVES:
This course introduces students to the foundational concepts, expressions, and institutions of Indian culture and heritage. Students will explore the richness of India's tangible and intangible legacies-its philosophies, arts, knowledge systems, and monuments-while engaging with key values like unity in diversity, sustainability, and respect for tradition. The course emphasizes civic responsibility, heritage preservation, and the contemporary relevance of India's cultural identity.

MODULE-1

UNIT-1 **0L + 12T + 0P + 0SL = 12 Hours**

CULTURE AND HERITAGE – MEANINGS AND DIMENSIONS:

Definitions: Culture and Heritage; Distinction between tangible and intangible heritage; Core features of Indian culture (Sanskriti): unity in diversity, continuity, and pluralism; Role of heritage in shaping identity and nation-building.

UNIT-2

HISTORICAL ROOTS OF HERITAGE:

Harappan Civilization and its heritage: city planning, trade, and script; ***Vedic culture*** and religious texts as cultural heritage; Cultural impact of Buddhism and Jainism; Bhakti and Sufi traditions as living intangible heritage; Local traditions and oral culture across regions.

PRACTICES:

- Comparative poster/chart on tangible vs intangible heritage.
- Case study presentation: a local or regional tradition.
- Group presentation on an ancient site recognized by UNESCO.
- Write a short essay: "Why preserve heritage in the digital age?"

MODULE-2

UNIT-1 **0L + 18T + 0P + 0SL = 18 Hours**

TANGIBLE HERITAGE – ART, ARCHITECTURE, AND MONUMENTS:

Architectural diversity: Temples, Mosques, Churches, Stupas, Forts, Palaces, and Rock-cut caves; Notable heritage sites: Hampi, Qutub Minar, Taj Mahal, Konark, Charminar, Lepakshi Temple, Amaravati Stupa, and Victoria Memorial; Conservation challenges: environmental threats, pollution, neglect, urbanization, and insensitive tourism; Role of ASI and UNESCO World Heritage recognition.

UNIT-2

INTANGIBLE HERITAGE – KNOWLEDGE, FESTIVALS, AND VALUES:

Indian classical music and dance as living heritage; Traditional systems: Ayurveda, Yoga, and traditional ecological knowledge; Community-based practices: folk traditions and seasonal festivals; Core cultural values: non-violence, nature worship, family systems, and community life.

UNIT-3

HERITAGE PRESERVATION IN MODERN INDIA:

Heritage sense during the freedom struggle (e.g., revival of Khadi, Swadeshi art); National institutions preserving heritage: IGNCA, INTACH, and ASI; Digital documentation and cultural mapping initiatives; Youth and civic responsibility in heritage conservation.

PRACTICES:

- Virtual tour/report on a heritage site.
- Storyboarding: heritage in daily life.
- Role-play or debate: "Heritage vs. Modernity – Finding the Balance."
- Case studies of successful community preservation efforts.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply key concepts of Indian culture and heritage to appreciate its role in shaping societal values and identity.	Apply	6, 11
2	Analyze the evolution and characteristics of India's tangible and intangible heritage across regions and time.	Analyze	5, 6, 8
3	Evaluate contemporary challenges and strategies in preserving heritage through institutional and civic efforts.	Evaluate	6, 7, 8
4	Evaluate how cultural values can inform ethical decision-making, sustainability practices, and inclusive societal participation.	Evaluate	6, 7, 8

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOKS:

1. B. B. Satpathy, Indian Culture and Heritage, DDCE, Utkal University, 2015.
2. NIOS Course Material: Indian Culture and Heritage (223).
3. A.L. Basham, The Wonder That Was India, Vol.1, Rupa Publications, 1954.

REFERENCE BOOKS:

1. S. Radhakrishnan, Indian Philosophy: Volume I, 2nd ed., Oxford University Press India, 2008
2. Romila Thapar, The Penguin History of Early India: From the Origins to AD 1300, Penguin Books India, 2003.
3. John Keay, India: A History-Revised and Updated, Harper Collins, 2012.
4. Michel Danino, The Lost River; On the Trail of the Sarasvatī, Penguin Books India, 2010.

SKILLS:

- ✓ *Cultural awareness and ethical reasoning.*
- ✓ *Visual, analytical, and storytelling skills.*
- ✓ *Heritage appreciation and preservation thinking.*
- ✓ *Inclusive and civic-minded citizenship.*

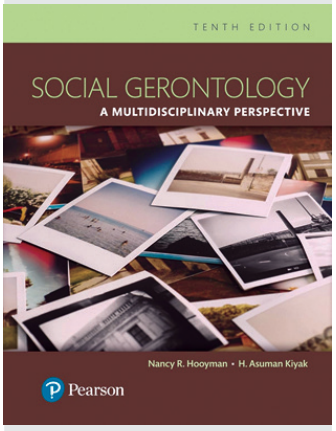


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25SS105 GERONTOLOGY

Hours Per Week :

L	T	P	SL	C
0	2	0	0	1

PREREQUISITE KNOWLEDGE: Basic understanding of Indian society, family structure, and human development from middle school social studies.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides an introduction to gerontology - the study of aging. This course helps students explore what it means to grow old in India, the role of the elderly in Indian families and society, and how we can care for and support older adults. It aims to help students understand the complexities of aging, challenge common stereotypes, and appreciate the diversity of the older adult population in India. Students will learn about real-life challenges faced by older people and ways to build a more inclusive, respectful society.

MODULE-1

UNIT-1 **0L+12T+0P+0SL = 12 Hours**

UNDERSTANDING AGING: BIOLOGICAL AND PSYCHOLOGICAL PERSPECTIVES:

Defining Gerontology: Introduction to the field, and its importance, with a note on the demographic shift in India; Theories of Aging: Biological theories (e.g., cellular clock theory, free radical theory) and psychological theories (e.g., Erikson's stages of psychosocial development, activity theory) and their applicability in the Indian context; Physiological Changes with Age: Common changes in body systems (e.g., cardiovascular, musculoskeletal, nervous) and common health challenges faced by older adults in India; Cognitive Changes with Age: Memory, attention, problem-solving, and the concept of cognitive reserve, with a focus on cognitive health in the Indian population; Mental Health in Later Life: Common challenges (e.g., depression, anxiety) and promoting well-being, considering cultural factors influencing mental health in older Indians.

UNIT-2

SOCIAL ASPECTS OF AGING AND DIVERSE EXPERIENCES IN INDIA:

Social Theories of Aging: Disengagement theory, continuity theory, socioemotional selectivity theory, and their relevance to Indian societal structures; Ageism and Stereotypes: Understanding prejudice and discrimination against older adults in India, challenging negative perceptions and cultural notions of aging; Family and Social Relationships: Role of the joint family system, nuclear families, friendships, and community networks in later life in India; Diversity in Aging in India: Exploring differences based on gender, rural/urban divide, socioeconomic status, religion, and regional variations in experiences of aging; Work, Retirement, and Leisure: Transitions in later life, challenges of formal and informal employment in old age, and the importance of meaningful engagement and traditional leisure activities.

PRACTICES:

- Create a "Grandparent Appreciation Poster" with stories or photos.
- Class discussion on respecting older people.
- Interview a grandparent or senior neighbour and share their experiences.

- Group discussions on common stereotypes about older adults in India and how to challenge them.

MODULE-2

UNIT-1 **0L+18T+0P+0SL=18 Hours**

HEALTH, WELLNESS, AND CARE IN LATER LIFE IN INDIA:

Promoting Healthy Aging: Lifestyle factors (nutrition, traditional diets, yoga, exercise), preventive care and traditional Indian health practices (e.g., Ayurveda); Common Health Conditions: Chronic diseases (e.g., arthritis, diabetes, heart disease) prevalent in India and their management, including access to healthcare; Healthcare Systems for Older Adults in India: Public and private healthcare, challenges of access and affordability, role of family caregivers, and emerging long-term care options; Palliative and End-of-Life Care: Understanding hospice, advance directives, and grief in the Indian cultural context; Technology and Aging: Assistive technologies, telehealth, and their role in supporting older adults in India, considering digital literacy and access.

UNIT-2

AGING IN INDIAN SOCIETY: POLICIES AND COMMUNITY SUPPORT:

Aging-in-Place and Age-Friendly Communities: Concepts and practical applications in Indian cities and villages; Social Policies and Programs for Older Adults in India: National Policy on Older Persons, Maintenance and Welfare of Parents and Senior Citizens Act, government schemes and initiatives; Intergenerational Relationships: Benefits of connecting different age groups, role of grandparents, and promoting intergenerational harmony in India; Ethical Considerations in Gerontology in India: Autonomy, informed consent, elder abuse prevention, and legal protections for older adults; Global Aging: Demographic trends and challenges worldwide, with a specific focus on India's rapidly aging population and its implications.

UNIT-3

INDIAN HERITAGE AND CULTURE IN AGING:

Culture and Civilization: Key characteristics of Indian culture and their influence on perceptions and experiences of aging; Visual Arts: Architecture, Sculpture, and UNESCO Heritage Sites as reflections of historical attitudes towards elders and their contributions; Performing Arts: Classical Music, Dance, and Literature as expressions of wisdom, tradition, and the life cycle in India; Traditional Wisdom and Aging: Exploring concepts like ***"vanaprastha"*** and the reverence for elders in Indian philosophy and texts; Role of Spirituality and Religion: Impact of spiritual and religious practices on well-being in later life in India.

PRACTICES:

- Research on age-friendly initiatives in Indian communities.
- Debate on the ethical considerations surrounding emerging technologies in aging, particularly for the Indian demographic.
- Analysis of how Indian cultural heritage reflects and influences the concept of aging.
- Visit to a local elder care centre or old age home.
- Make a "Healthy Aging Tips" chart.
- Skit or role play on how to care for elders at home or in public.

SKILLS:

- ✓ *Awareness of elderly rights and welfare mechanisms.*
- ✓ *Respect for cultural values and traditions towards elderly in India.*
- ✓ *Critical analysis of aging theories, stereotypes, demographic trends and policies with in Indian context.*
- ✓ *Understanding the diverse needs of older adults in India and societal responses.*
- ✓ *Interpretation of research data related to aging populations in India.*
- ✓ *Designing community-oriented aging interventions.*

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Understand and apply core concepts and theories of gerontology.	Apply	6, 7, 8, 9
2	Analyze the social determinants of aging and the diverse experiences of older adults in India.	Analyze	6, 7, 8, 9
3	Evaluate the healthcare needs and policy frameworks relevant to older adults in India.	Evaluate	6, 7, 8, 9
4	Design and assess community-based solutions for elder care.	Create	6, 7, 8, 9

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOKS:

1. Hooyman, N. R., & Kiyak, H. A., Social Gerontology: A Multidisciplinary Perspective. Pearson, 2018.
2. Quadagno, J., Aging and the Life Course: An Introduction to Social Gerontology. McGraw-Hill Education, 2017.

REFERENCE BOOKS:

1. Irudaya Rajan, S., India's Elderly: Burden or Asset? Routledge, 2016.
2. Prakash, I. J., Aging in India: A Sociological Perspective. Rawat Publications, 2009.
3. Kumar, A., Elderly in India: A Socio-Economic Profile. Serials Publications, 2011.
4. Moody, H. R., & Sasser, B. R., Aging: Concepts and Controversies. Sage Publications, 2018.
5. Paltasingh, Tattwamasi, & Tyagi, Renu, Caring for the Elderly: Social Gerontology in the Indian Context. Sage Publications, 2015.
6. Victor, Christina R., The Social Context of Aging: A Textbook of Gerontology. Routledge, 2005.

25ME102 DO IT YOURSELF

Hours Per Week :

L	T	P	SL	C
0	0	2	0	1

PREREQUISITE KNOWLEDGE: Nil

COURSE DESCRIPTION AND OBJECTIVES:

This course is to empower the learners with hands-on experience in designing, fabricating, and testing small-scale engineering systems or components The course objective is to emphasize creativity, innovation, and real-world problem solving through “learning by doing.”

MODULE-1

0L+0T+12P+0SL = 12 Hours

FUNDAMENTALS OF CARPENTRY AND PLUMBING

UNIT-1

CARPENTRY:

Introduction to carpentry tools, use of tools for making simple objects like wooden shelf, picture frame, planters box, coat rack, floating shelves. Ś***ilpa*** Śā***stra (Woodwork & Craftsmanship)***

UNIT-2

PLUMBING:

Introduction to Plumbing tools, use of tools for making simple objects like fitting of taps, faucets, shower heads, replacement of washers in taps, simple pipe joints, few installations

PRACTICES:

- Design and construct a stable and functional 3-tier bookshelf made from plywood or reclaimed wood using dowel joints and varnished for finish using measuring tape, saw, chisel, drill, wood glue.
- Construct a portable and sturdy 3-legged stool made of teak or pine with dowel or mortise joints using Mallet, wood rasp, chisel, clamps.
- Fabricate a safe and user-friendly wooden organizer for books with crate-style bins fixed on a frame with rounded corners for safety using sandpaper, saw, wood router
- Craft a custom wall art panel or divider using CNC-routed or hand-carved geometric or nature-themed panels using CNC router or jigsaw, sander, varnish.
- Make a small desktop stand with precision-cut grooves for gadgets and accessories to hold mobile phones, pens, and sticky note using Coping saw, file, drill press.
- Replace a worn-out rubber washer inside a leaking compression tap using adjustable spanner, flat-head screwdriver, replacement washer, PTFE tape.
- Install a downpipe from the terrace with filtration and diversion to a recharge well using PVC downpipes, filter mesh, first-flush diverter, underground pipe, stones.
- Install a new wall-mounted faucet to replace a broken head, using standard plumbing fittings, pipe wrench, adjustable spanner, pipe sealant tape.
- Plan and install a PVC-based water line with two tap outlets and one washing machine point using T- and elbow joints using PVC cutter, solvent cement, marker, sandpaper.
- Install outlet piping from a 500L overhead water tank to supply a bathroom, kitchen, and garden tap using gravity flow using HDPE pipes, ball valves, elbows, thread tape, support brackets.

SKILLS:

- ✓ *Accurate measuring, marking, and cutting of wood.*
- ✓ *Identification and use of plumbing tools and fittings.*
- ✓ *Operating FDM 3D printers.*
- ✓ *Safe handling of electrical tools and live circuits.*
- ✓ *Disassembly and reassembly of small appliances.*

MODULE-2

0L+0T+18P+0SL = 18 hours

DIY JOBS RELATED TO ADVANCED APPLICATIONS

UNIT-1

3D PRINTING:

Introduction to the concept of FDM – 3D printing, printing of objects like spoon, mobile stand, pen holder, key chain

UNIT-2

HOME APPLIANCES:

Introduction to tools used in assembling/repairing home appliances, assembling of ceiling fan, change of capacitor of fan, checking of faulty winding in electrical appliances, Preventive Maintenance of a Split AC System

.UNIT-3

HOUSE WIRING:

Introduction to tools used in house wiring, fixing of tube light, fan, checking and repairing of a faulty fuse, fixing of a faulty switch/socket,

PRACTICES:

- List the steps you would follow to set up the printer, calibrate the build plate, and load the filament. What safety precautions should you take when handling the 3D printer?
- You need to print a simple personalized keychain with your initials using a 3D printer.
- You want to print a simple carabiner hook that will be used for carrying tools or equipment.
- You want to create a small, functional lock mechanism using 3D printing.
- You need to create a functional smartphone stand to hold your phone at an angle for watching videos or video calls.
- You have purchased a ceiling fan and wish to assemble and install it without professional help. List the sequence of steps you would follow to safely assemble and install the fan. What safety precautions must you take before beginning the installation?
- Your refrigerator seems to be cooling less effectively than usual. What simple checks and maintenance tasks can you perform yourself before calling a technician? Include cleaning, defrosting, and ventilation considerations.
- You have bought a new flat-screen TV and wall mount. What are the steps to assemble and mount it safely? What precautions must be taken regarding wire management and socket access?
- A tube light in your kitchen is flickering continuously and eventually stops working. After replacing the tube and starter, the issue persists. Describe the steps you would take to identify and fix the problem. What are the possible faulty components? How would you confirm if the ballast needs replacement?
- A room in your house loses power suddenly while other areas are unaffected. You suspect a blown fuse. How would you confirm this? Explain how to safely check and replace the fuse. What precautions should be taken to avoid future fuse failures?
- A wall switch that controls your living room light is no longer working — the light does not turn on even after changing the bulb. How would you go about checking the switch? What steps would you follow to replace it yourself, and how would you ensure it's safe to operate afterward?
- A socket in your bedroom is loose and occasionally sparks when plugging in devices. What are the risks involved, and how would you inspect and repair the socket? What are the signs that the socket needs replacement versus just re-tightening the connections?

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply basic carpentry techniques to measure, cut, shape, and assemble wooden components using hand and power tools	Apply	1,5,6,7,10
2	Demonstrate the ability to install, maintain, and troubleshoot common residential plumbing systems	Apply	1,5,6,7,11
3	Design and fabricate functional 3D models using slicing software and additive manufacturing techniques	Create	1,5,10,11
4	Inspect, diagnose, and repair common faults in household appliances	Analyze	1,4,6
5	Construct safe and functional single-phase electrical wiring circuits for lighting, switching, and plug points	Evaluate	1,2,6,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOK:

1. “Basic Electrical Installation Work” by Trevor Linsley.

REFERENCE BOOKS:

1. “3D Printing and Additive Manufacturing: Principles and Applications” by Chee Kai Chua & Kah Fai Leong.
2. “Ultimate Guide: Plumbing” by Merle Henkenius
3. “Carpentry Complete” by Andy Engel.

II

YEAR

B.Tech.

COMPUTER SCIENCE AND ENGINEERING

I SEMESTER

25MT204	-	Probability and Statistics
25MT205	-	Discrete Mathematical Structures
25CS201	-	Data Structures
25CS202	-	Web Technology
25CS203	-	Database Management Systems
25CS204	-	Object Oriented Programming Through Java
25CS205	-	Digital Logic Design
XXXXXXX	-	Life Skills - I
	-	NCC/NSS/SAC/E-Cell/Student Mentoring/Social activities/Publication

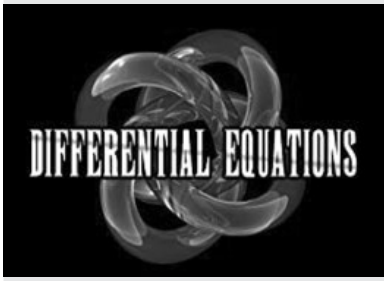
II SEMESTER

25MT203	-	Optimization Techniques
25CS801	-	Industry interface course
25CS206	-	Computer Organization and Architecture
25CS207	-	Design and Analysis of Algorithms
25CS208	-	Operating Systems
25CS209	-	Computer Networks
25CS210	-	Design Thinking and Engineering Orientation
25CS211	-	Field Project
	-	Open Elective - 1

COURSE CONTENTS

I SEM & II SEM

25MT204 PROBABILITY AND STATISTICS



Source: https://www.amazon.com/Differential-Equations/dp/B01H30X2JA

Hours Per Week :

L	T	P	SL	C
3	0	2	3	4

PREREQUISITE KNOWLEDGE: Basic knowledge in Mathematics and Statistics.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a comprehensive introduction to probability and statistical methods with applications in computing. It covers fundamental probability concepts, probability distributions, statistical inference, estimation techniques, and regression analysis. The course also introduces advanced statistical methods such as Principal Component Analysis (PCA) and Time Series Analysis. Students will develop a strong foundation in probabilistic modelling and statistical reasoning, which are essential for data analysis, machine learning, and decision-making in computing and engineering.

MODULE-1

UNIT-1 **9L+0T+6P+9SL=24Hours**

FUNDAMENTALS OF PROBABILITY

Introduction to probability and its applications in computing, Probability axioms, Conditional probability and independence, Bayes' theorem and its applications. Random variables: Distribution function, discrete random variable, probability mass function, continuous random variable, probability density function, Expectation, variance, and moments.

UNIT-2 **9L+0T+6P+9SL=24Hours**

PROBABILITY DISTRIBUTIONS

Discrete probability distributions: Uniform, Bernoulli, Binomial, Poisson, and Geometric.

Continuous probability distributions: Uniform, Normal, Exponential, Weibull, Cauchy, Logistic. Central Limit Theorem and its significance, Joint, marginal, and conditional distributions. Covariance and correlation.

PRACTICES:

- Implement probability axioms, conditional probability, and Bayes' theorem using programming (Python/R).
- Understand and visualize different probability distributions using real-world data.
- Compute statistical measures for discrete and continuous random variables.
- Analyze relationships between two random variables using joint, marginal, and conditional distributions.
- Apply tarka to evaluate the logical consistency of probabilistic models in data science, such as reasoning through the validity of Bayes' theorem for predictive algorithms in computing applications.

MODULE-2

UNIT-1 **9L+0T+6P+9SL=24Hours**

ESTIMATION AND STATISTICAL INFERENCE

Sampling techniques and sampling distributions, Point estimation and properties of estimators. Maximum Likelihood Estimation (MLE). Confidence intervals for population parameters. Hypothesis testing: Null

SKILLS:

- ✓ *Collect the data from various data sources and evaluate mean, median, mode mean deviation and standard deviation.*
- ✓ *Identify the areas which we can apply the probability theory.*

and alternative hypotheses, level of significance, p-value, critical value, Type I and Type II errors, Single sample and two sample tests for mean, proportion, variance and standard deviation.

UNIT-2 **9L+0T+6P+9SL=24Hours**

REGRESSION ANALYSIS

Simple linear model and assumptions, least-squares estimators and their properties, ML estimation of the parameters, statistical inference and prediction with the simple regression model, tests for normality, lack-of-fit testing, transformations on Y or X (Box-Cox, square root, log etc), Logistic regression.

UNIT-3 **9L+0T+6P+9SL=24Hours**

ADVANCED STATISTICAL METHODS

Principal Component Analysis (PCA) for Dimensionality Reduction, Introduction to Time Series Analysis and Forecasting.

PRACTICES:

- Perform point estimation, confidence intervals, and hypothesis testing on real-world datasets.
- Apply regression techniques for predictive modelling in computing applications.
- Reduce the dimensionality of large datasets while retaining essential features.
- Analyze and predict trends using time-series data in computing and finance.
- Integrate statistical methods with machine learning for practical applications.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Understand the fundamental concepts of probability, random variables, and probability distributions and apply them in computing applications.	Apply	1,2,
2	Analyze different probability distributions, their properties, and significance in real-world applications, including computing and machine learning.	Analyze	1,2,4,
3	Perform statistical inference, estimation, and hypothesis testing using different techniques and interpret the results effectively.	Evaluate	2,3,5,
4	Apply regression analysis techniques, including linear and logistic regression, and interpret results using statistical inference methods.	Apply	2,4,6,
5	Implement advanced statistical methods such as PCA and Time Series Analysis for data-driven decision-making in computing.	Analyze, Create	3,5,11,



TEXT BOOKS:

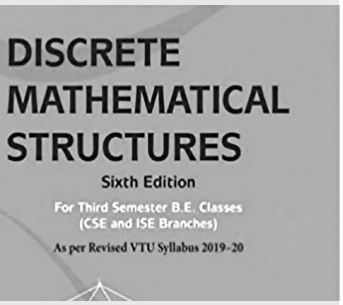
1. Sheldon M. Ross, Introduction to Probability and Statistics for Engineers and Scientists, 5th Edition, Academic Press, 2019.
2. Jay L. Devore, Probability and Statistics for Engineering and the Sciences, 9th Edition, Cengage Learning, 2015.

REFERENCE BOOKS:

1. Morris H. DeGroot, Mark J. Schervish, Probability and Statistics, 4th Edition, Pearson, 2011.
2. Casella, George, Roger L. Berger, Statistical Inference, 2nd Edition, Cengage Learning, 2002.
3. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.

Image source: https://www.google.com/search?q=PROBABILITY+%26+STATISTICS+normal+graph+IMAGE

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Source: https://www.amazon.in/Discrete-Mathematical-Structures-Dr-D-S-C/dp/9388478398

25MT205 DISCRETE MATHEMATICAL STRUCTURES

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Basic understanding of set theory, basic algebra, and logic.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the fundamental concepts of discrete mathematics that are essential in computer science. It includes the study of algebraic structures, logic, graph theory, Boolean algebra, and combinatorics. The objective is to provide a solid foundation for students to understand theoretical concepts and apply them in computer science applications such as algorithm design, database theory, and artificial intelligence.

MODULE-1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

ALGEBRAIC STRUCTURE

Binary operation, Relation, Semi Group, Monoid, Group Poset, Hasse Diagram, Lattice.

UNIT-2 **6L+6T+0P+6SL=18 Hours**

MATHEMATICAL LOGIC

Statements and notation, Connectives, Well-formed formulas, Truth table, Tautology, Equivalence Implication, Quantifiers, Predicate Calculus, Inference theory, Logic Gates operations.

PRACTICES:

- Constructing Hasse diagrams for given Posets
- Solving problems on logical equivalences
- Creating truth tables for compound propositions
- Applying predicate logic to simple problems
- Implementing basic logic gates in simulations
- Verifying tautologies using truth tables

MODULE-2

UNIT-1 **6L+6T+0P+6SL=18 Hours**

NORMAL FORMS

CNF, DNF, PDNF, PCNF, Conversion of CNF to DNF and vice versa. Minimization of Boolean Function (Karnaugh Maps).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

GRAPH THEORY

Definition of graph, Incidence, Degree, Walk, Path, Circuit, Special types of graphs, Connected graph, Weighted graph, Matrix representation of graph, Graph Isomorphism, Euler and Hamiltonian path and circuit, Eulerian graph, Hamiltonian graph, Planar graphs, Bipartite graph, Tree, Spanning Trees, Binary Trees, Decision Trees, Applications in AI.

UNIT-3 **6L+6T+0P+6SL=18 Hours**

GRAPH ALGORITHMS

Shortest path algorithms (Dijkstra's algorithm and Floyd-Warshall's algorithm), DFS, BFS algorithm.

Minimal spanning tree (Kruskal's algorithm and Prim's algorithm), Graph coloring, Chromatic number, The four color problem.

PRACTICES:

- Graph traversal exercises (DFS, BFS)
- Karnaugh Map simplifications for Boolean expressions
- Implementing Dijkstra's algorithm in Python/C++
- Drawing and analyzing binary trees and decision trees
- Converting between CNF and DNF
- Designing minimal spanning trees using Kruskal's and Prim's algorithms

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Understand fundamental concepts of algebraic structures, relations and mathematical logic, including propositional and predicate calculus.	Analyze	1,11
2	Apply Boolean algebra techniques such as normal forms (CNF, DNF) and Karnaugh Maps for function minimization and logical simplifications.	Apply	1,3,5,
3	Apply combinatorial techniques, including permutations, combinations, recurrence relations and generating functions, to solve discrete mathematics problems.	Analyze	1,2,5,
4	Analyze different graph structures, spanning trees and their properties, including Eulerian and Hamiltonian circuits, isomorphisms and planarity.	Analyze	2,3,4
5	Evaluate graph algorithms and minimum spanning trees to optimize computational problems.	Evaluate	3,5,6,10,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



SKILLS:

- ✓ *Familiarity of concepts of statements, logic and truth tables.*
- ✓ *Analyze closed form of discrete numeric function.*
- ✓ *Know some basic properties of graphs, trees and related discrete structures.*

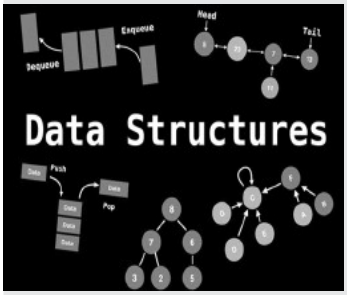
TEXT BOOKS:

1. Kenneth H. Rosen, “Discrete Mathematics and Its Applications”. McGraw-Hill, 7th Edition, 2012.
2. J.P. Tremblay and R. Manohar, “Discrete Mathematical Structures with Applications to Computer Science”. Tata McGraw-Hill, 1st Edition, 2001.

REFERENCE BOOKS:

1. C.L. Liu and D.P. Mohapatra, “Elements of Discrete Mathematics”. McGraw-Hill, 4th Edition, 2010.
2. Bernard Kolman, Robert C. Busby, and Sharon Cutler Ross, “Discrete Mathematical Structures”. Pearson, 6th Edition, 2008.
3. Susanna S. Epp, “Discrete Mathematics with Applications”. Cengage Learning, 4th Edition, 2011.

25CS201 DATA STRUCTURES



Source: https://www.youtube.com/watch?v=Qmt0QwzEmh0

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Programming in C.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a foundational understanding of essential data structures and their implementations. It covers core structures such as arrays, linked lists, stacks, queues, trees, and graphs, along with key algorithms for searching, sorting, and traversal. Emphasis is placed on selecting efficient data structures for problem-solving and analyzing their performance. The course establishes the basis for designing optimized algorithms and developing effective software solutions.

MODULE-1

UNIT-1 **6L+6T+ 6P+6SL=24 Hours**

DATA STRUCTURES BASICS

Data, Information, Data Types, Introduction to Data Structures, Storage Structures: Sequential and Linked Representations, Classification of Data Structures, Abstract Data Type (ADT), Applications of Data Structures

Searching: Linear Search, Binary Search

Sorting: Stable vs. Unstable Sorting, Internal vs. External Sorting, Selection Sort, Bubble Sort, Insertion Sort, Merge Sort, Quick Sort, Applications of Sorting.

UNIT-2 **6L+6T+ 6P+6SL=24 Hours**

LINKED LISTS

Linked List: Introduction, Representations, Types: Singly Linked List, Doubly Linked List, Circular Linked List -Representation, Operations, and Applications

PRACTICES:

RECURSION (BASIC)

- Compute the product of two numbers using recursion.
- Find the sum of the first N natural numbers using recursion.
- Find the factorial of a number using recursion.
- Compute the N-th Fibonacci number recursively.
- Compute a^n using recursion (power function).
- Convert a decimal number to binary using recursion.
- Print all factors of a number using recursion.
- Check whether a number is prime using recursion.
- Find the maximum digit product among numbers ≤ N using recursion.
- Sum two integers without using arithmetic operators (recursion + bitwise logic).

RECURSION (ADVANCED)

- Implement insertion sort recursively.
- Implement selection sort recursively.
- Implement bubble sort recursively.

SKILLS:

- ✓ *Experienced to Store data and various types of data to handle.*
- ✓ *Ordering and sorting of data.*
- ✓ *Indexing and Searching of required data from large data sequences.*
- ✓ *Exposed to various characteristics such as Linear or non-linear, Homogeneous or heterogeneous and Static and Dynamic.*

- Check whether an array is sorted using recursion.
- Find the sum of even elements in an array using recursion.
- Print alternate elements of a linked list using recursion.
- Reverse a doubly linked list using recursion.
- Find the middle element of a singly linked list using recursion.
- Generate all permutations of a list using recursion.
- Find all numbers ≤ N that are product of exactly two distinct primes (recursion).

SEARCHING ALGORITHMS

- Implement Linear Search.
- Implement Binary Search.

SORTING ALGORITHMS

- Implement Selection Sort (internal sorting).
- Implement Bubble Sort (stable sorting).
- Implement Insertion Sort (stable sorting).
- Implement Merge Sort (stable, external-friendly sorting).
- Implement Quick Sort (unstable sorting).

SINGLY LINKED LIST (SLL)

- Insert nodes (ascending order).
- Insert nodes (descending order).
- Search for a node in SLL.
- Delete a given node.
- Display list (forward).
- Display list (backward – recursion allowed).
- Count the number of nodes in SLL.
- Swap alternate nodes in SLL.
- Move last node to the front.
- Move first node to the end.

LINKED LISTS – ADVANCED PROGRAMS

- Merge two sorted linked lists into one sorted list.
- Print common elements of two sorted linked lists.
- Reverse a list using a stack.
- Reverse a list (recursive).
- Reverse a list (iterative).
- Reverse nodes in pairs (recursive).
- Reverse nodes in pairs (iterative).
- Check if a linked list is a palindrome.
- Detect and remove a loop in a linked list.
- Delete a node given only a pointer to it.

DOUBLY LINKED LIST (DLL)

- Insert a node in DLL.
- Display forward.
- Display backward.
- Search a node.
- Delete a node.
- Reverse a DLL.
- Reverse a DLL from position M to N.
- Count odd and even nodes.
- Merge two sorted DLLs.

- Remove adjacent duplicate nodes.

CIRCULAR LINKED LIST (CLL)

- Insert a node in circular DLL / CLL.
- Search a node.
- Display forward.
- Display backward.
- Delete a node in circular DLL/CLL.
- Move last node to the front of CLL.
- Print middle N nodes of CLL.
- Delete nodes greater than a given value.
- Delete adjacent duplicates in CLL.
- Find the sum of all nodes in CLL.

STACKS

- Implement two stacks using a single array.
- Reverse a stack using only push/pop operations.
- Evaluate postfix expression using a stack.
- Convert infix → postfix; infix → prefix.
- Check balanced parentheses using stack.
- Find the middle element in a stack.
- Remove the middle element from a stack.
- Implement stack using a linked list.
- Nearest greater element on the right (using stack).
- Sliding window maximum using a stack.

QUEUES

- Implement linear queue using array.
- Implement circular queue using array.
- Implement queue using linked list.
- Implement deque (double-ended queue).
- Application: Process scheduling using queue.
- Check equality of two inputs (characters/numbers).
- Determine greater/smaller among inputs.
- Identify if input is even/odd.
- Check divisibility by 3 and 5.
- Categorize input as positive, negative, or zero.

MODULE-2

UNIT-1 **6L+6T+ 6P+6SL=24 Hours**

STACK AND QUEUE

Stack: Stack ADT, Operations, Array & Linked Implementations, Applications: Expression Evaluation, Conversion of expressions: Infix to postfix, Infix to prefix.

Queue: Queue ADT, Operations, Array & Linked Implementations, Types: Linear, Circular, Dequeue, Applications of Queue

UNIT-2 **6L+6T+ 6P+6SL=24 Hours**

TREES

Basic Terminology, Types of Trees, Binary Trees: Introduction, Properties, Representations, Traversals, BST- Definition and operations, AVL Trees: Definition and operations, Heap Tree- Min Heap and Max Heap, Applications of Binary Trees.

UNIT-3 **6L+6T+ 6P+6SL=24 Hours**

GRAPHS

Graphs: Basic Terminology, Types of Graphs, Representations: Adjacency Matrix, Adjacency List, Graph Traversals: Breadth First Search, Depth First Search, Applications of Graphs.

Minimum Spanning Tree(MST): Prim's and Kruskal's Algorithms.

Single-Source Shortest Path Algorithms: Dijkstra's and Bellman-Ford

PRACTICES:

TREES (FUNDAMENTALS)

- Construct a binary tree using array representation.
- Construct a binary tree using linked representation.
- Implement preorder traversal.
- Implement inorder traversal.
- Implement postorder traversal.
- Implement level order traversal (using a queue).
- Implement BST insertion (iterative).
- Implement BST insertion (recursive).
- Implement BST search.
- Implement BST delete operation (delete leaf, one-child, two-child nodes)

TREES (INTERMEDIATE)

- Find the height of a binary tree.
- Count leaf nodes, internal nodes, and total nodes.
- Check if a binary tree is height-balanced.
- Create an Expression Tree and evaluate it.
- Construct an AVL tree and implement rotations (LL, RR, LR, RL).
- Insert nodes into an AVL tree and print the balanced tree.
- Implement Min-Heap construction.
- Implement Max-Heap construction.
- Perform heap insertion and heap deletion (heapify).
- Implement Heap Sort using Max-Heap.
- Implement Heap Sort using MIN-Heap.

GRAPH (CORE CONCEPTS)

- Implement graph using Adjacency Matrix.
- Implement graph using Adjacency List.
- Perform BFS traversal (iterative).
- Perform DFS traversal (recursive).
- Check if a graph is connected.
- Count the number of connected components (using DFS/BFS).
- Determine whether a given undirected graph is cyclic or acyclic.
- Find the degree of each vertex in a graph.

GRAPH (INTERMEDIATE)

- Find arrival and departure times of vertices during DFS.
- Identify whether a graph is bipartite.
- Determine the depth/height of a graph (from a source node).
- Implement Prim's Algorithm for Minimum Spanning Tree.
- Implement Kruskal's Algorithm for Minimum Spanning Tree (use Union–Find).
- Implement Dijkstra's Algorithm for shortest path (no negative edges).
- Implement Bellman–Ford Algorithm (handles negative edges).
- Detect negative weight cycle using Bellman–Ford.

GRAPH (ADVANCED APPLICATIONS)

- Print the shortest path tree produced by Dijkstra's algorithm.
- Print the shortest path tree produced by Bellman–Ford.
- Determine the reachability from a given source to all nodes.
- Check whether a directed graph is strongly connected.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Explore and apply fundamental data structures and operations (searching, insertion, deletion, traversal).	Apply	1, 2,
2	Select and use appropriate data structures to solve computational problems.	Apply	1,3,4,
3	Analyze the efficiency of algorithms and justify appropriate data structure choices.	Analyze	1, 2,
4	Design and develop solutions using advanced structures like trees and graphs.	Create	1, 2,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOKS:

1. Reema Thareja, "Data Structures Using C", 2nd Edition, Oxford University Press, 2014.
2. Seymour Lipschutz, "Data Structures with C", 1st Edition, McGraw Hill Education, 2017.

REFERENCE BOOKS:

1. Ellis Horowitz and Sartaj Sahni, "Fundamentals of Data Structures", illustrated edition, Computer Science Press, 2006.
2. Richard F. Gilberg and Behrouz A. Forouzan, "Data Structures: A Pseudocode Approach with C", 2nd Edition, CENAGE Learning, 2005.
3. R G Dromey and Pearson, "How to solve it by Computer", 2nd edition, Impression edition, 1998.



Source: https://www.dreamstime.com/web-development-coding-programming-internet-technology-business-concept-web-development-coding-programming-internet-technology-image121903546

25CS202 WEB TECHNOLOGY

Hours Per Week :

L	T	P	SL	C
0	2	2	0	2

PREREQUISITE KNOWLEDGE: OOPs through JAVA.

COURSE DESCRIPTION AND OBJECTIVES:

This course offers web development concepts, such as static and dynamic web page design, and provides internet programming knowledge, web servers, application servers, and design methodologies using object-oriented concepts. This course aims to build web applications using JSP, PHP, AngularJS, and NodeJS with client and server-side scripting technologies spanning multiple domains.

MODULE-1

UNIT-1 **0L+6T+6P+0SL=12 Hours**

INTRODUCTION:

HTML: Creating structured documents(T1-PP:26-28), Links and navigation(T1-PP:69-82), Tables(T1-PP:131-156), Forms(T1-PP:160-197), and Frames(T1-PP:203-223).

HTML 5: Introduction to HTML5, The HTML5 Canvas, HTML5 audio and Video;(T2-PP:557-629).

CSS: Cascading Style Sheets, CSS Properties. (T1-PP:265-340).

JavaScript: Learning JavaScript- adding scripts to your page, DOM, variables, operators, functions, conditional statements, Looping, Events, Built-in objects, form, and regular expression validation. (T2-PP:500-589).

UNIT-2 **0L+6T+6P+0SL=12 Hours**

JDBC: What is JDBC, system requirements, types of JDBC Drivers, creating database tables, connecting to a database, executing SQL statements, processing result sets, and making changes to a result set. (T2-PP:165-206 & RefT2-PP:347-484).

PRACTICES:

- Design a webpage having four frames named a) Top, b) Center, c) Bottom and d) Left. The top frame should contain the company logo and title. The bottom frame should contain copyright information. The left frame should contain various links like Home, Products, Services, Branches, About, etc., When clicked on respective links, the content should display on the center frame.
- Design a catalog page that should contain the details of all the books available on the website in a table. The details should contain the following: a) Snapshot of Cover Page b) Author Name c) Publisher. d) Price. e) Add to cart button.
- Design a timetable schedule for your current semester using the Table tag.
- Design a HTML page for Student Registration Form using Form Elements that includes Form, input-text, password, radio, checkbox, hidden, button, submit, reset, label, text area, select, option and file upload.
- Design a HTML web page with at least two <h1>, two images, two buttons, and appropriate CSS to display,
  - o All <h1> with font – size 12pt, and bold in Verdana font using Inline CSS.
  - o All <img> with border color yellow, thickness 10px using Document Level CSS.
  - o All <inputtype='button'> should change background color to red on mouse over them using External CSS.
- Design a HTML page having a text box and four buttons viz Factorial, Fibonacci, Prime and Palindrome. When a button is pressed, an appropriate java script function should be called to display the following:
  - o Factorial of that number.
  - o Fibonacci series up to that number.

- o Prime numbers up to that number.
- o Is it a palindrome or not?

- Design a web page that contains a color palette, and when the user moves the mouse to a particular area, it changes the background color of the web page.
- Design a registration page to validate the following fields using JavaScript.
  - o Make sure the user name starts with an uppercase letter
  - o The user name must have at least one digit
  - o Ensure that the Email is valid
  - o Ensure that thepasswordlengthisbetween8to20characters
  - o Make sure the password contains at least one upper case letter, one lower case, and one special character, excluding [. (dot), ,(comma), ;(semicolon), : (colon)].
- Design a web page to display the videos on-page, on user selection, using frames and HTML5 tags.
- Design a web page to display different types of objects using HTML5 Canvas.
- Design a web application to validate entered username and password through JDBC connection program and display user information on successful login, and provide a profile editing option to the user. Else, display an error message.
- Develop a JSP application that creates a user upon successful signup, updates user information upon successful login, displays user information on the home screen, and provides a logout button.
- Make an HTML form that collects the last name. Send the name to the JSP page. If there is an employee with that last name, show full details of him or her (just show the first employee if there are multiple people with the same name). If there is no employee with that last name, say “no employee records available.”.

SKILLS:

- ✓ *Perform client-side validation using Java Script and Angular JS.*
- ✓ *Store and retrieve data using Node JS.*
- ✓ *Generate dynamic web pages using JSP and PHP.*
- ✓ *Develop a web application or website for any real-time requirements.*

MODULE-2

UNIT-1 **0L+6T+6P+0SL=12 Hours**

JSP: JSP Processing, Generating Dynamic Content using Scripting Elements, Implicit JSP Objects, Sharing Data between JSP pages, JSP application design with JDBC. (T3-PP:459-613).

UNIT-2 **0L+6T+6P+0SL=12 Hours**

PHP: Introduction to PHP, Expressions, and control flow in PHP, functions and objects, Arrays, Accessing MySQL using PHP, Form Handling, Cookies, Sessions, and Authentication. (T2-PP:35-162).

UNIT-3 **0L+6T+6P+0SL=12 Hours**

ANGULAR AND NODE JS:

Angular JS: Introduction, Expressions, Modules, Directives, Controllers, Filters, Events, Forms, Form Validation. (T4-PP:397-447).

Node JS: Introduction, Setup Dev Environment, Modules, Node Package Manager, Creating Web server, File System, Events, Express.js, Accessing MySQL from Node.js. (T4-PP:39-185).

PRACTICES:

- Develop a JSP application to create a user on successful signup and update user information on successful login and display user information on the home screen and provide a logout button.
- Make an HTML form that collects the last name. Send the name to the JSP page. If there is an employee with that last name, show full details of him or her (just show the first employee if there are multiple people with the same name). If there is no employee with that last name, say “no employee records available.”
- Design a web page using PHP, upload the image into the web page, and display the image when the user clicks on the view button.
- Design a personal Information form, Submit & Retrieve the form data using $_GET(),
- $_POST() and $_REQUEST() Variables.
- Design a login page to validate the username and password through MySQL. If login is successful, display user information on the home page and modify user information on the edit page using sessions. When the user logs out, destroy all user-related sessions.
- Design a web page to accept payment data from the user and process the payment. on successful payment, display details on the screen. A Session should be set while doing payment, up to 10 minutes after that link/payment page should be destroyed irrespective of the user’s payment.
- Design a web application to validate the user registration page using AngularJS.
- Design a search engine using AngularJS. On key press, display data on the web page.

- Design a web page to validate user name and password using Node.js and PHP. When the user clicks on the login button, the server checks the data availability in data database. If the data matches, a successful login page is returned. Otherwise, a failure message is shown to the user.
- Design a web application to display the active duration of the user, i.e., the time between login and logout.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Develop structured web pages using HTML, HTML5, CSS, and JavaScript, including form validation and multimedia elements.	Apply	1,
2	Implement basic client-side scripting and data validation using JavaScript and regular expressions.	Apply	1,
3	Utilize JDBC and PHP to perform database operations, manage sessions, and implement user authentication in web applications.	Apply	2,
4	Evaluate the use of JSP and Node.js for building scalable, server-side components of a web application.	Analyse	2,
5	Design and integrate multi-tier web applications using AngularJS, JSP, Node.js, and PHP.	Create	3,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOKS:

1. Jon Duckett, “Beginning Web Programming with HTML, XHTML, and CSS”, 2nd Edition, Wiley Publishing, Inc, 2008.
2. Robin Nixon, “Learning PHP, MySQL & JavaScript WITH JQUERY, CSS & HTML5”, 4th Edition, O’Reilly, 2015.
3. Joel Murach , Michael Urban,“Murchurs Java Servlets and JSP”,3rd Edition, Mike Murach & Associates
4. Brad Dayley,” Node.js, MongoDB and AngularJS Web Development, Pearson Education,2014.

REFERENCE BOOKS:

1. Paul Deitel, Harvey Deitel, Abbey Deitel, “Internet & World Wide Web How to Program”, 5th Edition, Pearson Education, 2012.
2. Kishori Sharon, “Java APIs, Extensions and Libraries: With JavaFX, JDBC, jmod, jlink, Networking, and the Process API”,2nd Edition,2018.
3. R G Dromey and Pearson, “How to solve it by Computer”, 2nd edition, Impression edition, 1998.

25CS203 DATABASE MANAGEMENT SYSTEMS



Source: https://www.youtube.com/watch?v=IDpB9zF8LBw

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Knowledge of basic Discrete Mathematics / Discrete Structures like - Sets, relations, functions, Propositional logic, Boolean algebra and logic (important for query optimization and relational algebra).

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the core principles and techniques of Database Management Systems (DBMS), with a focus on Relational Database Management Systems (RDBMS) widely used in business, scientific, and engineering applications. It covers key topics including ER modeling, relational algebra, SQL, normalization, and the relational model to support effective database design. The course also addresses essential system concepts such as transaction processing, concurrency control, and recovery. The objective is to provide students with the theoretical foundation and practical skills needed to design, query, and manage relational databases using modern tools and techniques.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

DATABASE SYSTEM CONCEPTS

Databases And Database Users: Introduction and Characteristics; Advantages of DBMS, and database users. (T1 – PP: 3 – 17).

Database System Concepts and Architecture: Data models, Schemas and instances; Three-Schema architecture and data Independence; Database languages and interfaces; The database system environment; Centralized and Client-Server architectures for DBMS. (T1 – PP: 31 – 46)

Conceptual Data Modeling and Database Design: Entity types, Entity sets, Attributes and keys; Relationship types, Relationship sets, Roles and structural constraints; Weak entity types; Relationship types. (T1 – PP: 59 – 79)

UNIT-2 **6L+6T+6P+6SL=24 Hours**

RELATIONAL DATABASE DESIGN

Relational Database Design by ER–To-Relational Mapping: Relational Database design using ER-to-Relational mapping. (T1 – PP: 290 – 297)

The Relational Data Model and Relational Database Constraints: Relational model concepts; Relational model constraints and Relational database schemas. (T1 – PP: 150 – 160).

Relational Algebra: Unary relational operations - SELECT and PROJECT; Relational algebra operations from set theory; Binary relational operations - JOIN and DIVISION. (T1 – PP: 239 – 251)

SQL: SQL data definition and data types; specifying constraints in SQL, Basic retrieval queries in SQL; INSERT, DELETE, and UPDATE statements in SQL. (T1 – PP: 177 – 198)

PRACTICES:

- Design an ER diagram for a Library Management System, identifying entity types, attributes, and relationships.
- Construct an ER model for a Car Rental System, including weak entities and constraints.
- Convert the ER diagram of a Student-Course Enrollment System into a relational schema.
- Development of Relational Database schemas for Company/Student/Sailors using DDL

SKILLS:

- ✓ *Develop E-R model for real life applications.*
- ✓ *Design of relational databases for real world applications.*
- ✓ *Devise queries using relational algebra and SQL.*
- ✓ *Analyze transaction processing, concurrency control and recovery techniques*

constructs of SQL.

- Apply various DML Commands such as select, insert, update etc. of SQL on Relational Database.
- Design of Relational Database schemas by specifying different types of Constraints.
- Apply various Relational Database operators (Arithmetic, Logical &comparison) and string-matching constructs of SQL.
- Expressing queries using Aggregate Functions of SQL on Relational Database.
- Implement PRIMARY KEY, FOREIGN KEY, and CHECK constraints in the Orders and Customers tables.
- Queries on Relational Database using GROUP BY, HAVING and ORDER BY clauses of SQL.

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

NORMALIZATION

Complex Queries, Triggers, Views: More complex SQL retrieval queries; Specifying constraints as assertions and actions as triggers; Views (virtual tables) in Pl/SQL. (T1 – PP: 207 – 228)

Basics of Functional Dependencies and Normalization for Relational Databases: Informal design guidelines for relation schemas; Functional dependencies-inference rules, equivalence and minimal cover; Normal forms based on primary keys; Boyce-Codd normal form; Properties of relational decompositions, multi valued dependency, join dependencies. (T1 – PP: 459 – 494).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

Introduction To Transaction Processing Concepts and Theory: Introduction to transaction processing; Transaction and system concepts; Desirable properties of transactions; Characterizing schedules based on serializability. (T1 – PP: 745 – 764).

UNIT-3 **6L+6T+6P+6SL=24 Hours**

Concurrency Control Techniques: Two-phase locking techniques for concurrency control, concurrency control based on timestamp ordering. (T1 – PP: 781 – 792)

Database Recovery Techniques: Recovery concepts; Shadow paging; The ARIES recovery algorithm. (T1 – PP: 813 – 827)

Indexing Structures for Files and Physical Database Design: Single level and multi-Level indexing; Dynamic multi-level indexing using B-trees and B+ trees.

PRACTICES:

- Design and Development of company database and expressing Nested queries using SQL.
- Design and Development of student database and specifying queries using set operations.
- Design and Development of sailor's database and specifying queries using different types of JOINs.
- Implementation of PL/SQL programs with Control Structures.
- Implementation of PL/SQL programs with Procedures.
- Implementation of PL/SQL programs with Function.
- Implementation of PL/SQL programs with Triggers.
- Creation and dropping of VIEWS.
- Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values. F = {CH $\rightarrow$ G, A $\rightarrow$ BC, B $\rightarrow$ CFH, E $\rightarrow$ A, F $\rightarrow$ EG} is a set of functional dependencies (FDs) so that F+ is exactly the set of FDs that hold for R. How many candidate keys does the relation R have?
- Apply various DCL and TCL constructs of SQL on Relational Database.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply the fundamental concepts of database systems to identify database users, system components, and architecture.	Apply	1,
2	Analyze real-world applications to identify entities, relationships, and constraints for ER modeling.	Analyze	1,2,3,
3	Analyze database schemas to identify functional dependencies and normalize them using appropriate normal forms.	Analyze	1, 2, 4,
4	Design and implement database applications using PL/SQL, views, triggers, and stored procedures to handle real-world problems.	Create	3,5,7,9,
5	Analyze transaction schedules to determine serializability and identify concurrency issues.	Analyze	2,4,7,11,
6	Evaluate different concurrency control techniques such as two-phase locking and timestamp ordering to ensure database consistency.	Evaluate	2,4,6,7,11,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOKS:

1. Ramez, Elmasri and Shamkant B. Navathe, “Fundamentals of Database Systems”, 7th Edition, Pearson Education, 2016.
2. Raghu Rama Krishnan and Johannes Gehrke, “Database Management Systems”, 3rd Edition, Tata McGraw Hill, 2013.

REFERENCES :

1. Abraham Silberschatz, Henry F.Korth and S.Sudarshan, “Database System Concepts”, 7th edition, Tata Mc Graw Hill,2019.
2. Allen G. Taylor “Database Development for Dummies” 1st Edition, 2011
3. C. J. Date “Introduction to Database Systems” 7th Edition, Addison Wesley, 2003.



Source: https://www.datasciencecentral.com/wp-content/uploads/2021/10/866750746 2.jpeg

25CS204 OBJECT ORIENTED PROGRAMMING THROUGH JAVA

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Basics of Programming.

COURSE DESCRIPTION AND OBJECTIVES:

This course offers a solid foundation in object-oriented programming using Java, emphasizing the application of core programming concepts to real-world scenarios. Students will explore principles such as inheritance, polymorphism, abstraction, exception handling, and interfaces. The course also introduces practical modules on file handling, Java collections with sorting mechanisms, and multithreading with synchronization techniques. Through hands-on programming exercises and problem-based learning, students will gain the skills required to build modular, efficient, and scalable Java applications for modern software development.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

INTRODUCTION TO JAVA

Introduction to Java: History, features, JDK, JVM, Program Structure (T1 – Pp: 1–20), Data types, variables, literals, operators, expressions (T1 – Pp: 21–58), Input/output using Scanner, System. in, and System.out (T1 – Pp: 59–66), Conditional statements (if, switch), loops (for, while, do-while) (T1 – Pp: 67–96), Arrays: 1D and 2D arrays, basic array operations (T1 – Pp: 97–118), Introduction to methods: method declaration, invocation, parameter passing, return values, Understanding static vs. instance members (T1 – Pp: 119–138), Introduction to Strings, String methods, String Tokenizer (T1 – Pp: 139–168).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

PRINCIPLES OF OOP

Principles of OOP: Encapsulation, Inheritance, Polymorphism, Abstraction (T2 – Pp: 1–22), Defining classes and objects, methods, method overloading (T2 – Pp: 23–46), Constructors: default, parameterized, and constructor overloading, this keyword (T2 – Pp: 47–64), static variables and methods, Access modifiers: private, protected, public, default, final keyword (T2 – Pp: 65–90), packages and simple class organization (T1 – Pp: 325–340).

PRACTICES:

- Design a program to read marks of 5 subjects, calculate the average, and assign a letter grade. Use functions for each task: readMarks(), calculateAverage(), and assignGrade(). Display a remarks message like “Excellent”, “Good”, “Needs Improvement”.
- Create a console-based billing system that takes total purchase amount and applies discount: ₹1000 → 5%, ₹3000 → 10%, ₹5000 → 15%. Use functions for input, discount logic, and final output.
- Accept n integers from the user and classify each as prime or composite. Also count and display the total number of prime numbers entered.
- Track attendance for 5 students over 7 days using a 2D array (1 = Present, 0 = Absent). Count

and display each student's attendance percentage.

- Design a menu-based number tool with options to:
  a) Check palindrome

  b) Print factors

  c) Display digit sum

  d) Reverse number
- Create a basic ATM simulation:
  Start with balance ₹10,000

  Options: Check balance, Withdraw, Deposit, Exit

  Handle invalid input and over-withdrawals gracefully

  Each operation should be a method. Use switch-case for menu flow.
- Take a sentence from the user and compute:
  Word count

  Frequency of each vowel

  Most frequent letter

  First and last word
- Create a program to generate a price list. Display Product Name, Quantity, and Price in aligned tabular format using System.out.printf().
- Create a class BankAccount with account number, balance, and methods deposit(), withdraw(), and displayBalance(). Perform basic transactions.
- Create a Student class with overloaded constructors to initialize student data in different ways (e.g., with name only, with name and marks, etc.).
- Use a static variable to count the number of objects created for a class Product. Display the count after each object creation.
- Create a class Book with attributes: id, title, and price. Allow the user to add books, search by id, and display all books.
- Create an Employee class and maintain an array of 5 employees. Display highest paid employee using a separate method.
- Accept a string with multiple integers separated by space, comma, or |, tokenize them using StringTokenizer, and display the sum.
- Build a console menu (e.g., recharge system or billing app) using class methods for each menu option. Use OOP concepts for modularity.

SKILLS:

- ✓ *To analyse and develop algorithm for real life problems using Java.*
- ✓ *Experience with developing and debugging programs in different IDEs.*
- ✓ *Develop multi-threaded applications.*
- ✓ *Creating web applications*

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

Inheritance, Interfaces, and Exception Handling: Types of Inheritance, super keyword (T1 – Pp: 341–368), method overriding, Abstract classes vs Interfaces (T1 – Pp: 369–398), multiple inheritance via interfaces, Defining, implementing, and extending interfaces (T1 – Pp: 399–412), Java packages: creating, importing, and access control (T1 – Pp: 413–430), Exception handling: try-catch-finally, throw, throws, and custom exceptions (T1 – Pp: 431–460).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

Java Collections and File Handling: Collections: Overview, List (ArrayList, LinkedList), Set (HashSet, TreeSet), Map (HashMap, TreeMap), Iterators, enhanced for-loop, Generics (T1 – Pp: 461–508), Sorting: Comparable and Comparator for custom object sorting (T1 – Pp: 509–524), File Handling: File, FileReader, BufferedReader, FileWriter, Scanner (T1 – Pp: 525–560).

UNIT-3 **6L+6T+6P+6SL=24 Hours**

Multithreading and Synchronization: Threads vs. Processes, lifecycle, creating threads (Thread class, Runnable) (T1 – Pp: 561–580), Thread control: sleep, join, yield (T1 – Pp: 581–590), Synchronization: synchronized methods/blocks, wait, notify, notifyAll (T1 – Pp: 591–608).

PRACTICES:

- Create an abstract class Shape with an abstract method area(). Extend it in Circle, Rectangle, and Triangle classes. Override the method and compute respective areas.
- Define an interface BankAccount with methods deposit() and withdraw(). Implement it in a SavingsAccount class. Enforce withdrawal limits using interface-based design.
- Create a class Arithmetic with a method add(). Derive a class Adder and reuse the inherited method without overriding. Show method accessibility.
- Create a custom exception NotEligibleToVoteException. Accept user age and throw this exception if age < 18.
- Accept two numbers from the user and handle divide-by-zero using try-catch and display meaningful messages.
- Organize a simple menu-driven telecom recharge system into custom packages: com.telecom.services and com.telecom.util.
- Create a class Student with name, marks, and branch. Store multiple student objects in an ArrayList and sort by marks using Comparator.
- Read a text input and print all unique words (ignore case) using HashSet. Use StringTokenizer or split().
- Implement a phonebook using HashMap<String, String> where the key is a name and the value is a phone number. Include options to add, search, and remove contacts.
- Read student details from a file and display them. Allow appending new entries and rewriting the file with updated records.
- Read a paragraph from a file. Display word counts for all words with length ≥ 4 and frequency ≥ 10. Sort by frequency, then alphabetically.
- Use synchronized, wait, and notify to print first, second, third in the correct order, each from a different thread.
- Implement a multi-user ticket booking system where threads book limited seats. Prevent overbooking using synchronization.
- Simulate concurrent deposit and withdrawal operations on a single account from multiple threads. Use synchronized to maintain correct balance.
- Implement the classic producer-consumer problem using a shared buffer, one thread producing and another consuming items.
- Create two threads: one prints even numbers and the other prints odd numbers up to a given range in synchronized sequence.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Design and implement structured Java programs using control flow, arrays, and modular methods to solve computational and algorithmic problems.	Apply	1, 2,
2	Model real-world entities using classes and objects, applying principles of abstraction and encapsulation.	Apply	1,3,
3	Apply object-oriented principles such as inheritance, interfaces, and exception handling to build robust and reusable software components.	Apply	1, 2,3,
4	Manipulate and manage data using Java Collections and File I/O mechanisms with sorting and search logic.	Analyze	1, 2,4,
5	Construct multi-threaded applications demonstrating concurrency control and inter-thread communication.	Create	1, 2,5,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOKS:

1. Herbert Schildt, Java: The Complete Reference, 12th Edition, McGraw-Hill Education, 2021.
2. M.T. Somashekara, D. S. Guru, K.S. Manjunatha, Object-Oriented Programming with Java, PHI Learning, 2017.

REFERENCE BOOKS:

1. E. Balagurusamy, Programming with Java, 6th Edition, McGraw-Hill Education, 2019.
2. Mark Lassoff, Java Programming for Beginners: Learn the Fundamentals of Programming with Java, Packt Publishing, 2017.
3. Oracle Java Documentation & Tutorials – https://docs.oracle.com/javase/tutorial/

.



Source: https://collegeacademy.in/images/ex-blogs/deld.jpg

25CS205 DIGITAL LOGIC DESIGN

Hours Per Week :

L	T	P	SL	C
2	0	2	2	3

PREREQUISITE KNOWLEDGE: Number systems, Logic Gates.

COURSE DESCRIPTION AND OBJECTIVES:

This course covers number systems, logic gates, Boolean algebra, and simplifications using SOP & POS forms. It explores K-map minimization, prime implicants, and combinational circuit design, including adders, multiplexers, and decoders. Sequential circuits, flip-flops, shift registers, and counters like mod-n and ring counters are introduced.

MODULE-1

UNIT-1 **6L+0T+6P+6SL=18 Hours**

BASICS OF LOGIC FUNCTIONS AND SWITCHING ALGEBRA

Review of Number systems(T1-PP 1-10), Complements(T1-PP 11-16), Binary Codes(T1-PP 22-30, Logic gates(T1-PP 83-86& 144-158). Switching Algebra: Boolean algebra(T1-PP 95-123), Canonical and Standard Forms – SOP & POS forms, algebraic simplifications and realization using basic and Universal gates(T1-PP 124-143).

UNIT-2 **6L+0T+6P+6SL=18 Hours**

MINIMIZATION OF SWITCHING FUNCTIONS AND LOGIC DESIGN

Minimization of Switching Functions: Simplification of switching functions with Karnaugh Map of 3,4 variables for both min and max terms(T1-PP 170-190), Don't care conditions, (T1-PP 198-201) Limitations of K-map, AOI and OAI implementations(T1-PP 202-223).

Logic Design: Adders - Half adder, Full Adder, Parallel binary adder, Subtractor - Half Subtractor, Full subtractor, Parallel Adder/Subtractor(T1-PP 147-170).

PRACTICES:

- Logic gates
- 1-bit Adders and Subtractors
- Parallel binary adder
- Parallel Adder/Subtractor

MODULE-2

UNIT-1 **6L+0T+6P+6SL=18 Hours**

COMBINATIONAL LOGIC DESIGN

Code converters, Decoder, Logic implementation using Decoder, Encoder, limitations of encoder, Priority encoder, Multiplexer, Logic implementation using Multiplexer, Demultiplexer(T1-PP 172-188).

UNIT-2 **6L+0T+6P+6SL=18 Hours**

SEQUENTIAL LOGIC DESIGN -I

Basics: Classification of sequential circuits(T1-PP 245-247).

Latches and Flip Flops: Latches and flip-flops – SR, D, JK, T, Triggering types and circuits, Master-slave flip-flop, Flip flop conversions(T1-PP 248-260).

Registers- Shift registers - SISO, SIPO, PISO, PIPO, Bidirectional shift register(T1-PP 326-337).

UNIT-3 **6L+0T+6P+6SL=18 Hours**

SEQUENTIAL LOGIC DESIGN -II

Counters- Design of Asynchronous and Synchronous Counters, Mod–N counters, random counters, Ring Counter, Johnson Counter(T1-PP 343-355).

PRACTICES:

- Multiplexer & Demultiplexer
- Encoder & Decoder
- Code Converter
- Flip flops
- Registers
- Counters

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply knowledge of Boolean algebra and logic gates to design and simplify digital circuits.	Apply	1,2,3,10,
2	Apply minimization techniques to optimize switching functions.	Apply	1, 2,3,10
3	Design combinational circuits like adders, multiplexers, and decoders etc., for practical applications.	Apply	1,2,3,4,5,10,
4	Develop sequential circuits using flip-flops and shift registers etc., for various digital systems.	Apply	1,2,3,4,5,10,
5	Design the sequential circuits like synchronous and asynchronous counters.	Apply	1,2,3,4,5,10

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



INDIAN KNOWLEDGE SYSTEM (IKS)

TEXT BOOK:

1. M. Morris Mano, “Digital Design”, 6th Edition, Pearson Education, 2018.
2. ZviKohavi, Niraj K. Jha, “Switching and Finite Automata Theory”, 3rd Edition, Cambridge University Press, 2009.

REFERENCE BOOKS:

1. John.M Yarbrough, “Digital Logic Applications and Design”, Thomson Learning, 2006.
2. Charles H.Roth. “Fundamentals of Logic Design”, 6th Edition, Thomson Learning, 2013.
3. Donald P.Leach and Albert Paul Malvino, “Digital Principles and Applications”, 6th Edition, TMH, 2006.

SKILLS:

- ✓ *Apply Boolean algebra techniques to simplify expressions*
- ✓ *Design efficient combinational digital circuits using gates, multiplexers, and decoders etc.*
- ✓ *Design efficient sequential digital circuits using flip flop.*
- ✓ *Implement and analyze finite state machines, shift registers, and counters to develop solutions for various digital applications.*



Source: https://oworkers.com/8-process-optimization-techniques-how-to-get-started/

25MT203 OPTIMIZATION TECHNIQUES

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Basic knowledge of probability, linear algebra, and calculus is essential; programming skills are helpful for applying optimization techniques effectively.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the fundamental concepts of Optimization Techniques and aims to make learners aware of the importance of optimization in real scenarios. The most important objective is to provide an understanding of various classical and modern methods for solving constrained and unconstrained problems in both single and multivariable cases. Finally, the basic ideas behind evolutionary algorithms like Ant Colony Optimization and Particle Swarm Optimization could be discussed for further study.

MODULE-1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

INTRODUCTION TO CLASSICAL METHODS AND LINEAR PROGRAMMING PROBLEMS

Terminology:

Design Variables – Constraints(T-I P.g no-63), Objective Function, Problem Formulation, Linear Programming Problem(T-I P.g no-123) – Simplex Method (T-I P.g no-119)– Concept of Duality(T-I P.g no-195).

General Transportation Problem:

The transportation table finding in initial basic feasible solution, North-West corner method, Least cost method, Row minima method, Column minima method.(T-I P.g no-360).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

SINGLE VARIABLE OPTIMIZATION

Problems, Optimality Criterion(T-II Pp no 302–304) – Bracketing Method(T-II Pp no 304–306) – Region Elimination Methods(T-II Pp no 306–308): Interval Halving Method(T-II Pp no 306–308) – Fibonacci Search Method(T-II Pp no 308–310) – Golden Section Method(T-II Pp no 310–312).

Gradient-Based Methods: Newton-Raphson Method(T-II Pp no 312–314) – Bisection Method(T-II Pp no 314–316) – Application to Root Finding(T-II Pp no 316–318).

MULTIVARIABLE OPTIMIZATION, ALGORITHMS OPTIMALITY:

Criteria – Unidirectional Search.Direct Search Methods: Hooke-Jeeves Pattern Search Method(T-II Pp no 343–345).Gradient-Based Methods(T-II Pp no345-–347): Cauchy's Steepest Descent Method(T-II Pp no345—347) – Newton's Method – Quasi-Newton Methods – L-BFGS(T-II Pp no347—353).

PRACTICES:

PROBLEMS ON STACKS & QUEUES

- A company makes two products (X and Y) using two machines (A and B). Each unit of X that is produced requires 50 minutes of processing time on machine A and 30 minutes on machine B. Each unit of Y that is produced requires 24 minutes on machine A and 33 minutes on machine B.
- At the start of the current week, there are 30 units of X and 90 units of Y in stock. Available processing time on machine, A is forecast to be 40 hours, and on machine B, it is forecast to be 35 hours.
- The demand for X in the current week is forecast to be 75 units and for Y, it is 95 units. The company policy is to maximize the combined sum of the units of X and Y in stock at the end

of the week.

- Formulate the problem of deciding how much of each product to make in the current week as a linear program. Solve this linear program graphically.
- Solve using the Simplex method the following problem:

  Maximize $Z = f(x, y) = 3x + 2y$

  Subject to: $2x + y \le 18, 2x + 3y \le 42, 3x + y \le 24, x \ge 0, y \ge 0$

  Luminous lamps have three factories – F1, F2, and F3 with production capacity of 20, 30, and 50 units respectively. The company needs to transport these lamps to four warehouses – W1, W2, W3, and W4 with demands of 40, 30, 20, and 10 units respectively. The unit transportation cost (in Rs.) between factories and warehouses are given below. Find an initial basic feasible solution of the given transportation problem using the Least Cost Method.

Faculty	Warehouse				Supply
	W1	W2	W3	W4	
F1	1	2	1	4	30
F2	3	3	2	1	50
F3	4	2	5	9	20
Demand	20	40	30	10	

- A mobile phone manufacturing company has three branches located in the cities of Jaipur, Udaipur, and Mumbai. The company has to transport mobile phones to three destinations—Kanpur, Pune, and Delhi. The availability from Jaipur, Udaipur, and Mumbai is 40, 60, and 70 units respectively. The demand at Kanpur, Pune, and Delhi are 70, 40, and 60 units respectively. The transportation cost is shown in the matrix below (use the Least Cost Method to find a basic feasible solution (BFS)).

Source	Kanpur	Pune	Delhi	Supply
Jaipur	4	5	1	40
Udaipur	3	4	3	60
Mumbai	6	2	8	70
Demand	70	40	60	170

	D1	**D2**	**D3**	**D4**	**Supply**
S1	11	13	17	14	250
S2	16	18	14	10	300
S3	21	24	13	10	400
Demand	200	225	275	250	

Find Solution using Row Minima Method

Find Solution using Column Minima Method

- Find the positive root of the following equation by bisection method. $f(x) = \sin\left(\frac{x}{2\pi}\right) - \cos^2(x)$
- Consider figure below. The cross sectional Area A of a gutter with equal base and edge length of 2 is given by $A = 4\sin(\theta)(1 + \cos(\theta))$ using an initial interval of $[0, 2\pi]$, find the interval after 3 iterations. Use an initial interval $\varepsilon = 0.2$.



SKILLS:

- ✓ *Project scheduling process skills including defining project activities, and estimation of time and resources*
- ✓ *Resource optimization skills to adjust the project schedule as per the demand and supply issues of project resources*
- ✓ *Feasibility Analysis.*

MODULE-2

UNIT-1 **6L+6T+0P+6SL=18 Hours**

STOCHASTIC OPTIMIZATION PROBLEMS

Notion of regret, online to batch conversion, Methods offering vanishing regret - OGD, EG, OMD Convex Sets and Functions: Affine and convex sets, convexity preserving operations, separating and supporting hyper-planes, generalized inequalities, Operations preserving convexity, conjugate function,(T-IIPpno779-784)

Quasi-convex functions, Log-concave and log-convex functions, Convexity with respect to generalized inequalities(T-II Pp no784—786).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

NON-CONVEX OPTIMIZATION PROBLEMS

Applications - sparse recovery, affine rank minimization, low-rank matrix completion, Convex approaches –relaxation-based methods, Non-convex approaches – projected gradient descent, alternating minimization(T-II Pp no339—343)

Algorithms: Unconstrained problems, equality constrained problems.

UNIT-3 **6L+6T+0P+6SL=18 Hours**

EVOLUTIONARY AND METAHEURISTIC OPTIMIZATION METHODS

Evolutionary Algorithms: Learn how nature-inspired techniques like Genetic Algorithms (GA) work. (T-II Pp no 693—737. Swarm Intelligence: Understand Particle Swarm Optimization (PSO), inspired by birds flocking together. (T-II Pp no 693—737).

PROBLEMS ON LINKED LISTS

- Show that each convex cone is indeed a convex set.
- Consider the linear system $0 \le x_i \le 1$, for $i = 1, 2, ..., n$ and let P denote the solution set. Explain how to solve a linear programming problem max $\left(c^T x : x \in P\right)$. What if the linear system was $a_i \le x_i \le b_i$ for $i = 1, 2, ..., n$. Here we assume $a_i \le b_i$ for each i.
- Can projected gradient descent (PGD) be used to obtain a stationary solution?
- Show that strong smoothness does not imply convexity by constructing a nonconvex function f: R→R that is 1-SS.
- Use Genetic Algorithm to solve a simple Knapsack Problem (choose items within weight limit to get max value).
- Use Particle Swarm Optimization to find the lowest point of a curved surface (optimize a math function).

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Understand the classical, linear programming and transportation problem along with game theory	Analyze	1,2,3,
2	Analyze real world problems around you.	Design	1,4,
3	Apply optimization to real-world problems	Apply	2,5,
4	Implement dynamic programming in real world complex problems	Analyze, Apply	1,4,6,
5	Evaluate and optimize solutions for complex real-world problems using advanced techniques like reinforcement learning.	Apply	2, 4, 6,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):

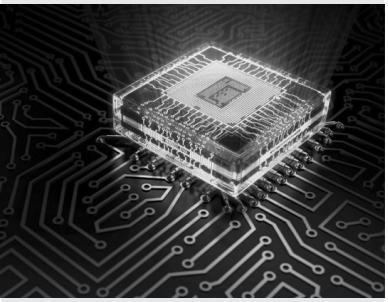


TEXT BOOKS:

1. Kanti Swarup, Man Mohan and P.K.Gupta “Optimization Research, Sultan Chand & Sons. 2005.
2. S.S.Rao “Engineering Optimization Theory and Practice” 4th Edition, Wiley publishers, 2009.

REFERENCE BOOKS:

1. M.C Bhuvaneswari, “Application of Evolutionary Algorithms for Multi-Objective “Optimization in VLSI and Embedded System” Spinger, 2014.
2. Ashlock D, “Evolutionary Computation for Modeling and Optimization” Spinger 2006
3. Kalyanmoy Deb “Optimization for Engineering Design”, Prentice Hall of India 2nd Edition 2012.
4. A. Ravindran and K.M Rogsdell, G.V. Reklaites, “Engineerung Optimization Applications”, Wiley, 2nd Edition, 2006
5. E. J. Haug and J.S Arora, “Applied Optimal Design”, Wiley Publishers, 1979.



Source: https://machinelearningmedium.com/assets/images/computer-architecture.jpg

25CS206 COMPUTER ORGANIZATION AND ARCHITECTURE

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Digital Logic Design.

COURSE DESCRIPTION AND OBJECTIVES:

This course covers the basics of modern Computer Organization and Microprocessor. The emphasis is on understanding the design of computer and its components. The student will learn the concepts of data representation, micro-operations, memory and input output organizations. It also deals with the basic architecture, various addressing modes, pin definitions, interfacing of 8086 with memory and peripheral devices.

MODULE-1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

INTRODUCTION, RTL, DATA REPRESENTATION & COMPUTER ARITHMETIC

Introduction, Register Transfer language, Register Transfer, Bus and Memory Transfers, Data Representation- Fixed Point Representation, Floating Point Representation.

Computer Arithmetic: Fixed-point arithmetic operations such as Addition and Subtraction, Multiplication Algorithms, and Division Algorithms.

UNIT-2 **6L+6T+0P+6SL=18 Hours**

MICRO OPERATIONS & BASIC COMPUTER ORGANIZATION AND DESIGN

Micro operations: Arithmetic, Logic, Shift Micro Operations, Arithmetic Logic Shift Unit.

Basic Computer Organization and Design: Computer Register, Computer Instructions, Memory Reference Instructions. Register Reference Instructions, Input Output Reference Instructions, Interrupt and Instruction Cycle.

PRACTICES:

- Design a Common bus system for eight registers with eight bits each using multiplexers.
- Design a Common bus system for four registers with four bits each using three state gate buffers.
- A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers.
- How many selection inputs are there in each multiplexer? What size of the multiplexers are needed? How many multiplexers are there in the bus?
- Perform arithmetic operations (+42) + (-13) and (-42) – (-13) in binary using signed 2's complement representation for negative numbers.
- Find the product using the Booth Multiplication Algorithm.

  a. (9) x (13) b. (9) x (-13) c. (-9) x (13) d. (-9) x (-13)
- Perform the division of 27 and 4 using the Division algorithm.
- Design a 4-bit combinational circuit decrementer using 4 full adder circuits.
- Register A holds the 8-bit binary 11011001. Determine the B operand and the logic micro-operation to be performed to change the value in A to:

  a) 01101101 b) 11111101

MODULE-2

UNIT-1 **6L+6T+0P+6SL=18 Hours**

MEMORY AND INPUT-OUTPUT ORGANIZATION

Memory Organization: Memory Hierarchy, Main Memory and its interfacing, Auxiliary Memory, Associative Memory, Cache Memory.

Input-Output Organization: Peripheral Devices, Input-Output Interface, Asynchronous Data Transfer, Modes of Transfer, Priority Interrupts. Direct Memory Access (DMA).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

8086 MICROPROCESSOR

8086 microprocessor Architecture, Register model, Memory segmentation, Physical address generation, Addressing modes, Instruction set, Interrupts of 8086. Pin configuration of 8086.

UNIT-3 **6L+6T+0P+6SL=18 Hours**

MEMORY AND I/O INTERFACING TO 8086

8086 Interfacing with static RAM and ROM chips, Address decoding techniques, 8255 PPI, Programmable Communication Interface (8251 USART), Programmable interrupt controller (8259).

PRACTICES:

- Programs on different Data Transfer Instructions using 8086.
- Arithmetic operations: Addition, Subtraction, Multiplication, Division using 8086.
- Programs to analyze different addressing modes of 8086.
- Program to sort the array of given numbers in ascending order.
- Interfacing of 8086 with RAM and ROM.
- Interfacing of 8086 with various IO devices using 8255 PPI.
- Interfacing of 8086 with 8251 USART.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Analyze the computer organization and system bus.	Analyze	1,2,3,9,10
2	Design different digital circuits required to perform the micro operations.	Apply	1,2,3,9,10
3	Design interface circuits for memory and peripheral, DMA, and communication devices. Compare various modes of transfer.	Analyze	1,2,3,4,5, 6,7, 9,10,
4	Analyze the architecture of the 8086 microprocessor and its addressing modes.	Analyze	1,2,3,4,5, 6,7, 9,10,
5	Analyze various interfaces for the 8086 microprocessor.	Analyze	1,2,3,4,5, 6, 7, 9,10

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





TEXT BOOK:

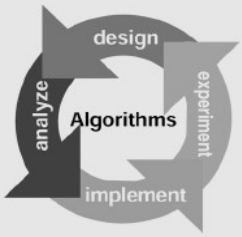
1. M. Morris Mano, "Computer System Architecture", 3rd Edition update, Pearson, 2017.
2. Douglas V.Hall, "Microprocessors and Interfacing", 2nd edition, Tata McGraw Hill, 2017.

REFERENCE BOOKS:

1. William Stallings, "Computer Organization & Architecture: Designing for Performance", 11th Edition, Pearson, 2019.
2. Advanced Microprocessors and Peripherals – A. K. Ray and K.M. Bhurchandani, TMH, 2nd Edition 2006.

SKILLS:

- ✓ *Learn different data representations.*
- ✓ *Design digital circuitry for implementing different operations.*
- ✓ *Identify the types of memories and their uses.*
- ✓ *Study various data transfer mechanisms in digital computer and I/O.*



Source: https://www.facebook.com/Design-and-Analysis-of-Algorithms-1553902878155564/

25CS207 DESIGN AND ANALYSIS OF ALGORITHMS

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Programming for problem solving, Discrete Mathematical Structures, Data Structures.

COURSE DESCRIPTION AND OBJECTIVES:

This course offers the basic knowledge required to analyze the asymptotic performance of algorithms. In addition, this course provides the knowledge required to solve different problems using suitable design strategies such as the greedy method, divide and conquer, dynamic programming, backtracking and branch & bound. This course helps to understand the impact of the choice of data structures and algorithm design strategies on the performance. This course also provides the understanding of advanced graph applications and throws light on tractable and intractable problems.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

INTRODUCTION

Algorithm, Pseudo-code for expressing algorithms, Performance analysis – space and time complexity; Asymptotic notation - big oh notation, Omega notation, Theta notation and little oh notation; Randomized Algorithms; (T1-PP: 1-67)

Analysis of recursive algorithms through recurrence relations- substitution method, Recursion tree method, Masters Theorem. (R1-96-102, T2-53-66)

Disjoint sets: Disjoint set operations, Union and find algorithms. (T1-PP: 101-111).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

DIVIDE & CONQUER AND GREEDY METHOD

Divide and Conquer: General method, Applications - Binary search, Quick sort, Merge sort and Strassen's matrix multiplication. ; (T1-PP: 127-194)

Greedy Method: Applications - job sequencing with deadlines, Knapsack problem, Minimum cost spanning trees. ; (T1-PP: 197-220).

PRACTICES:

- Sort a given set of elements using the following methods and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n inputs. The elements can be read from a fi le or can be generated using the random number generator.
  a. Quick sort b. Merge sort
- Search for a given set of elements using the following methods and determine the time required to search the given element. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus no. of elements. The elements can be read from a fi le or can be generated using the random number generator.
  a. Linear Search b. Binary Search
- Implement the following using divide and conquer approach. To multiply two given square matrices. To multiply two given square matrices using Strassen's matrix multiplication.
- Design the Algorithm to solve Job sequencing with deadlines problem and Analyze its time complexity. Implement the above algorithm using Greedy method.

- Design the Algorithm to solve fractional Knapsack problem using Greedy method. Analyze the time complexity and implement the above algorithm.
- Design the Algorithm to find minimum spanning tree and its cost for an undirected graph. Analyze the time complexity and implement the above algorithm.

MODULE-2

UNIT-1 **6L+6T+6SL+6P=24 Hours**

DYNAMIC PROGRAMMING AND BACKTRACKING

Dynamic Programming: General method, Applications - optimal binary search trees, Matrix chain multiplication, 0/1 knapsackproblem, Reliability Design, All pairs shortest path problem, Travelling sales person problem, Longest Common Sequence Problems (T1-PP:252-300).

UNIT-2 **6L+6T+6SL+6P=24 Hours**

BACKTRACKING

Backtracking: General method, Applications - N-Queen problem, Sum of subsets problem, Graph colouring and Hamiltonian cycles. (T1-PP:339-375).

UNIT-3 **6L+6T+6SL+6P=24 Hours**

BRANCH & BOUND AND P, NP, NP - HARD AND NP-COMPLETE

Branch and Bound: General method, Applications- Travelling sales person problem, 0/1 knapsack problem using LC branch and bound solution and FIFO branch and bound solution. (T1-PP:379-411)

P, NP, NP - HARD and NP-Complete: Basic Concepts - Non-Deterministic Algorithms - The Classes NP-Hard and NP Complete- NP Hard Problems- Clique Decision Problem-Cook's Theorem. (T1-PP:495-518)

PRACTICES:

- Design the Algorithm to find all pairs shortest path problem by using dynamic programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to find optimal binary search tree and its cost by using dynamic programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to find optimal order of matrix chain multiplication and its cost using dynamic programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to find optimal route for travelling sales person problem and its cost by using dynamic Programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to solve N-queens problem by using backtracking approach and Analyze its time complexity. Implement the above algorithm.
- Design the Algorithm to solve sum of subsets problem using backtracking approach and Analyze its time complexity. Implement the above algorithm.
- Design the Algorithm to solve 0/1 Knapsack problem using Branch and Bound method. Analyze the time complexity and Implement the above algorithm.

SKILLS:

- ✓ *Analyze the given algorithm concerning space and time complexities and compare it with other algorithms.*
- ✓ *Develop algorithms for solving problems using divide and conquer, greedy, dynamic programming, backtracking and branch & bound techniques.*
- ✓ *Application of existing design strategies to solve real-world problems.*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Analyze the efficiency of a given algorithm using time and space complexity theory. Understanding algorithmic design strategy like divide and conquer approach.	Analyze	1, 2, 11
2	Apply greedy algorithm Strategy for suit able problems and argue the correctness of such algorithms with respect to the global optimization.	Apply	1, 2,3, 5, 11
3	Apply the dynamic programming paradigm and identify the kind of problem best suited to solve using dynamic programming.	Apply	1, 2, 3, 5, 11
4	Compare and contrast the design principles of branch and bound with backtracking strategy.	Apply	1, 2,3,5, 11
5	Investigate computational complexity of different class of problems.	Analyze	1, 2,4,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOKS:

1. Ellis Horowitz,SatrajSahni and Rajasekharan,“Fundamentalsof Computer Algorithms”, 2nd Edition, Galgotia publications, 2006.
2. Thomas H. Coremen, Charles E. Leiserson and Ronald L. Rivest, “Introduction to Algorithm”, 2nd Edition, MIT press Ltd., 2014.

REFERENCE BOOKS:

1. Anony Levitin, “Introduction to Design and Analysis of Algorithms”, 3rd Edition, Pearson Education, 2016.
2. Donald E. Knuth, “The Art of Computer Programming”, 2nd Edition, Addison Wesley Publishing Company, 1998.
3. Ronald L. Graham, Donald E. Knuth and Oren Patashnik, “Concrete Mathematics”, 2nd Edition, Addison wesley Publishing Company,1998.
4. Dasgupta, Papadimitriou and Vazirani, “Algorithms”, 1st Edition, McGraw-Hill publishers, 2008.
5. Weiss, “Data Structures and Algorithm Analysis”, 1st Edition, Addison-Wesley Publishing Company, 2016.

25CS208 OPERATING SYSTEMS



Source: https://www.123rf.com/stock-photo/operating_system.html

Hours Per Week :

L	T	P	SL	C
2	0	2	2	3

PREREQUISITE KNOWLEDGE: Knowledge of computers fundamentals, Computer organization & Digital logic and its design.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims at concepts and principles of Operating Systems, its overall responsibility in acting as an interface between the system's hardware components and the user. Further, it also helps students to understand the different scheduling policies, process synchronization mechanisms, deadlock handling mechanisms and memory management techniques.

MODULE-1

UNIT-1 **6L+0T+6P+6SL=18 Hours**

LINUX FILE SYSTEM

Introduction to LINUX File System: The LINUX file System, File System Hierarchy, File system Commands, File Attributes, File Permissions.

Basic commands: cal, date, echo, bc, script, passwd, who, uname, tty, stty, pwd, cd, mkdir, rmdir, ls, cat, cp, rm, mv, more, file, wc, od, chmod, vi. (T1, Pg. No: 45 - 78)

Filters: cmp, comm, diff , head, tail, find, cut, paste, sort, uniq. (T1, Pg. No: 95 - 96)

Regular Expressions: grep, egrep, fgrep, Sed- line addressing, context addressing, text editing, substitution. (R4, Pg. No: 354 – 356, 409)

Shell Script: Creating shell script, Shell Input, Shell Output, Basic operators, Conditional and Iterative statements. (R4, Pg. No: 547 – 567);

UNIT-2 **6L+0T+6P+6SL=18 Hours**

PROCESS SCHEDULING

Introduction to Operating System: What Operating System do; Operating System Structure; (T2, Pg. No: 03) Operating System Operations: Process Management, Memory Management, Storage Management. (T2, Pg. No: 18 – 29)

System Security: Security Problem, Program Threats, System and Network Threats, User Authentication. (T2, Pg. No: 621 – 650)

Process concept-overview, Process Scheduling, Operations on Process; Inter Process Communication; (T2, Pg. No: 101 – 123)

Process (CPU) Scheduling-Scheduling Criteria, Scheduling Algorithms; Multiple-Processor scheduling; (T2, Pg. No: 183 – 205**)**.

PRACTICES:

- Use the cat command to create a fi le containing the following data. Call it mytable.txt use tabs to separate the fields.

  1425 ravi 15.65

  4320 ramu 26.27

  6830 sita 36.15

  1450 raju 21.86

SKILLS:

- ✓ *Manage open-source operating systems like Ubuntu, Fedora etc.*
- ✓ *Know the concepts of Processes scheduling and File Systems.*
- ✓ *Identification of different disk scheduling methodologies.*

a. Use the cat command to display the fi le, mytable.txt.

b. Use the vi command to correct any errors in the fi le, mytable.txt.

c. Use the sort command to sort the fi le mytable.txt according to the first field.

d. Call the sorted fi lemytable.txt (same name)

e. Print the fi lemytable.txt.

f. Use the cut &paste commands to swap fields 2and 3mytable.Call itmytable.txt (same name)

g. Print the new fi le, mytable.txt.

- Write a shell script that takes a command–line argument and reports on whether it is directory, a file, or something else.
- Write a shell script that accepts one or more file name as arguments and converts all of them to uppercase, provided they exist in the current directory.
- Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
- Write a shell script that computes the total and average marks of a student according to the following;
  - If average marks ≥ 69 then result is -Distinction.
  - If average marks ≥ 59 and ≤ 70 then result is – First Class.
  - If average marks ≥ 49 and ≤ 60 then result is – Second Class
  - If average marks ≤50 then result is -Pass.
  - Note that any subject marks ≤ 40 then result is -Fail.
  - Accept student name and six subject marks through the keyboard.
- Write an interactive fi le-handling shell program. Let it offer the user the choice of copying, removing, renaming, or linking files. Once the user has made a choice, have the program ask the user for the necessary information, such as the fi le name, new name and so on.
- Write a shell script, which receives two file names as arguments. It should check whether the two file contents are same or not. If they are same then second fi le should be deleted.
- Write a shell script that accepts a file name starting and ending line numbers as arguments and displays all the lines between the given line numbers.
- Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
- Write a shell script to biggest among three numbers.
- Write a shell script to find whether the entered number is prime or not.
- Write a shell script to find the whether the entered number is Armstrong or not.
- Write a shell script to find the factorial of the given number.
- Implementation of new process creation and its communications.
- Implement of thread creation and deletion.
- Implementation of FCFS scheduling.
- Implementation of SJF and RR Scheduling.

MODULE-2

UNIT-1 **6L+0T+6P+6SL=18 Hours**

PROCESS SYNCHRONIZATION & DEADLOCKS

Process Synchronization: The critical-section problem; Peterson's solution; Synchronization hardware; Semaphores; Classical problems of synchronization: Producer-Consumer Problem, Readers-Writers Problem, Dining Philosophers Problem, Monitors. (T2, Pg. No: 225 – 252).

DEADLOCKS

Deadlocks: Deadlock characterization; Methods of handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery. (T2, Pg. No: 285 – 305).

UNIT-2 **6L+0T+6P+6SL=18 Hours**

MEMORY MANAGEMENT

Memory Management: Basic concept to memory management, Swapping, Contiguous Memory Allocation, Paging, Structure of the Page Table, Segmentation. (T2, Pg. No: 315 – 345)

Virtual Memory Management: Demand Paging, Page Replacement: Optimal, First in First Out (FIFO), Second Chance (SC), Not recently used (NRU), Least Recently used (LRU), Allocation of Frames. (T2, Pg. No: 357 – 385).

UNIT-3 **6L+0T+6P+6SL=18 Hours**

SECONDARY STORAGE STRUCTURE

Secondary Storage Structure: Over view of mass-storage structure, Directory & Disk Structure, RAID Structure, disk scheduling; (T2, Pg. No: 505 – 532)

File System Interface - File concept, File-system structure, Allocation Methods, Access Methods, File-System Mounting, File Sharing, Protection, Free Space Management. (T2, Pg. No: 461 – 482).

PRACTICES:

- Implementation of producer consumer problem.
- Implementation of Banker's algorithm for Dead lock avoidance.
- Assume that you have a page-reference string for a process with m frames (initially all empty). The page-reference string has length p, and n distinct page numbers occur in it.

  a) What is a lower bound on the number of page faults?

  b) What is an upper bound on the number of page faults?
- Consider the following page-replacement algorithms. Rank these algorithms on a five-point scale from "bad" to "perfect" according to their page-fault rate. Separate those algorithms that suffer from Belady's anomaly from those that do not.

  a) LRU replacement. b) FIFO replacement. c) Optimal replacement. d) Second-chance replacement.
- Develop the program to find the total no. of page faults would occur for the following replacement algorithms, assuming one, two, three, four, five, six, and seven frames for the gives reference string?

  Page reference string: 1, 2, 3, 4,2, 1, 5, 6, 2, 1, 2, 3,7, 6, 3, 2, 1, 2, 3, 6.

  Remember that all frames are initially empty, so your first unique pages will cost one fault each.

  - LRU replacement.
  - FIFO replacement.
  - Optimal replacement.
- What is the minimum number of page faults for an optimal page replacement strategy for The reference string 1, 2, 3,4, 5, 3, 4, 1, 6,7, 8, 7, 8, 9, 7, 8,9, 5, 4, 5,4, 2. with four page frames?
- Consider a demand-paged computer system where the degree of multiprogramming is currently fixed at four. The system was recently measured to determine utilization of the CPU and the paging disk. Three alternative results are shown below. For each case, what is happening?

  a) Can the degree of multiprogramming be increased to increase the CPU utilization? Is the paging helping?

  b) CPU utilization 13 percent; disk utilization 97 percent.

  c) CPU utilization 87percent; disk utilization 3 percent.

  d) CPU utilization 13 percent; disk utilization 3 percent.
- Implementation of Disk scheduling algorithm–FCFS.
- Implementation of Disk scheduling algorithm–SSTF and SCAN.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply regular expressions and LNIX text processing tools to search, filter, manipulate text data and develop the shell programming.	Apply	1, 2,
2	Apply the concepts of process scheduling algorithms and process synchronization techniques to derive the efficiency of resource utilization.	Apply	1, 2, 3,
3	Analyze process synchronization issues and evaluate deadlock conditions to identify and recommend appropriate solutions for safe process execution.	Analyze	1,2,4,
4	Apply memory and virtual memory management techniques such as paging, segmentation, and page replacement algorithms to manage and optimize system memory usage.	Apply	3,5,7,9
5	Analyze secondary storage structures and file system designs to evaluate the performance and efficiency of disk scheduling algorithms, directory organization, and space management techniques.	Analyze	2,4,7,11
6	Evaluate the performance of different memory allocation and replacement algorithms under varying workloads and system conditions.	Evaluate	2,4,6,7,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





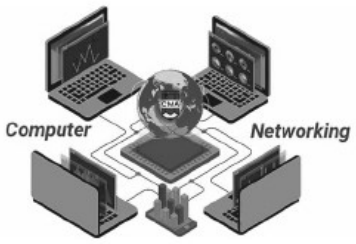
TEXT BOOKS:

1. Sumitabha Das, Unix concepts and applications, TMH Publications, 4th Edition, July 2017.
2. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, “Operating System Concepts”, 9th Edition, John Wiley & SonsInc, 2013.

REFERENCE BOOKS:

1. Richard. Stevens and Stephen A Rago, “Advanced Programming in the Unix Environment”,3rd Edition, Addison-Wesley, 2013.
2. William Stallings, “Operating Systems-Internals and Design principles” PHI, 7th Edition, 2012.
3. Gary J. Nutt. Addison-Wesley, “Operating Systems: A Modern Perspective”, 2nd Edition, 2001.
4. B.A. Forouzan & R.F.Giberg, - Unix and shell Programming, Thomson, 1st Edition, New Delhi,2003.

25CS209 COMPUTER NETWORKS



Source: https://snabaynetworking.com/what-is-computer-network-and-its-types/

Hours Per Week :

L	T	P	SL	C
3	0	2	3	4

PREREQUISITE KNOWLEDGE: Java Programming, UNIX Commands.

COURSE DESCRIPTION AND OBJECTIVES:

The goal of this course is introduces the fundamental concepts, architecture, protocols, and technologies used in modern computer networks. It provides a layered approach to understanding how data is transmitted across networks, focusing on the OSI and TCP/IP models, data communication, routing, switching, network security, and wireless networking. Emphasis is placed on both theoretical foundations and practical implementations, enabling students to analyze, configure, and troubleshoot network systems effectively.

MODULE-1

UNIT-1 **9L+0T+6P+9SL=24 Hours**

INTRODUCTION TO COMPUTER NETWORKS AND INTERNET

Understanding of Networks,Network Hardware,Network software,OSI,TCP/IP ReferenceModels, Topology, Example Networks:ARPANET,INTERNET.

PHYSICAL LAYER-Guided Transmission Media:Twisted Pair,Coaxial Cable,Fiberoptics, Wireess transmission.

UNIT-2 **9L+0T+6P+9SL=24 Hours**

DATALINK LAYER AND LOCAL AREA NETWORKS

Introduction to Datalink layer Services,Error Detection and Error Correction Techniques, Elementary Data Link Protocols,sliding window Protocols.

THE MEDIUM ACCESS SUBLAYER - Channel Allocation Problem,Multiple Access Protocols, Ethernet,Switching,WirelessLAN,Bluetooth.

PRACTICES:

- Study of Network devices in detail.
- Repeater
- Hub
- Switch
- Bridge
- Gateway
- Router
- Implement the data link layer framing methods such as character count,character stuffing,bit stuffing.
- Implement on a data set of characters the three CRC polynomial CRC 12, CRC 16 ,CRC CCIP
- Find all the IP addresseson your network using Unicast,Multicast,Broadcast on your Network.
- Apply the commands such as ping,Tracert, ipconfig,Pathping,telnet,FTP,ARP,Hostname,Nbtstat, Netdiag and Nslookup to solve various problems.

SKILLS:

- ✓ *Investigate syntax and semantics of a regular and context free languages.*
- ✓ *Develop the problem understanding solving ability.*
- ✓ *Design optimized solutions for a language.*

MODULE-2

UNIT-1 **9L+0T+6P+9SL=24 Hours**

NETWORK LAYER

Introduction to Forwarding and Routing,Network service Models,Routing Algorithms,Congestion Control Algorithms,Internetworking,the network layer in the internet(IPV4 and IPV6)

UNIT-2 **9L+0T+6P+9SL=24 Hours**

TRANSPORT LAYER

Transport Layer Service,Elements of Transport Protocol,Multiplexing and demultiplexing Connection oriented Tansport(TCP),connection less Transport(UDP)

UNIT-3 **9L+0T+6P+9SL=24 Hours**

APPLICATION LAYER

DNS, E-Mail, World Wide Web:Architectural Overview,HTTP,SNMP,FTP,SMTP,TELNET.

PRACTICES:

- Implement Dijsktra's algorithm to complete the shortest path through a graph.
- Simulate the Distance Vector Routing Algorithm for a network of 4 nodes with given initial distance tables and display how the distance tables update after each iteration.
- Take an example subnet of hosts. Obtain broadcast tree for it.
- Use Packet tracer software to bulid network topology and configure using Distance vector routing and link state routing protocols
- Study of different types of Network cables(copper and Fiber) and Prepare cables(Straight and Cross) to connect two or more systems.

  i) Use a crimping tool to connect jacks.

  ii) Use a LAN tester to connect the cables

  iii) Install and configure Network devices:HUB,Switch,Routers(Consider both manageable and non-manageableswitches).Perform logical configuration of the system and set the bandwidth of different ports.

  iv) Install and configure Wired and Wireless NIC and transfer files between systems in wired and wireless LAN.consider both adhoc- and infrastructure modes of operation

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Understand the principles and models of computer networks, including OSI and TCP/IP layers.	Analyze	1,2,5,6,
2	Evaluate different physical layer media and switching methods.	Evaluation	1,2,5,6,
3	Demonstrate various protocols involved in datalink layer operations.	Apply, Analyze	1,2,5,
4	Analyze various design issues,Protocols and functionalities of network layer.	Apply	1,2,
5	Understanding transport layer functionalities and applying multuplexing techniques.	Apply	1,2,5,6,
6	Apply different protocols to perform end-to-end delivery and interaction with users.	Apply	1,2,3,5,6,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):

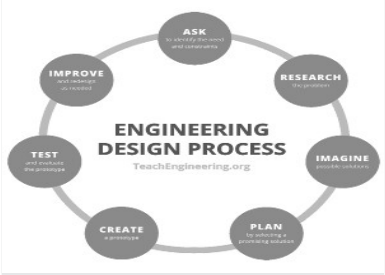


TEXT BOOK:

1. S. S. Sastry, “Introductory Methods of Numerical Analysis”, 5th ed., PHI Learning, 2012.
2. M. K. Jain, “Numerical Methods for Scientific and Engineering Computation”, New Age International, 8th Ed., 2022.

REFERENCE BOOKS:

1. P. Kandasamy, “Numerical Methods”, S. Chand, 2nd ed., 2015.
2. C. Ray Wylie, Louis C. Barrett, “Advanced Engineering Mathematics”, 6th Edition, McGraw-Hill Book Co., New York, 1995.
3. Gupta C.B., Singh S.R., and Mukesh Kumar, “Engineering Mathematics for Semester I & II”, McGraw Hill Education (India) Pvt. Ltd., 2015.
4. H. K. Das, “Advanced Engineering Mathematics”, S. CHAND & COMPANY PVT. LTD, 2008.



Source: https://www.teachengineering.org/activities/view/ced-2680-engineering-design-process-thinking.

25CS210 DESIGN THINKING AND ENGINEERING ORIENTATION

Hours Per Week :

L	T	P	SL	C
0	0	2	0	1

PREREQUISITE KNOWLEDGE: The course requires a basic understanding of Computer Science Engineering concepts, problem-solving techniques, and an interest in user-centric design and system development. Familiarity with computational thinking and collaborative work is also beneficial.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the principles of design thinking as a systematic framework for problem-solving and innovation in engineering, emphasizing user-centric solutions. It provides hands-on experience in designing and prototyping software systems and fosters interdisciplinary collaboration. The course explores computational thinking, system design, and emerging trends like AI, IoT, and big data, with a focus on ethics and sustainability.

The primary objective of this course is to introduce students to the principles of design thinking as a structured framework for addressing complex engineering challenges. By focusing on user-centric methodologies, students will learn to empathize with end-users, define problems accurately, and generate innovative solutions. This course aims to enhance creativity, critical thinking, and collaborative problem-solving skills specific to software and system design. Students will explore the role of computer science engineering in multidisciplinary solutions and its applications in various fields like AI, IoT, and big data. Additionally, the course emphasizes the importance of hands-on learning through prototyping, iterative testing, and user feedback. It also seeks to instill an understanding of ethical considerations, sustainability, and inclusivity in technology design and engineering practices.

MODULE-1

UNIT-1 **0L+6T+0P+0SL=6 Hours**

DESIGN THINKING PRINCIPLES AND USER-CENTRIC DESIGN

Getting Under Your Skin, How Design Thinking Is About More Than Style –t1- Page 19,Converting Need into Demand, Putting People First – t1-Page 45,A Mental Matrix, “These People Have No Process!” – t1-Page 69,Building to Think, The Power of Prototyping – t-Page 93,Returning to the Surface, The Design of Experiences – t-Page 115,Spreading the Message, The Importance of Storytelling – t1 -Page 135.

UNIT-2 **0L+6T+0P+0SL=6 Hours**

Design Thinking Meets the Corporation, Teaching to Fish – t1-Page 161,The New Social Contract, We’re All in This Together – t1-Page 183,Design Activism, Inspiring Solutions with Global Potential –t1- Page 209,Designing Tomorrow—Today – t1-Page 233,Redesigning Design – t1-Page 249.

PRACTICES:

- Research thoroughly to understand user needs and challenges.
- Use empathy to create user personas and map user journeys.
- Follow the design thinking process: Empathize, Define, Ideate, Prototype, and Test.
- Encourage collaborative brainstorming and test prototypes with users.
- Prioritize usability, accessibility, and inclusivity in all designs.

MODULE-2

UNIT-1 **0L+6T+0P+0SL=6 Hours**

ENGINEERING ORIENTATION AND EMERGING TRENDS

Introduction – t2-Page 7, You Can Do It! – t2-Page 8, What is "Success"? – t2-Page 11, Keys to Success in Engineering Study – t2-Page 18, Models for Viewing Your Education –t2- Page 24,Structure Your Life Situation –t2- Page 28.

UNIT-2 **0L+6T+0P+0SL=6 Hours**

Introduction – t2- Page 37,What is Engineering? – t2- Page 38. The Engineering Design Process – t2- Page 40,Case Study: Human-Powered Helicopter – t2- Page 43,Rewards and Opportunities of an Engineering Career – t2- Page 51,Engineering Disciplines – t2- Page 55,Engineering Job Functions – t2- Page 59,Employment Opportunities – t2- Page 63,Important Fields for the Future – t2- Page 67,Sustainability – t2- Page 71,Engineering as a Profession – t2- Page 75

UNIT-3 **0L+6T+0P+0SL=6 Hours**

Introduction – t2- Page 272,Organization of Engineering Education – t2-Page 273,The Role of Community Colleges in Engineering Education – t2- Page 275, The Engineering Education System – t2- Page 278, Academic Advising – Page 280, Academic Regulations – t2- Page 283,Engineering Ethics – t2- Page 291, Student Conduct – t2- Page 292,Opportunities for Graduate Education in Engineering – Page 293,Engineering as Preparation for Further Education – t2- Page 294.

PRACTICES:

- Apply computational thinking to design scalable, secure, and efficient systems.
- Focus on ethical practices like data privacy, AI fairness, and inclusivity.
- Stay updated with advancements in AI, IoT, and big data, applying user-centric design principles.
- Use effective data visualization techniques to present insights clearly.
- Continuously refine designs based on user feedback and real-world needs.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply design-thinking principles to user-centric software and system development.	Apply	1,2,5
2	Use prototyping tools and frameworks to develop and test technology solutions.	Create	3,4,6
3	Understand the role of empathy, ethics, and sustainability in CSE design.	Analyze	2,4,7
4	Collaborate effectively to solve real-world problems using iterative methods..	Analyze	1,5,6
5	Apply engineering principles and design thinking to emerging trends like AI, IoT.	Apply	3,6,7

SKILLS:

- ✓ *Critical and creative thinking for problem-solving and innovation.*
- ✓ *Empathy to understand user needs and challenges.*
- ✓ *Collaboration skills to work effectively in teams.*
- ✓ *Basic technical knowledge in computational thinking and system design.*
- ✓ *Prototyping and testing to develop and refine solutions.*

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOK:

1. Brown, Tim. “Change by Design: How Design Thinking Creates New Alternatives for Business and Society”, Revised Edition, Harper Business, 2009.
2. Studying Engineering: A Road Map to a Rewarding Career by Raymond B. Landis, Steffen Peuker, and Jennifer Mott Edition: 5th (2023).

REFERENCE BOOKS:

1. Gero, John S. & Maher, Mary Lou. “Designing Design: How Design Thinking Transforms the Practice of Engineering”, First Edition, Springer, 2019.
2. Brown, Tim. “Design Thinking: A Method for Creative Problem Solving”, First Edition, Harper Design, 2015.

III

YEAR

B.Tech.

COMPUTER SCIENCE AND ENGINEERING

I SEMESTER

25EN301	-	Soft Skills Laboratory
25CS301	-	Artificial Intelligence
25CS302	-	Machine Learning
25CS303	-	Cryptography and Network Security
25CS304	-	Software Engineering
25CS305	-	Theory of Computation
	-	Department Elective - 2
	-	NCC/NSS/SAC/E-Cell/Student Mentoring/Social Activities/Publication

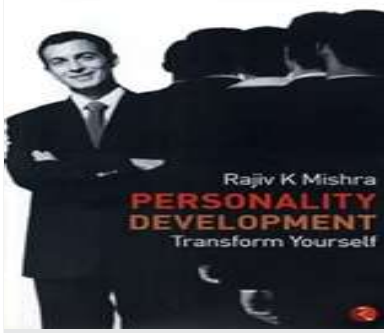
II SEMESTER

25TP301	-	Quantitative Aptitude and Logical Reasoning
25EN302	-	Professional Communication Skills
25CS306	-	Privacy Preserving and Intrusion Detection
25CS307	-	Parallel & Distributed Computing
	-	Department Elective - 2,3,4
	-	Open Elective - 2
25CS308	-	Inter Department Project

COURSE CONTENTS

I SEM & II SEM

25EN301 SOFT SKILLS LABORATORY



Source: https://img.bookchor.com/images/cover/596/9788129103451.jpg

Hours Per Week :

L	T	P	SL	C
0	0	2	0	1

PREREQUISITE KNOWLEDGE: Basics of personality traits and qualities of a good human being and motivation for self-improvement. .

COURSE DESCRIPTION AND OBJECTIVES:

This course is designed to equip undergraduate engineering students with essential soft skills that are critical for success in both academic and professional settings. The course will provide a comprehensive foundation in key soft skills such as personality development communication, teamwork and leadership, time management, and getting ready to industry to face different kinds of interviews and develop strategies to win the interviews. Through a combination of theoretical learning and practical exercises, students will enhance their ability to interact effectively with peers, professors, and professionals like interview board members in a variety of settings. Emphasis will be placed on real-world applications, helping students build confidence and develop the interpersonal skills necessary for effective collaboration, innovation, and career advancement This course will help the students, to develop interpersonal and intrapersonal skills, analyze and improve their personality traits. In addition to the above, the learners to be aware of the needs of the industry to focus on preparing for job placement/ industry ready.

MODULE-1

UNIT-1 **0L+0T+12P+0SL=12 Hours**

PERSONALITY DEVELOPMENT-1

Analyzing self for personality development through Strengths, Weaknesses, Opportunities and Threats or Challenges. Atma-Vichara (self-inquiry)

Creating self-awareness through Open Arena, Blind spots, Facade and Unknown areas of individual (Jagrat, Swapna, Sushupti, Turiya and the importance of Atma-Bodha.)

The importance of team building skills and various skills required to build a team along with examples (Panchatantra and Mahabharata).

UNIT-2 **0L+0T+12P+0SL=12 Hours**

PERSONALITY DEVELOPMENT-2

Importance of goal setting, Edwin Locke's theory of goal setting, setting goals, SMART technique for goal setting, dividing goals. Purusharthas (Dharma, Artha, Kama, Moksha)

Need for the stress management, ways to manage stress, stress management skills and techniques, Stress management strategies. (Dhyana meditation and Pranayama for holistic well-being and emotional balance).

Major themes of time management, benefits of time management, effective time management, POSEC method, Covey's Time management. Dinacharya and Ritucharya.

PRACTICES:

- SWOT analysis
- Johari window
- Team building skills
- Goal setting
- Stress management
- Time management.

SKILLS:

- ✓ *Balancing social and emotional intelligence quotients though analyzing interpersonal and intrapersonal skills.*
- ✓ *Solving personal and professional life hiccups with confidence and maturity.*
- ✓ *Career planning with clear personal and professional goals.*
- ✓ *Preparing tailor made resume and face various job interviews with enriched personality traits.*

MODULE-2

UNIT-1 **0L+0T+12P+0SL=12 Hours**

GETTING READY TO INDUSTRY:

Industry needs: dress sense, appearance, creating an image, image management and etiquettes at work place.

Samskaras (refinement), ancient concepts of Aachara (right conduct), and Sadvritta (ideal behavior)

Various language functions, expressions and usage: informal, semi-formal and formal.

UNIT-2 **0L+0T+12P+0SL=12 Hours**

JOB RECRUITMENT PROCESS

Narrating about self, academic achievements, project works, family background (if required) Purusharthas and Svadharma (personal duty or career path).

Participating group discussions, various roles in group discussion, taking turns in the discussion, expressing opinions, agreeing, disagreeing, leading using SPELT technique, consolidating the information and arriving to a conclusion. Shastrartha, Sabha traditions of discourse.

Practicing HR questions.

UNIT-3 **0L+0T+12P+0SL=12 Hours**

DOCUMENT PREPARATRION TO THE NEXT LEVEL:

Collecting the data, finding a suitable template, presenting the objectives, education qualification, projects, co and extracurricular activities, achievements, certifications, hobbies and personal information.

Presenting ambitions, aspirations, successes and persuading.

PRACTICES:

- Image management
- Essential language functions and expressions
- Self-introduction
- Group discussion
- Facing interviews
- Resume preparation
- Statement of purpose

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Introspect on personal traits (for example strengths, weaknesses, known and unknown areas, hidden etc.,) and emerge as a balanced personality with improved self-awareness and self-worth.	Analyze	11
2	Learn the importance of the use of formal language in their professional as well as personal areas to represent their personality by demonstrating gender sensitive language at workplace in the cross cultural environment..	Apply	9
3	Be able to prepare a resume and gain the confidence to face an interview.	Create	10
4	Possess the interpersonal skills to conduct himself/herself effectively in everyday professional and social contexts.	Apply	8
5	Be able to bring professionalism into his/her daily activities.	Apply	8,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOKS:

1. Edward Holffman, Ace the corporate personality, McGraw Hill,2001.
2. Adrian Furnham, Personality and intelligence at work, Psychology Press, 2008.

REFERENCE BOOKS:

1. John Adair Kegan Page, “Leadership for innovation” 1st edition, Kogan, 2007.
2. Krishna Mohan & NP Singh, “Speaking English effectively” 1st edition, Macmillan, 2008.
3. Dr. S.P. Dhanvel, English and Soft skills, Orient Blackswan, 2011.
4. Rajiv K. Mishra, Personality Development, Rupa& Co. 2004.
   a. https://zety.com/blog/resume-formats for latest resumes.



Source: https://www.forbes.com/sites/bernardmarr/2020/08/03/3-important-ways-artificial-intelligence-will-transform-your-business-and-turbocharge-success/

25CS301 ARTIFICIAL INTELLIGENCE

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Probability and statistics.

COURSE DESCRIPTION AND OBJECTIVES:

The primary objective of this course is to introduce the basic principles, techniques, and applications of Artificial Intelligence. In addition to this, student will understand the building blocks of AI such as search, knowledge representation, inference, logic, and learning. This course enables the students to develop a small AI system for real time problems.

MODULE-1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

INTELLIGENT SYSTEMS

Introduction, what is AI, Examples of AI systems (T-PP:1-5), Brief history of AI Agent, Agents and environments, Structure of agents, the concept of rationality, the nature of environments, Types of agents, Agentic AI, problem solving approaches to typical AI problems (T-PP:34-58).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

PROBLEM SOLVING

State Space Problem; Searching: Uninformed search, Informed Search: Solving problems by searching: Heuristic functions, Hill climbing, Best First Search, A* algorithm, AO* algorithm (T-PP:64-102), Searching game trees: Min Max Search, Alpha Beta pruning (T-PP:161-170).

PRACTICES:

- In the classical vacuum cleaner problem, we have two rooms and one vacuum cleaner. There is dirt in both the rooms and it is to be cleaned. The vacuum cleaner is present in any one of these rooms. Find the solution, how we can reach to reach a state in which both the rooms are clean and are dust free.
- In this problem, three missionaries and three cannibals must cross a river using a boat which can carry at most two people, under the constraint that, for both banks, that the missionaries present on the bank cannot be outnumbered by cannibals. The boat cannot cross the river by itself with no people on board. Find the solution, how to solve the given problem.
- You are given two jugs, a 4-gallon one and a 3-gallon one, a pump which has unlimited water which you can use to fill the jug, and the ground on which water may be poured. Neither jug has any measuring markings on it. Find the solution, how can you get exactly 2 gallons of water in the 4-gallon jug?
- There is a farmer who wishes to cross a river but he is not alone. He also has a goat, a wolf, and a cabbage along with him. There is only one boat available which can support the farmer and either of the goat, wolf or the cabbage. So at a time, the boat can have only two objects (farmer and one other). But the problem is, if the goat and wolf are left alone (either in the boat or onshore), the wolf will eat the goat. Similarly, if the Goat and cabbage are left alone, then goat will eat the cabbage. The farmer wants to cross the river with all three of his belongings: goat, wolf, and cabbage. What strategy he should use to do so?
- Either place a block that doesn't have other blocks stacked on top of it on another block with the same behaviour, or on the table. The initial and the goal state are described by the exact position of each block. Find the solution, how to solve the given problem.

- Given a 3×3 board with 8 tiles (every tile has one number from 1 to 8) and one empty space. The objective is to place the numbers on tiles to match the final configuration using the empty space. We can slide four adjacent (left, right, above, and below) tiles into the empty space. Find the solution, how to solve the given problem by using using A* search algorithm.
- The rules of tic-tac-toe on the 3 × 3 field are as follows. Before the first turn all the field cells are empty. The two players take turns placing their signs into empty cells (the first player places Xs, the second player places Os). The player who places Xs goes first, the another one goes second. Find the solution, how to solve the given problem where the winner is the player who first gets three of his signs in a row next to each other (horizontal, vertical or diagonal).
- In crypt arithmetic problem, the digits (0-9) get substituted by some possible alphabets or symbols. The task in crypt arithmetic problem is to substitute each digit with an alphabet to get the result arithmetically correct. Find the solution, how to solve the given problem, where we can perform all the arithmetic operations on a given crypt arithmetic problem.

SKILLS:

- ✓ *Analyze Intelligent systems.*
- ✓ *Apply problem solving techniques.*
- ✓ *Interface various knowledge representation.*
- ✓ *Create a dynamic planning.*

MODULE-2

UNIT-1 **6L+6T+0P+6SL=18 Hours**

KNOWLEDGE REPRESENTATION AND PLANNING

Propositional logic: Inference in propositional logic, Resolution, Forward chaining, Backward chaining, First order logic: Reasoning patterns in First order logic, Resolution, Forward chaining, Backward chaining (T-PP:234-243 & PP-322-356), The planning problem: Planning with state space search, Partial order planning, Planning graphs (T-PP:366-386).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

LEARNING

Forms of learning: Supervised Learning, Unsupervised learning (T-PP:693-707), Reinforcement learning, Ensemble Learning, Expert System (T-PP:748-752).

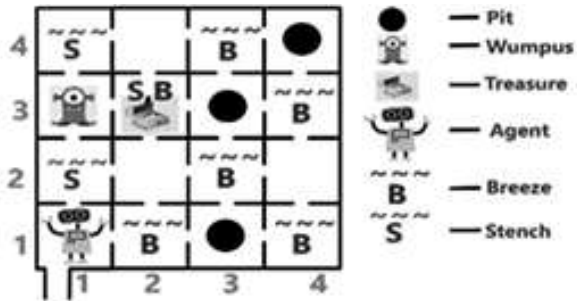
UNIT-3 **6L+6T+0P+6SL=18 Hours**

GENERATIVE AI

Introduction to Generative AI, Introduction to LLMs (T-PP:860-864), Introduction to Retrieval Augmented Generation (RAG), Transformers Architecture.

PRACTICES:

- With logic programming, compare expressions and find out unknown values.
- The Wumpus world is a cave with 16 rooms (4×4). Each room is connected to others through walkways (no rooms are connected diagonally). The knowledge-based agent starts from Room [1, 1]. The cave has – some pits, a treasure and a beast named Wumpus. The Wumpus cannot move but eats the one who enters its room. If the agent enters the pit, it gets stuck there. The goal of the agent is to take the treasure and come out of the cave. The agent is rewarded, when the goal conditions are met. The agent is penalized, when it falls into a pit or being eaten by the Wumpus. Some elements support the agent to explore the cave, like -The Wumpus's adjacent rooms are stench. -The agent is given one arrow which it can use to kill the Wumpus when facing it (Wumpus screams when it is killed). – The adjacent rooms of the room with pits are filled with breeze. -The treasure room is always glittery. Find the Wumpus presented room.



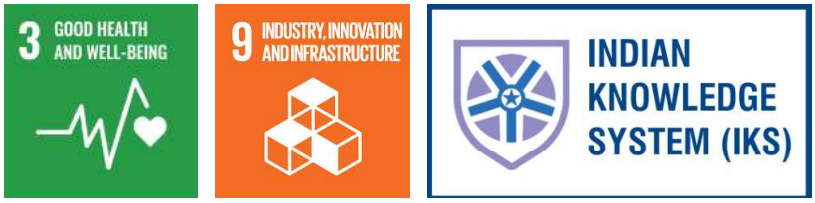
- you are on one side of a river with a wolf, a goat, and a cabbage. You want to transport all three to the other side of the river, but you can only transport one object at a time. You cannot leave the wolf and the goat alone, or the cabbage and the goat alone; you are the only thing keeping them from eating each other. How can you transport everything from one side of the river to the other? Formulate it in terms of a Planning Domain Definition Language (PDDL).
- Impliment the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
- Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.
- Implement k-nearest neighbors classification using python.
- Implement linear regression using python.
- Implement the naïve Bayesian classifier for a sample training dataset. Compute the accuracy of the classifier, considering few test data sets.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply AI search Models and Generic Search strategies for problem solving.	Apply	1,3
2	Inspect and analyze Logic for representing Knowledge and Reasoning of AI systems and Conduct investigation and implement project using AI learning techniques.	Analyze	2,
3	Apply and evaluate the searching strategies to achieve the goal for a given situation..	Apply	6
4	Design different learning algorithms for improving the performance of AI systems.	Apply	4,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



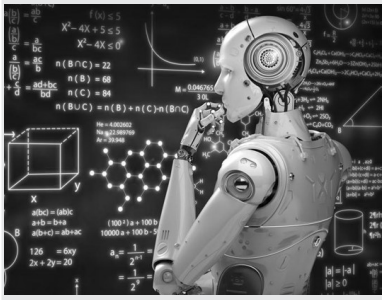
TEXT BOOK:

1. S. Russel and P. Norvig, “Artificial Intelligence – A Modern Approach”, 4th Edition, Pearson Education, 2010.

REFERENCE BOOKS:

1. David Poole, Alan Mackworth, Randy Goebel, “Computational Intelligence : a logical approach”, Oxford University Press,2013.
2. G. Luger, “Artificial Intelligence: Structures and Strategies for complex problem solving”, 4th Edition, Pearson Education, 2008.
3. J. Nilsson, “Artificial Intelligence: A new Synthesis”, Elsevier Publishers. 2012.

25CS302 MACHINE LEARNING



Source: https://www.forbes.com/sites/kalevleetaru/2019/01/15/why-machine-learning-needs-semantics-not-just-statistics/

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Probability & Linear Algebra, Python language.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a broad introduction to various machine learning concepts including Supervised learning (parametric/non-parametric algorithms, support vector machines, neural networks) and Unsupervised learning (clustering, dimensionality reduction) methods. Students will get an understanding of various challenges of Machine Learning and will be able to decide on model complexity. Numerous case studies introduced in this course allow the students to apply machine-learning algorithms in computer vision, medical imaging, audio, and text domains. Laboratory experiments of this course will introduce students to advanced Machine Learning Python libraries such as Scikit-Learn, Mat plot lib, and many other recent ML-related APIs. The course is designed such that the students get enough hands-on experience with a major focus on the practical implementation of theoretical concepts.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

INTRODUCTION

What is machine learning? Machine learning applications; Types of Learning: Supervised learning, unsupervised learning, and reinforcement learning. (T1,pp.1-9)

Model Training Essentials: Loss function, Types of loss function,(T1, pp.33–36) Bias–Variance Trade-off (T2,pp. 67–69), Regularization, types of regularization techniques (T1, pp. 151–153), Hypothesis Testing and Variable Selection, Re-sampling methods: Sub sampling and Upsampling, SMOTE, ADASYN (Adaptive Synthetic); Cross Validation (validation set, Leave-One-Cut (LOO), k-fold strategies),(R2,pp.198-215) and bootstrap; Evaluation measures-Error functions, Confusion Matrix, Accuracy, Precision and Recall, F1 Score (R3,pp.86-89).

Regression Analysis: Linear Regression, Simple and Multiple Linear Regression (R2, pp.63-92), Polynomial Regression (R1, pp.290), Logistic Regression, Multi nominal Regression (R1,pp-119-127). Ordinary Least Squares Method (R2,pp.45-48), Model Shrinkage-Ridge, and LASSO regression (R2, pp.237-241).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

FEATURE SELECTION

Feature Selection Strategies: Problem statement and Uses, Types of feature selection Filter method: Chi-Square test, Wrapper method: forward/backward selection algorithms, Embedded method: Elastic net.(R3,pp.38-47).

Dimensionality Reduction: Matrix factorisation(T1, pp.561-569), Principal components analysis, Linear discriminant analysis (T3, pp.207-226).

PRACTICES:

- Apply the following tasks to any given dataset:
  a. Load and visualize data.
  b. Check out and replace missing values.
  c. Encode the Categorical data.
  d. Splitting the dataset into Training and Test set.

SKILLS:

- ✓ *Statistical data analysis.*
- ✓ *Classify / Cluster data.*
- ✓ *Tool usage for developing ML applications.*

e. Splitting the dataset into k-folds.

f. Feature scaling.

- Working with imbalanced datasets:

  a. Load an imbalanced dataset and visualize imbalance in the data as a bar plot.

  b. Balance the dataset using:

  i. Random Over sampling.

  ii. Random Under sampling.

  iii. SMOTE.

  e. Report your observations on the performance of models trained using balanced and imbalanced data

- House price prediction:

  a. Create a model that predicts a continuous value (price) from input features (square footage, number of bedrooms and bathrooms).

  b. Implement a univariate Model using Least Squares and plot best-fit line.

  c. Implement a multivariate Model using Least Squares and plot best-fit line.

  d. Retrieve model error and model coefficients.

  e. Observe Variance Inflation Factor (VIF).

  f. Implement Ridge regression model.

  g. Implement LASSO regression model.

  h. Report your observations on the above models for house prediction.

- Heart disease prediction:

  a. Implement a logistic regression model to predict whether an individual is suffering from heart disease or not.

  b. Evaluate and compare model performance using the following validation approaches:

  i. Validation set approach.

  ii. K-fold cross validation.

  iii. Stratified K-fold cross validation.

  iv. LOO strategy.

  c. Plot Confusion matrix.

  d. Report performance of the model in terms of the following metrics:

  i. Accuracy.

  ii. Precision-Recall.

  iii. F1 Score.

  e. Report your observations and explain when to use what type of measures.

- Implement the Polynomial Regression algorithm to fit data points. Select the appropriate data set for your experiment and draw graphs.
- Perform effective feature selection in a given dataset using any one of the feature selection techniques.
- Dimension Reduction:

  a. Load a dataset.

  b. Apply dimension reduction using:

  i. Principal Component Analysis

  ii. Linear Discriminant Analysis

  c. Apply the model on data with reduced dimension.

  d. Compare and contrast model performance in each case.

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 hours**

CLASSIFICATION: Classification Types: Binary, Multi-class, and Multi-label Classification,(T2,pp.254-257) CART (Classification and Regression Trees)(T2,pp.216),k-NN (k-Nearest Neighbor classifier) (T3 pp64-66)and Fuzzy k-NN, Learning Vector Quantization (LVQ),(R3, pp.219).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

NEURAL NETWORKS AND ENSEMBLE METHODS

Perceptron Learning Algorithm: Delta rule, Gradient Descent and Stochastic Gradient Descent. Multi-layer Perceptron Learning: Backpropagation and, Activation Functions: Types and Applications (T3, pp.255-263), Ensemble Methods: Ensemble Learning Model Combination Schemes, Voting and Error-Correcting Output Codes, Bagging: Random Forest Trees, Extra Tree Classifier, Boosting: AdaBoost, XGBoost, Gradient Boosting, Stacking (T3, pp.183-202).

UNIT-3 **6L+6T+6P+6SL=24 Hours**

Clustering Techniques: Fuzzy C-Means Clustering (R4, pp.578-581), Graph-Based Clustering: Minimum Spanning Tree (MST) (R4, 611-615), Spectral Clustering (R5. Pp.396-413), Self-Organizing Maps (SOMs) (R3. pp.619), Cluster Evaluation Methods, Challenges and Major Methodologies in Clustering (R4, pp.423-430).

PRACTICES:

- Implement the CART classifier for a sample training data set stored as a.csv file. Compute the accuracy of the classifier, considering few test data sets.
- Assuming a set of spam or not-spam mails that need to be classified, use the Artificial Neural Network (ANN) classifier model to perform this task. Calculate the accuracy, precision, and recall for your data set.
- Implement Fuzzy-kNN.
- Implement Boosting algorithm to classify the iris data set. Print both correct and wrong predictions. Python ML library classes can be used for this problem.
- Demonstrate the working of the Extra Tree classifier. Use an appropriate data set for building the extra tree and apply this knowledge to classify a new sample?
- Implement and build models using the following Ensemble techniques.
  a. Bagging.
  b. Boosting
  c. Stacking.
- Build a model to perform Clustering using Fuzzy-C-means after applying PCA and determining the value of C using the Elbow method.
- Unsupervised Modeling:
  a. Cluster the data using the following models:
    i. Spectral Clustering.
    ii Self-Organizing Maps (SOMs).
  b. Compare and contrast model performance in each case.
- Implement any Recommendation system.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply a wide variety of learning algorithms such as Probabilistic, Discriminative and Generative algorithms for a given application.	Apply	1,2,3
2	Design an end-to-end Machine-learning model to realize solutions for real-world problems.	Create	1, 2, 3, 5
3	Implement various machine learning models using advanced ML tools.	Apply	2, 3, 4
4	Analyze and evaluate the performance of various machine learning model approaches on different kinds of data.	Analyze & Evaluate	1, 2, 4
5	Develop expertise in ensemble methods and their application in improving predictive performance.	Apply	2, 3, 4

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOKS:

1. "Pattern Recognition and Machine Learning" – Christopher M. Bishop, Springer, 2006.
2. "Machine Learning" – Tom M. Mitchell, McGraw-Hill Education, 1997.
3. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron (2nd Edition, 2019).

REFERENCE BOOKS:

1. "The Elements of Statistical Learning"by Trevor Hastie, Robert Tibshirani, and Jerome Friedman, Springer, 2nd Edition, 2009.
2. "Introduction to Statistical Learning" (ISLR) by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, Springer, 2nd Edition, 2021.
3. "Data classification algorithms and applications" by Charu C.Agarwal, CRC Press,2015.
4. "Introduction to Data Mining" by Pang-Ning Tan, Michael Steinbach, and Vipin Kumar (2nd Edition, 2018)
5. "A tutorial on spectral clustering" by von Luxburg, U, Springer, Stat Comput 17, 395–416 (2007).

25CS303 CRYPTOGRAPHY AND NETWORK SECURITY



Source: https://www.brainkart.com/subject/CRYPTOGRAPHY-AND-NETWORK-SECURITY-PRINCIPLES-AND-PRACTICE_136/

Hours Per Week :

L	T	P	SL	C
2	0	2	2	3

PREREQUISITE KNOWLEDGE: Computer networks.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on the modern concepts of network security using various cryptographic algorithms and underlying network security applications. It enables to understand various symmetric and asymmetric cryptographic techniques. It focuses on providing security services in applications such as e-mail functioning, web security and secure electronic transactions protocol and system security.

MODULE - 1

UNIT-1 **6L+0T+6P+6SL=18 Hours**

INTRODUCTION

Introduction to Computer and Network Security Concepts: Computer Security Concepts, Security attacks, Security services, Security mechanisms, A model for network security.

Classical Encryption Techniques: Symmetric cipher model, Substitution techniques, Transposition techniques.

UNIT-2 **6L+0T+6P+6SL=18 Hours**

SYMMETRIC CRYPTOGRAPHY

Symmetric Ciphers: Block cipher principles, Data encryption standard, Strength of DES, AES cipher, Multiple encryption and Triple DES, Block cipher design principles, RC4.

PRACTICES:

- Implement Substitution and Transposition Ciphers
  - Ceaser cipher
  - Play fair cipher
  - Hill cipher
  - Rail fence cipher
- Implement Symmetric Cipher
  - S-DES
  - RC4
- • Implement Asymmetric Cipher
  - RSA
  - Diffie-Hellman

MODULE-2

UNIT–1 **6L+0T+6P+6SL=18 Hours**

ASYMMETRIC CRYPTOGRAPHY

Asymmetric Ciphers and Cryptographic Hash Functions: Principles of public key cryptosystems, RSA algorithm, Diffie-Hellman Key Exchange, Message Authentication requirements, Authentication functions, Message authentication Codes, Hash functions, Digital signature standard.

SKILLS:

- ✓ *Design various security services for appropriate applications.*
- ✓ *Identifying the appropriate firewall, password management and anti-virus models for specific applications.*
- ✓ *Test and resolve threats and malfunctions in network.*
- ✓ *Apply different security mechanisms for web applications.*
- ✓ *Build authentication system for security protocols.*

UNIT–2 **6L+0T+6P+6SL=18 Hours**

SECURITY APPLICATIONS

Network Security Applications: Kerberos, X.509 authentication service.

E-Mail Security: Pretty good privacy, S/MIME.

IP Security Overview: IP security architecture, Authentication header, encapsulating security payload, Combining security associations, Internet Key Exchange(IKE).

UNIT–3 **6L+0T+6P+6SL=18 Hours**

WEB AND SYSTEM SECURITY

Web Security: Secure socket layer and transport layer security, HTTPS, Secure Shell (SSH)

System Security: Intruders, Intrusion detection, Malicious software, Firewalls

Security Practices: Block Chains, Cloud Security and IOT Security

PRACTICES:

- Configure IP Address in a system in LAN (TCP/IP Configuration)
- Configure DNS to establish interconnection between systems
- Demonstrate Study Tool to Configure Firewall and VPN.
- Adding users, setting permissions
- Demonstrate Intrusion detection using any Tool.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply cryptographic techniques in various security service solutions effectively in everyday professional and social contexts.	Apply	1,2, IKS
2	Analyze the usage of secure protocols to safeguard sensitive data using internet.	Analyze	1,2,6, IKS
3	Usage of tools to Identify abnormal activity in network communication to take appropriate action.	Apply	1,3,5, IKS
4	Apply various security protocols to safe guard the data internet using SSL/TCL.	Apply	1,2,6, IKS
5	Evaluate advanced security practices in real-world applications.	Evaluate	1,2,4,5, IKS

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOKS:

1. William Stallings, “Cryptography and Network security”, 7th Edition, Pearson Education, 2017.

REFERENCEBOOKS:

1. 1. William Stallings “Network Security Essentials Applications and Standards”, 2nd Edition, Pearson Education, 2009.
2. Eric Malwald, “Fundamentals of Network Security”, 4th Edition, Pearson Education, 2010.
3. Buchmann, “Introduction to Cryptography”, 2nd Edition, Pearson Education, 2009.
4. Charlie Kaufman, “Radis Perlman and Mike Speciner, Network Security - Private Communication in a Public World”, 1st Edition, Pearson Education, 2009.
5. Network Security, Firewalls And VPNs, J. Michael Stewart, Jones & Bartlett Learning, 2013.



Source: https://artoftesting.com/software-engineering

25CS304 SOFTWARE ENGINEERING

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Data base management systems, OOPs through Java.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on the concepts of software life cycle, role of process models and methods to prepare software requirement specification document. In addition to that, it also imparts knowledge of design, development and testing of software. The objective of this course is to enable the student to develop efficient, cost effective, feasible software as per user requirements.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

INTRODUCTION

Introduction to Software Engineering: Introduction to Software and Software engineering, Software characteristics, Software project, Software myths, Project Planning, Scheduling and Management.

Generic View of Process: Software Engineering - A layered technology, A process framework, Software Development Life Cycle (SDLC), The Capability Maturity Model Integration (CMMI).

Process Models: Conventional Model, Agile process models - Unified process model, Extreme Programming, Scrum.

UNIT-2 **6L+6T+6P+6SL=24 Hours**

REQUIREMENTS ENGINEERING

Requirements Engineering: Functional and Non-functional requirements, User requirements, System requirements, Requirement engineering tasks, formal requirements specification and verification, Feasibility Study.

Building the Analysis Model: Data modeling - Data objects, Attributes, Relationships, Cardinality and modality. Class based modeling - Identify analysis classes, specify attributes and Define operations.

Design Engineering: Design model, Design concepts. Creating an Architectural Design-Architectural styles and patterns.

Performing User Interface Design: Golden rules; User interface analysis and design.

PRACTICES:

Laboratory session of this course is designed in such a way that the student should complete three projects of the given type by performing the below experiments.

- Development of software requirements specification using Mind-Map tool.
- Project planning using Gantt charts.
- Project estimation using metrics.
- Capture Use Case Scenarios and model UML Use Case Diagrams.
- Model the UML state chart and Activity diagrams.
- Model the UML Class and Sequence diagrams.

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

TESTING

Testing Strategies: A strategic approach to software testing, Unit testing, Integration testing, Validation testing, System testing.

Testing Tactics: Black-Box and White-Box testing techniques, Art of debugging. Product

Metrics: Metrics for analysis model; Metrics for design model, Metrics for source code; Metrics for testing; Metrics for maintenance.

UNIT-2 **6L+6T+6P+6SL=24 Hours**

RISK AND QUALITY MANAGEMENT

Risk Management: Software risks, Risk identification; Risk projection; Risk refinement, Reactive vs Proactive risk strategies, RMMM.

Quality Management: Quality concepts, Formal technical reviews, Statistical Software Quality Assurance.

Computer-Aided Software Engineering (CASE): Use of appropriate CASE tools- Requirement engineering tools, Project planning tools, Testing tools.

UNIT-3 **6L+6T+6P+6SL=24 Hours**

SOFTWARE MAINTENANCE AND TRENDS

Project Scheduling, Software maintenance, Characteristics of Software Maintenance, Software Reverse Engineering, Restructuring, Forward Engineering, Software Process Improvement - Approaches to SPI.

Ethics in Software Engineering: Introduction to ethics, Codes of ethics, Ethics in cybersecurity decisions, Ethical concerns in design decisions, Ethical impact of software on different communities.

PRACTICES:

Laboratory session of this course is designed in such a way that the student should complete three

- Projects of the given type by performing the below experiments.
- Estimate the test coverage and Structural complexity of product using metrics.
- Develop the test cases for all the functional requirements of projects selected.
- Perform the functional testing using Selenium tool.

LIST OF PROJECTS:

Project-1: A Point-Of-Sale (PoS) System: A POS system is a computerized application used to record sales and handle payments; it is typically used in a retail store, it includes hardware components such as a computer and bar code scanner, and software to run the system. It interfaces to various service applications, such as a third-party tax calculator and inventory control. These systems must be relatively fault tolerant; that is, even if remote services are temporarily unavailable they must still be of capturing sales and handling at least cash payments. A POS system must support multiple and varied client-side terminals and interfaces such as browser, PDAs, touch-screens.

Project-2: Online Bookshop Example: Following the model of amazon.com or bn.com, design and implement an online bookstore.

Project-3: A Simulated Company: Simulate a small manufacturing company. The resulting application will enable the user to take out a loan, purchase a machine, and over a series of monthly production runs, follow the performance of their company.

Project-4: A Multi-Threaded Airport Simulation: Simulate the operations in an airport. Your application should support multiple aircrafts using several runways and gates avoiding collisions/ conflicts. Landing: an aircraft uses the runway, lands, and then taxis over to the terminal. Take-Off : an aircraft taxies to the runway and then takes off.

Project-5: An Automated Community Portal: Business in the 21st Century is above all BUSY. Distractions are everywhere. The current crop of “enterprise intranet portals” is often high noise and low value, despite the large capital expenditures it takes to stand them up. Email takes up 30 - 70% of an employee’s time. Chat and Instant Messaging are either in the enterprise or just around the corner.

SKILLS:

- ✓ *Analyze Intelligent systems.*
- ✓ *Apply problem solving techniques.*
- ✓ *Interface various knowledge representation.*
- ✓ *Create a dynamic planning.*

Meanwhile, management is tasked with unforeseen and unfunded leadership and change-agent roles as well as leadership development and succession management. What is needed is a simplified, repeatable process that enhances communications within an enterprise, while allowing management and peers to self-select future leaders and easily recognize high performance team members in a dynamic way. Additionally, the system should function as a general-purpose content management, business intelligence and peer-review application. Glasscode's goal is to build that system.

Project-6: Content Management System: The goal is to enable non-technical end users to easily publish, access, and share information over the web, while giving administrators and managers complete control over the presentation, style, security, and permissions. Features: Robust Permissions System, Templates for easy custom site designs, Total control over the content, Search engine friendly URL's, Role based publishing system, Versioning control, Visitor profiling.

Project-7: An Auction Application: Several commerce models exist and are the basis for several companies like eBay.com, pricellne.com etc. Design and implement an auction application that provides auctioning services. It should clearly model the various auctioneers, the bidding process, auctioning etc.

Project-8: A Notes and File Management System: During one's student years and professional career one produces a 1 lot of personal notes, documents. All these documents are usually kept 1 on papers or individual fi les on the computer. Either way the bulk of the information is often erased corrupted and eventually lost. The goal of this project is to build a distributed software application that addresses this problem. The system will provide an interface to create, organize and manage personal notes through the Internet for multiple users. The system will also allow users to collaborate by assigning permissions for multiple users to view and edit notes.

Project-9: Library Management System (LMS): The goal is to enable students and librarians to easily access and manage the library and run it smoothly. Each physical library item - book, tape cassette, CD, DVD, etc. could have its own item number. To support it, the items may be barcoded. The purpose of barcoding is to provide a unique and scannable identifier that links the barcoded physical item to the electronic record in the catalog. Barcode must be physically attached to the item, and barcode number is entered into the corresponding field in the electronic item record. Barcodes on library items could be replaced by RFID tags. The RFID tag can contain item's identifier, title, material type, etc. It is read by an RFID reader, without the need to open a book cover or CD/DVD case to scan it with barcode reader.

Project-10: Hospital Management System: Simulate to show and explain hospital structure, staff, and relationships with patients, and patient treatment terminology.

Project-11: Draft Software Requirement Analysis for the following Problem Statement: Fuel Delivery System: An unattended petrol (gas) pump system that includes a credit card reader. The customer swipes the card through the reader and then specifies the amount of fuel required. The fuel is delivered, and the customer's account debited.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Use basic concepts of software engineering for designing software product.	Apply	1,2
2	Compare different process models and identify appropriate process model based on project requirements.	Evaluate	2, 4,
3	Build Software Requirement Specification (SRS) document for any software product.	Apply	3, 5
4	Design of solutions using UML diagrams like Use case, Sequence diagrams etc.	Apply	3, 4, 5
5	Create an appropriate architecture for a given project that meets all quality constraints.	Create	5,
6	Apply different testing techniques to ensure bug free software and metrics to measure the software size, complexity, and budget etc.	Apply	4, 5, 11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):

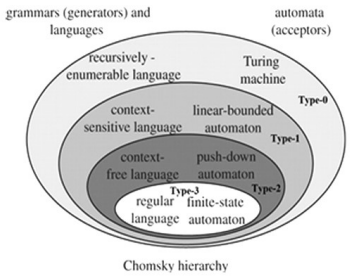


TEXT BOOK:

1. Roger S. Pressman, “Software Engineering, A practitioner’s Approach”, 6th Edition, McGrawHill International Edition, 2008.
2. Booch G., Rumbaugh J. and Jacobsons I, “The Unified Modeling Language User Guide”, 2nd Edition, Addison Wesley, 2005.

REFERENCE BOOKS:

1. Simon Sennet, Steve McRobb and Ray Farmer, “Object Oriented Systems Analysis and Design, 2nd edition, 2004.
2. Dr. Pankaj Jalote “Software Engineering: A Precise Approach”-edition 2010.
3. Ethics in Information Technology” – George W. Reynolds.
4. Software Engineering Ethics” – Lucy C. Suchman & Donald Gotterbarn (ACM/IEEE resources).



Source: https://sameer9247.wordpress.com/2016/11/15/theory-of-computation/

25CS305 THEORY OF COMPUTATION

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Knowledge of Set theory, graphs, trees and logic.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims to teach the student to identify different formal language classes and their relationships, strong theoretical foundation for designing compilers. In addition to this the student will be able to learn the techniques for information processing, design different grammars, automata and recognizers for different formal languages.

MODULE-1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

Introduction to Languages and Grammars: Alphabets - Strings - Operations on Languages, Automata and Grammars - Regular languages - Chomsky hierarchy of languages.

Finite Automata (FA): Deterministic Finite Automata (DFA) - How a DFA Process Strings - the language of DFA - Nondeterministic Finite Automata (NFA) - NFA with epsilon transition - Language of NFA- Equivalence of NFA and DFA - Minimization of DFA - Moore and Mealy machine - Equivalence of Moore and Mealy Machine - Applications and Limitation of FA.

UNIT-2 **6L+6T+0P+6SL=18 Hours**

Regular Expressions (RE): Definition - Operators of regular expression and their precedence - Algebraic laws for RE - RE to FA - FA to RE - Arden theorem - Pumping lemma for regular languages (RL) - Closure properties of RL.

Grammars: Classification of Grammars - Right and Left Linear Regular Grammars - Equivalence between RG and FA.

PRACTICES:

- Design DFA and NFA which accepts the following languages over the alphabet {0,1}, and also covert NFA to DFA. Give separate Automata for each and also write RE for the obtained automata.
  a) The set of all strings ends with 00.
  b) With three consecutive 0's.
  c) With 011 as a substring.
  d) Either begin or ends with 01.
  e) Strings whose fourth symbol from the right end is 1.
  f) Even number of 0's.
  g) number of 1 's are divisible by three.
- Design NFA to recognize the following set of strings.
  a) abc, abd, and aacd: Assume the alphabet is {a, b, c, d}.
  b) 0101,101 and 011: Assume the alphabet is {0,1}.
  c) ab, bc and ca: Assume the alphabet is {a, b, c}.

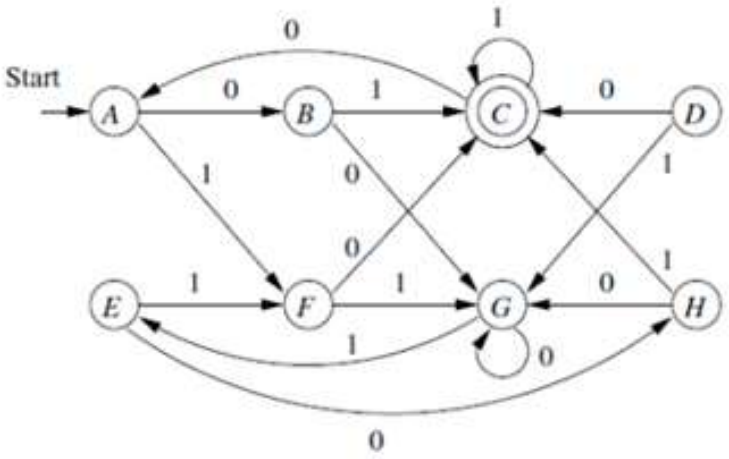
SKILLS:

- ✓ *Investigate syntax and semantics of a regular and context free languages.*
- ✓ *Develop the problem understanding solving ability.*
- ✓ *Design optimized solutions for a language.*

- Convert epsilon NFA to DFA.

	0	1	ε
→A	Φ	A	B
B	B	Φ	C
Ⓒ	B	A	Φ

- Minimize the following DFA



- Construct Mealy and Moore Machines and equivalent them for the residue (remainder) mod 3 of binary input.
- Construct Finite Automata for the following Regular Expressions.
  a) RE=ab(a+b)*
  b) RE=(a+ab)(ab+ab)*
- Prove that the following languages are not Regular.
  a) L={a^p | p is a prime number}
  b) L={b^n | n=i^2 and i>1}
  c) L={ WW^R | W is (a,b)*}
  d) L={$a^n b^{n+1}$ | n≥1}

MODULE-2

UNIT-1 **6L+6T+0P+6SL=18 hours**

Context Free Grammar (CFG): Context Free Grammars and Languages - Derivations - Parse trees - Ambiguity in grammar - Simplification of CFGs - Normal forms for CFGs: CNF and GNF - Closure properties of CFLs, Application of pumping lemma.

UNIT-2 **6L+6T+0P+6SL=18 Hours**

Push Down Automata (PDA): Definition of PDA - Language of PDA - Acceptance by final state - Acceptance by empty stack - Non-Deterministic PDA and Deterministic PDA - Equivalence of PDA and CFG - Two stack PDA.

UNIT-3 **6L+6T+0P+6SL=24 Hours**

Turing Machines (TM): Definition, Turing Machine as Acceptor and Transducer - Computable functions -Types of Turing Machines - Universal TM - Church Turing Thesis - Halting Problem of TMs - Universal Turing Machine - Recursive and Recursively Enumerable Languages - Undecidable problems - Post's

correspondence problem.

PRACTICES:

- Construct CFG for the following:
  a) L={$a^n b^n$ | n>1}
  b) L={WW^R | W is (a,b)*}
  c) L={a^p | p is a prime}
- Derive the strings 10001 using left most derivation and right most derivation and parse tree by using the following grammar. And show that grammar is ambiguous.
  S→ T000T
  T→ 0T|1T|∈
- Convert the following CFG to CNF.
  S→ ABC|Aa
  A→ a
  B→ d|∈
  C→ Aab|a
- Convert the following CFG to GNF.
  S→ AA | 0
  A→ SS | 1
- Prove that the following are not CFL.
  a) L={a^p | p is a prime number}
  b) L={b^n | n=i2 and i>1}
  c) L={ WW^R | W is (a,b)*}
  d) L={a^n b^{n+1} |n≥1}
- Convert the following language or PDA to CFG.
  a) L={$a^n b^n$ | n≥1} and
  b) δ(q,0,z)={(q,xz)}
  c) δ(q,0,x)={(q,xx)}
  d) δ(q,1,x)={(q,x)}
  e) δ(q,∈ , x)={(p,∈)}
  f) δ(p,1,x)={(p,xx)}
  g) δ(p,∈ , x)={(p, ∈)}
  h) δ(p, 1,z)={(p, ∈)}
- Construct PDA for the following Languages.
- a) L={$0^n 1^m$ | n≥m}
- b) L={$a^n b^n$ | n≥1}
- c) L= L={w | w∈ {a, b}*}
- d) L={w| n_a (w) > n_b (w)}
- e) L={$0^n 12^n$ | n>0}
- f) L=Where w^R is reverse of w
- g) L= Where w^R is reverse of w
- h) L={ww^R | w∈{a,b}*} Where w^R is reverse of w
- Construct PDA for the following Languages.
  a) L= {$a^n b^n c^n$ | n >1}
  b) L={$a^n b^m a^m b^n$ | n,m≥1}
- Construct Turing Machine for the following Languages.
  a) L={$a^n b^n$ | n ≥ 1}
  b) L={$0^{2n} 1^n$ | n >0 }

c) L={ww^r | w is (0+1)*}

d) L={$a^nb^nc^n$ | n≥1}.

e) Well balanced Parenthesis for example: () ()

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Explain the fundamental concepts of formal languages, automata, grammars, and their relationships in the Chomsky hierarchy.	Analyse	1,2,
2	Design finite automata (DFA, NFA, epsilon-NFA), regular expressions, and grammars for regular languages.	Apply	1, 2, 3
3	Analyze context-free grammars, perform derivations, and simplify grammars into normal forms.	Analyse	1, 2, 4
4	Construct pushdown automata and demonstrate equivalence with context-free grammars for various languages.	Apply	1, 2, 3
5	Evaluate computational capabilities of Turing machines and analyze decidability and undecidability of problems.	Evaluate	1, 2, 4, 5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):







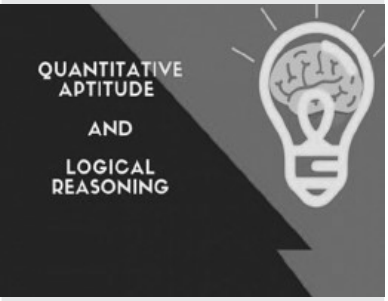


TEXT BOOKS:

1. Hopcroft and Ullman, “Introduction to Automata Theory, Languages and Computation”, 3rd Edition, Pearson/ Prentice Hall India, 2007.
2. Michael Sipser, “Introduction to Theory of Computation”, 3rd Edition, Course Technology, 2012.

REFERENCE BOOKS:

1. Zed A Shaw, Learn C the Hard Way: Practical Exercises on the Computational Subjects You Keep Avoiding (Like C), Addison Wesley, 2015.
2. Christoph Dürr, Sorbonne University, Jill-Jênn Vie, Inria, Competitive programming in Python, Cambridge University Press, 2020.
3. Kandar, Shyamalendu. Introduction to automata theory, formal languages and computation. Pearson Education India, 2013.



Source: https://images.app.goo.gl/kvtVgA8TkvDCqLhj7

25TP301 QUANTITATIVE APTITUDE AND LOGICAL REASONING

Hours Per Week :

L	T	P	SL	C
0	2	0	0	1

PREREQUISITE KNOWLEDGE: Basic Logical Thinking and Problem-Solving Ability.

COURSE DESCRIPTION AND OBJECTIVES:

The "Quantitative Aptitude & Logical Reasoning" course develops students' problem-solving, analytical, and logical thinking skills through topics like arithmetic, algebra, data interpretation, and puzzles. It prepares learners for aptitude tests and decision-making scenarios, enhancing their speed, accuracy, and confidence to excel in academics, placements, and real-world challenges.

MODULE-1

UNIT-1 **0L+12T+0P+0SL=12 Hours**

QUANTITATIVE APTITUDE-1

Number system, LCM & HCF of numbers, Percentage, Ratio and proportion and Partnership.

UNIT-2 **0L+12T+0P+0SL=12 Hours**

QUANTITATIVE APTITUDE-2

Profit, loss and discount, Simple Interest & Compound interest, Data Interpretation.

PRACTICES:

- Each concept would be taught in detail in the class.
- Followed by 10 problems solved in the class.
- Timer based questions solving during class room practice.
- Students would have to solve 10 additional problems as homework assignment in each concept.

MODULE-2

UNIT-1 **0L+18T+0P+0SL=18 Hours**

QUANTITATIVE APTITUDE-1

Time and work, Time & distance, Permutation & Combinations, Probability.

UNIT-2 **0L+18T+0P+0SL=18 Hours**

LOGICAL REASONING-2

Number series & Letter series, Analogy, Coding & decoding, Syllogisms, Puzzle test.

UNIT-3 **0L+18T+0P+0SL=18 Hours**

LOGICAL REASONING-3

Blood relations, Direction sense test, Order & Ranking, Seating Arrangements, Calendar & Clocks.

PRACTICES:

- Each concept would be taught in detail in the class.
- Followed by 10 problems solved in the class.
- Timer based questions solving during class room practice.
- Students would have to solve 10 additional problems as homework assignment in each concept..

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Develop the ability to solve arithmetic and reasoning problems quickly and accurately, without relying on paper-based calculations.	Apply	1, 3, 5,
2	Strengthen data interpretation skills for effective analysis and decision-making.	Analyze	1, 4, 5
3	Address the evolving demands of the current job market while preparing individuals for higher studies, including exams such as the CAT, GMAT, etc..	Evaluate	2, 5, 10
4	Enhance analytical skills to evaluate logical statements, arguments, and patterns..	Evaluate	2, 6
5	Develop proficiency in solving reasoning problems like puzzles, analogies, syllogisms, and seating arrangements.	Apply	2, 6, 8,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





TEXT BOOKS:

1. R. S. Aggarwal- Quantitative Aptitude for Competitive Examinations- S. Chand Publications- Edition-2023.
2. ARIHANT- A New Approach to Verbal & Non-Verbal Reasoning- Arihant Publication- Revised Edition-2021.

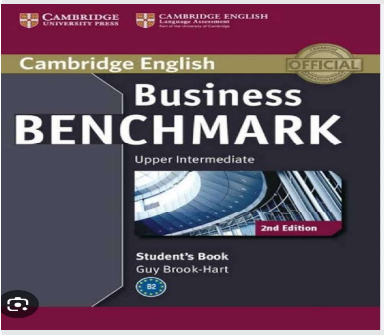
REFERENCE BOOKS:

1. Trishna Knowledge Systems- Quantitative Aptitude for Competitive Examinations- Pearson Publication- First Edition- 2013.
2. B.S. Sijwali & S. Sijwali- A New Approach to Reasoning- Arihant Publication- Edition- 2018.
3. R. S. Aggarwal- A Modern Approach to Verbal &Non-Verbal Reasoning-S. Chand Publication- Revised Edition- 2018.

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SKILLS:

- ✓ *Helps in developing and improving problem-solving skills.*
- ✓ *Flexing and honing logical abilities.*
- ✓ *Allow students to develop critical thinking skills.*



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25EN302 PROFESSIONAL COMMUNICATION SKILLS

Hours Per Week :

L	T	P	SL	C
0	0	2	0	1

PREREQUISITE KNOWLEDGE: Basic sentence formation, understanding contextual meanings, basic writing skills and moderate fluency in English.

COURSE DESCRIPTION AND OBJECTIVES:

To enhance language proficiency of the students in business contexts, focusing on listening, speaking, reading, and writing (LSRW) skills. Students will gain exposure to various forms of corporate communication to develop their speaking and writing abilities. Further, the course aims to bolster comprehension skills through targeted reading and listening activities, ensuring students are well-equipped to navigate professional environments effectively. The students also certified by Cambridge English in Linguaskill test, ranging from B1 to C2 levels.

MODULE-1

UNIT-1 **0L+0T+6P+0SL=6 Hours**

STAFF DEVELOPMENT AND TRAINING - JOB DESCRIPTION AND JOB SATISFACTION

Reading: skimming and scanning a text for effective and speedy reading and dealing with ideas from different sectors of corporate world in different business contexts.

Writing: understanding mechanics of writing and basics of report writing, letter writing and structures.

Listening: listening to develop comprehension skills, understand rhythm and accent and finding missing words or phrases in gapped sentences, Listening- short recordings.

Speaking: Using language expressions - giving opinion, suggestions. Ice breaking activities; discussing and sharing opinions, extended speaking and presenting on various business topics specially related to SDGs and IKS.

Svadharma (one's prescribed duty) is emphasized — aligning personal skills and nature (Swabhava) with the role in society/work.

Shiksha & Gurukul Parampara.

Karma Yoga and Trupti (Contentment).

Ancient texts emphasize balance of Artha (wealth), Dharma (duty), Kama (desires), and Moksha (liberation) → aligning with modern work-life balance and holistic job satisfaction.

The Purusharthas framework (Dharma, Artha, Kama, Moksha) gives a holistic model of employee motivation..

UNIT-2 **0L+0T+6P+0SL=6 Hours**

GETTING THE RIGHT JOB, MAKING CONTACT

Reading: Reading for specific details and understanding purpose of the text and author's point of view-matching with statements, reading business mails, understanding formal communication.

Writing: Understand business correspondence, tone structure and useful language phrases and expressions in written communication. (Integrates precision in written expression inspired by ancient Indian documentation methods like the Arthashastra and epistolary traditions)

Listening: Listening to various speakers to grasp tone and comprehend the text to understand the native speakers' accent.

Speaking: Learning language expressions and developing oral communication in business context-one-minute oral presentations and engaging in dialogues with a group and peers. Understanding business phone calls and tips and practicing phone conversation in a professional context.

Svadharma & Varna-Ashrama Dharma: Svadharma means performing work that aligns with one's nature (swabhava). This is like career alignment with skills, interests, and personality today.

Varna-Ashrama Dharma: Work roles in society were traditionally aligned with temperament, aptitude, and values. This is not about caste but about skill-based division of labour.

PRACTICES:

- Reading and Listening comprehension tests.
- Multiple Text Matching.
- Group Discussions topics can be selected from the point of view of SDGs and IKS, Sabha and Samvida (dialogue forums).
- One-minute presentation.
- Role-Play, practicing Business Vocabulary and grammar exercises.
- JAM sessions.
- Gapped and open gapped fill in short texts with multiple-choice cloze questions Matching statements with texts.
- Listening to comprehend long recordings, filling gaps.
- Business report writing structure, letter writing.
- Learning common expressions in business writing.
- Open gapped sentences and gapped sentences.
- Cohesive devices and linking words tasks in writing.

SKILLS:

- ✓ *Helps enhance problem-solving abilities.*
- ✓ *Improves skills in analyzing and interpreting data from graphs, charts, and tables.*
- ✓ *Encourages the development of critical thinking skills.*
- ✓ *Promotes systematic approaches to problem-solving and making sound decisions.*
- ✓ *Manage time efficiently while solving multiple questions in exams.*

MODULE-2

UNIT-1 **0L+0T+6P+0SL = 6 Hours**

BREAKING INTO THE MARKET - LAUNCHING A PRODUCT

Reading: Finding missing sentences in a long text, report reading and correcting errors, reading and finding answers for open gapped tasks, and gapped sentences.

Writing: writing business reports and proposals, mails, letter writing, modal verbs, Passive and active voices.

Listening: Listening long text on a topic, long monologue to comprehend and answering multiple choice questions.

Speaking: Role play and group discussions to exchange information.

UNIT-2 **0L+0T+6P+0SL = 6 Hours**

STAND AT A TRADE FAIR - BEING PERSUASIVE

Reading: Reading long texts and answering multiple choice questions focusing on comprehension skills, filling missing paragraphs in a long text.

Writing: Understanding structure of memo, note and developing memo and note in a professional context and write and improve online tasks.

Listening: Listening to conversations and finding missing word or phrase, listening to understand speakers' intention.

Speaking: Presenting business ideas, making formal requests, art of agreeing and disagreeing to develop oral skills through negotiation tasks.

Vak Shakti (Power of Speech) Ancient Indian texts emphasized effective, ethical, and convincing communication.

Rig Veda & Upanishads: Vak (speech) was considered divine energy — persuasion was about clarity,

truth, rhythm, and impact.

Natyashastra: Trained in rhetoric, gestures, and audience engagement, similar to persuasive communication in sales and trade fairs.

Arthashastra & Chanakya Neeti

Kautilya's Arthashastra gave strategies for trade, negotiation, pricing, and persuasion — advising rulers and merchants on how to win trust and influence.

It stressed three modes of persuasion:

Sama (gentle persuasion/conciliation)

Dana (offering/incentives)

Bheda (using logic and differentiation)

Danda (force — less relevant in trade fairs, but shows complete framework).

UNIT-3 **0L+0T+6P+0SL=6 Hours**

STARTING A BUSINESS – FINANCING A START-UP

Reading: Reading long texts and answerable multiple-choice questions focusing on scanning and inferencing.

Writing: Understanding the structure, language and tone of an enquiry letter for business purpose.

Listening: Listening to conversations and finding missing word or phrase, listening to understand speakers' intention. Also listen a long conversation and answer multiple choice questions

Speaking: Role play and one-minute talk on a business idea.

PRACTICES:

- Role-Play, practicing Business Vocabulary and grammar exercises.
- One-minute presentations. (topics can be selected from the point of view of SDGs and IKS)
- Doing reviews topics can encourage values from Indian epics and classical texts promoting dharma (ethical living) and loka-sangraha (universal wellbeing)
- Error correction
- Listening to comprehend native speaker's accent
- One long text with multiple choice questions
- Five short themed monologues and finding answer
- Grammar- learning prepositions and vocabulary.
- Taking individual long turns and follow-up discussion
- Interview-Collaborative task
- Understanding structure of a report writing and developing a business report
- Finding missing sentences
- Grammar- conditional sentences, clauses and linking ideas
- Missing Paragraph Text Completion
- Multiple choice question and answers in reading

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Possess comprehensive skills in listening and reading business texts in formal context.	Analyze	7
2	Communicate effectively both in their academic as well as professional environment.	Apply	10
3	Clear grasp on the register of business language and mechanics of writing.	Analyze	8
4	Possess the ability to write business reports and proposals clearly and precisely to succeed in their future.	Create	11
5	Make effective presentations and participate in formal discussions.	Create	10,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOKS:

1. Guy Brook Hart (2014) Cambridge English Business Bench Mark: Upper Intermediate, Second Edition: CUP.

REFERENCE BOOKS:

1. CUP (2002) Cambridge: BEC VANTAGE Practice Papers, CUP.
2. Schnurr, S (2013) Exploring Professional Communication: Language in Action. London: Routledge.
3. Seely, J (2005) The Oxford Guide to Effective Writing and Speaking. Oxford: OUP
4. Webresource:https://www.cambridgeenglish.org/exams-and-tests/linguaskill/information-about-the-test/practice-materials/.

25CS306 PRIVACY PRESERVING AND INTRUSION DETECTION

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Cryptography and Network Security.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims to understand modern concepts related to Intrusion Detection System.The Course compares alternative tools and approaches for Intrusion Detection through quantitative analysis to determine the best tool or approach to reduce risk from intrusion students undergoing this course can identify and describe the parts of all intrusion detection systems and characterize new and emerging IDS technologies according to the basic capabilities all intrusion detection system share.

MODULE - 1

UNIT-1 **8L+8T+0P+8SL=24 Hours**

INTRODUCTION

Basics concepts of Security, Introduction to Intrusions, Need of Intrusion Detection, Classification of Intrusion Detection Systems, sources of Vulnerabilities, Attacks against various security objectives, countermeasures of attacks.

UNIT-2 **8L+8T+0P+8SL=24 Hours**

INTRUSION DETECTION AND PREVENTION TECHNOLOGIES

Host-based intrusion detection system(HIDS),Network-based IDS, Information sources for IDS, Host and Network Vulnerabilities and Countermeasures, Intrusion detection techniques, misuse detection, pattern matching, rule-based and state-based anomaly detection: statistical based. machine learning based, mining-based hybrid detection.

IDS and IPS Architecture - Tiered architectures,Single-tiered,Multi-tiered,Peer-Peer.Sensor:sensor functions, sensor deployment and security.

PRACTICES:

- Installing Snort into the Operating System
- Configuring and Starting the Snort IDS
- Writing and Adding a Snort Rule
- Triggering an Alert for the New Rule.

MODULE - 2

UNIT-1 **8L+8T+0P+8SL=24 Hours**

AGENTS

Agent functions. agent deployment and security. Manager component: manager functions, manager deployment and security. Information flow in IDS and IPS, defending IDS/IPs, Case study on commercial and open-source IDS.

UNIT–2 **8L+8T+0P+8SL=24 Hours**

ALERT MANAGEMENT

Alert correlation, Pre-process, Correlation Techniques, Post-process, Alert Correlation architecture.

UNIT–3 **8L+8T+0P+8SL=24 Hours**

CORRELATION DATA FUSION

Cooperative Intrusion Detection, Cooperative Discovery of Intrusion chain, Abstraction-based Intrusion

Detection, Internet based communication and cooperation, agent based cooperation.

PRACTICES:

- Demo of Eavesdropping attack and its Prevnetion using SSH.
- Demonstrate how ro provide secure data storage,secure data transmission and for creating digital signatures(GnuPG).
- Setup a honey pot and monitor the honeypot on network(KF sensor)
- Installation of rootkits and study about the variety of option.
- Perform wireless audit on an access point or a router and decrypt WEP and WPA.(NetStumbler)
- Install any open-source IDS and study the logs.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Analyze appropriate situations and scenarios where intrusion detection may be applied to achieve an increased level of specific intrusion detection Systems.	Analyze	1, 2,
2	Design and implement basic IDS using signature-based or anomaly-based approaches.	Create	1, 2,
3	Apply Intrusion Detection tools and techniques inorder to improve their security posture.	Apply, Analyze	1, 2, 5,
4	Evaluate the Security of an organization for better performances.	Evaluate	1, 2,10,
5	Apply knowledge to the architecture,configuration and analysis of specific Intrusion Detection Systems.	Create	1, 2, 11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



TEXT BOOKS:

1. C.Endrof,E.Schultz and J.Mellander,Intrusion Detection & Prevention,McGraw-Hill/Osborne 2006.

REFERENCE BOOKS:

1. Ali A.Ghorbani,Network intrusion detection and prevention concepts and techniques,springer,2010.
2. J.M.Kizza,Computer Network Security,Springer,2005.
3. Chris sanders and Jason Smith,Applied Network Security Monitoring:Collectiom and Analysis,Syngress,2013.

25CS307 PARALLEL AND DISTRIBUTED COMPUTING

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Fundamental of programming and Operating Systems, Computer Networking, and Computer Organization and Architecture.

COURSE DESCRIPTION AND OBJECTIVES:

This course covers the challenges faced in constructing parallel and distributed applications, various implementation techniques, paradigms, architectures and parallel algorithms. Current trends in parallel and distributed computing like Open MP, POSIX Threads, Apache Hadoop (DFS).

MODULE - 1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

INTRODUCTION TO PARALLEL COMPUTING

The idea of Parallelism (T1-Pp:1), Power and potential of parallelism (T1-Pp:1), examining sequential and parallel programs (T1-Pp:1), Scope and issues of parallel and distributed computing (T1-Pp:2-6), Goals of parallelism (T1-Pp:6-8)

PARALLEL ARCHITECTURE: Pipeline architecture, Array processor, Multi-processor architecture, Systolic architecture, Dataflow architecture. Architectural classification schemes, Flynn's, Shores and Feng's classification schemes, Vector processing, GPU co-processing, Multiprocessor caches and Cache Coherence (T1-Pp:45), Non-Uniform Memory Access (NUMA).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

PARALLEL ALGORITHM DESIGN PRINCIPLES AND PROGRAMMING

Need for communication and coordination/synchronization (T1-Pp:27), Scheduling and contention (T1-Pp:31). Preliminaries on decomposition and its types (T1-Pp:86), task dependency graphs, Granularity, concurrency and task interaction (T1-Pp:86-94). Parallel algorithmic patterns like divide and conquer, Map and Reduce, Specific algorithms like parallel Merge Sort (T1-Pp:380-411).

PRACTICES:

- Identify Multiple Instruction Single Data, or MISD. How would an MISD system work? Give an example.
- Suppose a shared-memory system uses snooping cache coherence and write-back caches. Also suppose that core 0 has the variable x in its cache, and it executes the assignment x = 5. Finally suppose that core 1 doesn't have x in its cache, and after core 0's update to x, core 1 tries to execute y = x. What value will be assigned to y? Why?
- Consider a simplified version of bucket-sort. You are given an array A of n random integers in the range [1...r] as input. The output data consist of r buckets, such that at the end of the algorithm, Bucket i contains indices of all the elements in A that are equal to i.
- Describe a decomposition based on partitioning the input data (i.e., the array A) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.
- Describe a decomposition based on partitioning the output data (i.e., the set of r buckets) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.
- Consider seven tasks with running times of 1, 2, 3, 4, 5, 5, and 10 units, respectively. Assuming that it does not take any time to assign work to a process, compute the best- and worst-case speedup for a centralized scheme for dynamic mapping with two processes.

MODULE - 2

UNIT-1 **6L+6T+0P+6SL=18 Hours**

INTRODUCTION TO DISTRIBUTED SYSTEMS

Definition (T2-Pp:1), types and goals of the distributed systems, design issues, distributed system models (T2-Pp:7-40), synchronous vs asynchronous execution (T2-Pp:38), hardware and software concepts related to distributed systems, middleware models (T2-Pp:71-75).

UNIT–2 **6L+6T+0P+6SL=18 Hours**

DISTRIBUTED COMPUTING AND COMMUNICATION DESIGN PRINCIPLES

A Model of distributed executions, Models of communication networks, Global state of distributed system, Models of process communication. Communication and Coordination: Shared Memory, Consistency, Atomicity, Message Passing, Consensus, Conditional Actions, Critical Paths, Scalability, and cache coherence in multiprocessor systems, synchronization mechanism.

UNIT–3 **6L+6T+0P+6SL=18 Hours**

PARALLEL AND DISTRIBUTED PROGRAMMING FRAMEWORKS

Overview of CUDA, Open MP (T1-Pp:311-331), POSIX Threads (T1-Pp:280-294), Apache Hadoop(DFS), current trends in parallel and distributed computing.

PRACTICES:

- Give five types of hardware resource and five types of data or software resource that can usefully be shared. Give examples of their sharing as it occurs in practice in distributed systems.
- The host computers used in peer-to-peer systems are often simply desktop computers in users' offices or homes. What are the implications of this for the availability and security of any shared data objects that they hold and to what extent can any weaknesses be overcome through the use of replication?
- Consider two communication services for use in asynchronous distributed systems. In service A, messages may be lost, duplicated or delayed and checksums apply only to headers. In service B, messages may be lost, delayed or delivered too fast for the recipient to handle them, but those that are delivered arrive with the correct contents. Describe the classes of failure exhibited by each service. Classify their failures according to their effects on the properties of validity and integrity. Can service B be described as a reliable communication service?
- Illustrate distributed design through a substantial case study, examining in detail the design of the Google infrastructure, a platform and associated middleware that supports both Google search and a set of associated web services and applications including Google Apps.
- Implementation of the parallel algorithms (on a PC-cluster under Linux platform). The programs will be based on POSIX Thread, MPI programming, Hadoop, Apache Spark etc.

SKILLS:

✓ .

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply the fundamentals of parallel and parallel computing including architectures and paradigms.	Apply	1, 2, 4, 5
2	Analyze the various design principles of parallel algorithms.	Analyze	1, 2, 4, 5, 9,
3	Learn the intricacies of distributed programming.	Analyze	1, 2, 3, 6, 8,
4	Analyze and execute basic parallel and distributed applications using basic programming models and tools.	Analyze	1, 2, 6 9
5	Design and implement distributed algorithms and communication protocols to solve real-world problems.	Evaluate	1, 2, 4, 6, 10,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





TEXT BOOKS:

1. Ananth Grama, Anshul Gupta, and George Karypis, Vipin Kumar; Introduction to Parallel Computing; Addition Wesley, 2nd Edition;2003
2. Andrew S. Tanenbaum; Distributed Operating Systems, Pearson Publications, 3rd edition;2017.

REFERENCE BOOKS:

1. Introduction To Parallel Processing, M.Sasikumar, Dinesh Shikhare and P. Ravi Prakash, Randy Chow, T. Johnson, Distributed Operating Systems and Algorithms, Addison Wesley.
2. Distributed Systems Principles and Paradigms Andrew S. Tanenbaum Maarten Van Steen, 3rd Edition, 2017.
3. https://hpc.llnl.gov/training/tutorials/introduction-parallel-computing-tutorial.
4. https://www.geeksforgeeks.org/introduction-to-parallel-computing/.

IV

YEAR

COMPUTER SCIENCE AND ENGINEERING

B.Tech.

I SEMESTER

▶	25CS401	-	Computing Ethics
▶		-	Department Elective - 5
▶		-	Department Elective - 6
▶	25CS802	-	Work-in-Lieu of Course – Research paper publications/patents/presentations/ global certifications

II SEMESTER

▶	25CS403 / 25CS402	-	Project Work / Internship

COURSE CONTENTS

I SEM & II SEM

25CS401 COMPUTING ETHICS



Source: https://www.brainkart.com/subject/CRYPTOGRAPHY-AND-NETWORK-SECURITY-PRINCIPLES-AND-PRACTICE_136/

Hours Per Week :

L	T	P	SL	C
0	2	2	0	2

PREREQUISITE KNOWLEDGE: General awareness on ethical guidelines.

COURSE DESCRIPTION AND OBJECTIVES:

The course will provide students with an understanding on Engineering Ethics and the nature of moral issues and dilemmas faced by Computer Science and engineers in their professional lives. It will give them an awareness on professional rights and responsibilities of an engineer and acquaint them on the Code of Conduct and Ethics prescribed by professional bodies like IEEE, ACM etc. for its members.

MODULE-1

UNIT-1 **0L+6T+6P+0SL=12 Hours**

INTRODUCTION

Introduction: Why Computer Ethics(T1-Pp:1-12)? The Sociotechnical Systems Perspective, Sociotechnical Computer Ethics, Micro- and Macro-Level Analysis(T1-Pp:13-51).

UNIT-2 **0L+6T+6P+0SL=12 Hours**

Ethics and Information Technology-Privacy, Security, Bias, Accountability, Digital Divide.

Introduction: Doing Ethics, Ethical Theories and Concepts. (T1-Pp:52-92).

PRACTICES:

- Surprises about Social Networking
- RFID and caring for the Elderly People
- Turing does not need to know• Turnitin Dot Com
- Email Privacy and Advertising
- Workplace Spying
- Datamining and e-Business

MODULE-2

UNIT-1 **0L+6T+6P+0SL=12 Hours**

ETHICS IN IT CONFIGURED SOCIETIES

Technology as the Instrumentation of Human Action, Three Features of IT-Configured Activities, IT Configured Domains of Life, Democracy and the Internet(T1-Pp:93-134). Information Flow, Privacy and Surveillance (T1-Pp:135-151)

Introduction: Information Flow with and without Information Technology(T1-Pp:152-174.)

UNIT-2 **0L+6T+6P+0SL=12 Hours**

DIGITAL INTELLECTUAL PROPERTY

The complexities of Digital Property, Protecting Property rights in Software, Free and Open Source Software, PS Versus FOSS (T1-Pp:175-216).

Introduction: Law and Order in the Internet, Sociotechnical Order, Online Crime, Hackers and Hacker Ethic, Sociotechnical Security. (T1-Pp:217-253)

SKILLS:

- ✓ *Understand ethical principles and professional codes in computing.*
- ✓ *Analyze and resolve ethical dilemmas in IT and software development.*
- ✓ *Assess ethical challenges in AI, IoT, Blockchain, and Quantum Computing.*
- ✓ *Identify legal and social issues related to digital privacy, security, and intellectual property*

UNIT-3 **0L+6T+6P+0SL=12 Hours**

PROFESSIONAL ETHICS

Introduction: Why Professional Ethics, The Paradigm of Professions, Characteristics of Professions, Sorting out Computing and its Status as a Profession, Codes of Ethics, Professional Relationships, A Legal perspective on professionalism in Computing. (T1-Pp:254-301)

ETHICAL ISSUES IN EMERGING TECHNOLOGIES

Ethical concerns in AI(T2-Pp:1-35), IoT, Blockchain(T2-Pp:36-113), and Quantum Computing(T2-Pp:124-139).; issues of bias, privacy, and data security. Legal, social, and cultural implications of emerging technologies and their responsible application.

PRACTICES:

- Obtaining pirated Software Abroad
- Free Software that follows Proprietary Software
- Using Public Domain Software in Proprietary Software
- Wiki Warfare
- Yahoo and Nazi Memorabilia
- Bot Roast
- Software Safety
- Conflicts of Interest
- Security in a Custom Database

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	The importance of Computer Ethics and apply sociotechnical systems perspective to analyze ethical issues.	Apply	8
2	Identify and evaluate ethical concerns related to Privacy, Security, Bias, Accountability, and the Digital Divide in IT	Evaluate	8
3	Analyze ethical dimensions of IT-configured societies focusing on Information Flow, Privacy, and Surveillance.	Analyze	8
4	Demonstrate understanding of Digital Intellectual Property, Online Crime, and Hacker Ethics in the context of law and order.	Apply	8
5	Evaluate Professional Ethics and emerging ethical issues in AI, IoT, Blockchain, and Quantum Computing	Evaluate	7, 8

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





TEXT BOOK:

1. Deborah G. Johnson, Keith W. Miller and Prentice Hall, “Computer Ethics”, 4th Edition, Pearson, 2009.
2. Introduction to Emerging Technologies” Prepared by the Ministry of Science and Higher Education (MOSHE) in collaboration with Bahir Dar University, Adama Science & Technology University, and Addis Ababa Science & Technology University, 2019.

REFERENCE BOOKS:

1. Charles E. Harris, Michael S. Pritchard and Michael J. Rabins, “Engineering Ethics: Concepts and Cases”, 4th edition, Wadsworth Thompson Learning, 2009.
2. M. Govindarajan, S.Natarajan and V. S. Senthil Kumar, “Engineering Ethics”, Prentice Hall of India, Reprint 2013.

DEPT. ELECTIVES

B.Tech.

COMPUTER SCIENCE AND ENGINEERING

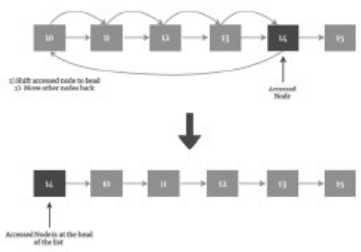
DEPARTMENT ELECTIVES

- 25CS803 - Advanced Data Structures
- 25CS804 - Advanced JAVA Programming
- 25CS805 - Digital Image Processing
- 25CS806 - Computer Graphics
- 25CS807 - Mobile-Adhoc Networks
- 25CS808 - Text Mining
- 25CS809 - Operating System Design
- 25CS810 - Cloud Computing
- 25CS811 - Deep Learning
- 25CS812 - Cloud Security
- 25CS813 - Software Project Management
- 25CS814 - Compiler Design
- 25CS815 - Numerical Algorithms
- 25CS816 - Simulation and Modelling

COURSE CONTENTS

I SEM & II SEM

25CS803 ADVANCED DATA STRUCTURES



Source: https://www.baeldung.com/cs/advanced-data-structures

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Data structures.

COURSE DESCRIPTION AND OBJECTIVES:

Advanced data structures are one of the most important disciplines of data science since they are used for storing, organizing, and managing data and information to make it more efficient, easier to access, and modify. The objective of this course is to improve students' skills in designing data structures and algorithms for various solutions, and problem design in large systems and applications such as databases, information retrieval systems, bioinformatics, and geographic information systems.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

HASHING

Introduction, Hash Function, Collision Resolution Techniques: Separate Chaining, Operations on Separate Chaining, Open Addressing: Linear Probing, Quadratic Probing, Double Hashing, Operations on Open Addressing, Rehashing. (T1-Pp:193-209).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

TREES

AVL Tree, Operations of AVL Tree(T1-Pp:144-157), Splay tree, Operations on Splay Tree (T1-Pp:551-565), Red Black Tree (RBT), Operations on RBT (T1-Pp:566-575), M-way search tree and operations (R1-Pp:423-425).

PRACTICES:

- Implement a hash table using separate chaining for collision resolution. Perform insert, search, and delete operations.
- Implement a hash table using linear probing for collision resolution. Perform insert, search, and delete operations.
- Implement a hash table using quadratic probing for collision resolution. Perform insert, search, and delete operations.
- Implement a hash table using double hashing for collision resolution. Perform insert, search, and delete operations.
- Implement the AVL Tree with operations like insert, delete, and balance. Demonstrate rotations during insertions and deletions.
- Implement a Splay Tree and demonstrate the splaying operation during insert, search, and delete.
- Implement a Red-Black Tree and perform insertions and deletions, ensuring the tree maintains its properties.
- Implement an M-way Search Tree and perform operations like insert, delete, and search. Demonstrate the handling of internal nodes with more than two children.

SKILLS:

- ✓ *Analyse the data structure required for various applications.*
- ✓ *Usage of trees, graphs, heaps and tries.*
- ✓ *Know various pattern searching algorithms.*

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

HEAPS AND PRIORITY QUEUES

Heaps: Introduction, types of heaps, Operations on heaps, heap implementation with an array(T1-Pp:245-252), heap sort technique to sort elements Mergeable heaps, Application of heap. (T1-Pp:300-303).

Priority Queues: Introduction, Types of priority queues, implementing max priority queue and min priority queue. (T1-Pp:257-259) & (R1-Pp:407-411).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

SORTING AND PATTERN SEARCHING

Sorting: Merge Sort, External Merge sort (T1-Pp:304-307, 336-338)

Pattern Searching: Pattern matching algorithms -Brute force algorithm, Knuth-Morris-Pratt algorithm (R3-Pp:35-42), Robin-Karp algorithm (R3-Pp:367-369).

UNIT-3 **6L+6T+6P+6SL=24 Hours**

GRAPHS AND TRIES

Graphs: Minimum Spanning Tree: Prim's Algorithm, Kruskal's Algorithm(T1-Pp:413-418)

Tries: Standard Tries, Operations on Standard Tries, Compressed Tries, Operations on Compressed Tries, Suffix tries. (R2-Pp:557-561).

PRACTICES:

- Implement Merge Sort to sort an array of integers.
- Implement External Merge Sort for handling large files that can't fit into memory.
- Implement the Brute Force algorithm for pattern matching in a text.
- Implement the Knuth-Morris-Pratt (KMP) pattern matching algorithm.
- Implement the Rabin-Karp algorithm for searching patterns in a text.
- Implement Prim's Algorithm to find the Minimum Spanning Tree (MST) of a graph.
- Implement Kruskal's Algorithm to find the Minimum Spanning Tree (MST) of a graph.
- Implement a Standard Trie with insert, search, and prefix operations.
- Implement a Compressed Trie and demonstrate the compression of nodes.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply hashing and collision resolution techniques to design efficient data retrieval systems.	Apply	1,2,3
2	Apply the concepts of advanced Trees for solving problems effectively.	Apply	1,2
3	Implement efficient pattern matching algorithms and apply heap concepts to build priority queues for optimized data processing.	Apply	1,2,3,5
4	Implement and analyze pattern matching and external sorting algorithms to solve large-scale data and text processing problems.	Analyse	1,2,3,4,5,
5	Analyse and Implement different types of tries, graphs and apply them to problem solutions.	Analyse	1,2,3,5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):

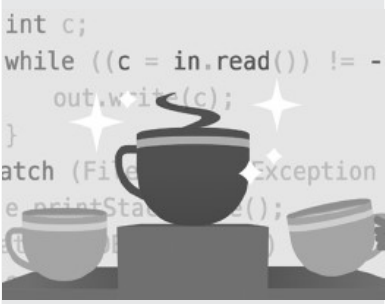


TEXT BOOKS:

1. Mark Allen Weiss, Data Structures and Algorithm Analysis in C++, 2nd Edition, Pearson, 2004.

REFERENCE BOOKS:

1. Data Structures: A Pseudocode Approach with C, 2nd Edition, R. F. Gilberg and B.A. Forouzan, Cengage Learning.
2. Fundamentals of Data Structures in C, 2nd Edition, E. Horowitz, S. Sahni and Susan Anderson Freed, Universities Press.
3. Text Algorithms, Efficient Algorithms on Texts by Maxime Crochemore and wojciech Rytter.



Source: https://www.kpl.gov/catalog/item/?i=ent://LYNDA/0/LYNDA:103608

25CS804 ADVANCED JAVA PROGRAMMING

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Java Programming.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a comprehensive introduction to advanced web application development using Java. It incorporates modern tools such as Git, Maven, Spring Frameworks, and Hibernate to build scalable enterprise-grade applications. The course emphasizes hands-on implementation and version control workflows, promoting real-world project readiness.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

Version Control with Git & Maven: Basics of Git: init, clone, commit, push, pull, branch, merge (T1 – Pp: 1–124). GitHub and collaborative workflows (T1 – Pp: 125–180).

Introduction to Maven: Project Object Model (POM), dependencies, lifecycle. Creating and managing Maven-based Java projects (T2 – Pp: 1–134).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

Java Web Frameworks: Introduction to Java Web frameworks(T4 – Pp: 1–22): What is Framework in Java, Advantages of Frameworks, popular Java Web Frameworks: Spring, Spring Boot, Hibernate, Java Server Faces, Google Web Toolkit etc. (T4 – Pp: 23–42, 97–110), Comparison of the web frameworks(T4 – Pp: 110–120).

ORM: What is Object Relational Mapping, How ORM Works, Features of ORM, Advantages, Java ORM- Hibernate, JAVA Persistence API (JPA), ORM implementation (T3 – Pp: 1–45).

PRACTICES:

- Create a local Git repository for a Java project, perform commits, and push it to a remote GitHub repository.
- Create feature branches, implement basic changes, and merge them using Git to simulate team collaboration.
- Set up a Maven project, add dependencies like MySQL connector and hibernate, and manage builds using lifecycle phases.
- Package a Java project into an executable JAR using Maven, and run the application.

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

Hibernate: Overview of Hibernate, Hibernate Architecture, Hibernate Mapping Types, Hibernate O/R Mapping, Hibernate Annotation, Hibernate Query Language (T3 – Pp: 46–160).

Struts: Struts framework, Struts features, Model 1vs Model 2 (MVC) Architecture, Core components of Struts, Configuration of Struts, Creating a Struts Application (R1 – Pp: 1–118).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

Spring: Overview of Spring, Spring Architecture, Inversion of Control (IOC) and Dependency Injection, XML Configuration on Spring, Creating a Spring Application. Spring MVC, flow of Spring Web MVC, Spring Web MVC Framework Example(T4 – Pp: 43–96, 121–189).

UNIT-3 **6L+6T+6P+6SL=24 Hours**

Spring: Overview of Spring, Spring Architecture, Inversion of Control (IOC) and Dependency Injection, Spring Boot: Overview of Spring Boot, Spring vs Spring MVC vs Spring Boot, SB architecture, Components of SB, Creating Spring Boot Application (R2 – Pp: 1–152).

PRACTICES:

- Develop a basic Hibernate application using XML mapping to connect and interact with a relational database.
- Re-implement the same application using Hibernate annotations for ORM configuration and object mapping.
- Build a web-based CRUD system using Hibernate and perform insert, update, delete, and view operations on entities.
- Design a login form using Struts with user validation and display appropriate messages on success or failure.
- Create and configure a full Struts application using struts-config.xml, ActionForm, and ActionServlet classes.
- Build a Spring MVC application to manage a student database with Create, Read, Update, and Delete operations.
- Build a Spring Boot application for a product catalog that uses embedded Tomcat.
- Use Spring Data JPA to implement database access methods without writing SQL, supporting full CRUD functionality.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply version control workflows and project build automation using Git and Maven to streamline Java enterprise application development.	Apply	1,5
2	Compare and evaluate popular Java web frameworks based on architectural style, features, and applicability in enterprise scenarios.	Analyze	1,2,6
3	Develop ORM-based applications using Hibernate and JPA for relational database interaction.	Apply	1,2,5,
4	Design secure and modular web applications using Struts and Spring MVC frameworks.	Create	1,2,3
5	Construct enterprise apps using Spring Boot and implement persistence with Spring Data JPA.	Create	1,2,3,5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





SKILLS:

- ✓ *To know the importance of java web frame-works to develop enterprise java applications.*
- ✓ *To experience developing a web application using MVC architecture.*
- ✓ *Design and develop CRUD applications using different java frameworks.*
- ✓ *To derive hands on experience of developing enterprise applications using advanced web frameworks like Spring and Spring Boot.*

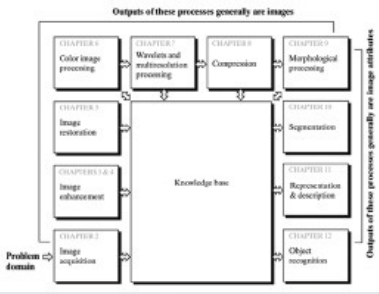
TEXT BOOKS:

1. Jon Loeliger & Matthew McCullough, Version Control with Git, 2nd Edition, O'Reilly Media, 2012.
2. Tim O'Brien, Maven: The Definitive Guide, O'Reilly Media, 2008.
3. Jeff Linwood and Dave Minter, Beginning Hibernate, 5th Edition, Apress, 2020.
4. Craig Walls, Spring in Action, 6th Edition, Manning Publications, 2022.

REFERENCES BOOKS:

1. Sharanam Shah, Vaishali Shah. "Structs2 for Beginners", 3rd Edition, SPD, 2014.
2. Iuliana Cosmina et al., Pro Spring Boot 3, Apress, 2023.

25CS805 DIGITAL IMAGE PROCESSING



Source: https://www.researchgate.net/figure/Fundamental-steps-in-digital-image-processing_fig10_333856607

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Probability and Statistics.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on imparting knowledge about the aspects of Image Processing and its applications. The main objective of the course is to learn digital image fundamentals, image transforms, image enhancement, restoration and compression, morphological image processing, representation and description.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

FUNDAMENTALS OF IMAGE PROCESSING

Fundamental steps in digital image processing (T1-pp-41-43), Components of image processing system (T1-pp-44-46), A simple image formation model, Image sampling and quantization (T1-pp-63-65), Basic relationships between pixels (T1-pp-79-82), types of digital images. Basic mathematical tools used in image processing (T1-pp-83-98), Vector and matrix operations (T1-pp-106-108), Image transforms (T1-pp-109), Introduction to Fourier Transform and DFT (T1-pp-225-229), FFT (T1-pp-303-308), Discrete Cosine Transform (DCT) (T1-pp-487-491), Discrete Wavelet transforms.

UNIT-2 **6L+6T+6P+6SL=24 Hours**

IMAGE ENHANCEMENT IN THE SPATIAL AND FREQUENCY DOMAINS

Basic gray - level transformations (T1-pp-122-130), Histogram processing (T1-pp-133-140), Basics of spatial filtering (T1-pp-153-154), smoothing spatial filters (T1-pp-164-165), sharpening spatial filters(T1-pp-175-189), Combining Spatial Enhancement methods (T1-pp-191-195), the basics of filtering in the frequency domain (T1-pp-260-266), Image smoothing in frequency domain filters (T1-pp-272-283), Image sharpening in frequency domain filters (T1-pp-284-292), selective filtering (T1-pp-296-301).

PRACTICES:

- Develop a module to enhance the image by using image arithmetic and logical operations. (T2-pp-337).
- Develop a module for an image enhancement using kernel operations (T2-pp-68-74).
- Develop a module for gray level slicing with and without background (T2-pp-68-74).
- Develop a module for image enhancement using histogram equalization (T2-pp-81-84)
- Develop a module to filter an image using low pass & high pass filter in spatial domain. Compare the performance of both filters (T2-pp-89-107)
- Develop a module for smooth an image using low pass & high pass filters in frequency domain. Compare the performance of both filters (T2-pp-115-120)

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

IIMAGE RESTORATION AND COMPRESSION

A model of image degradation/restoration, Noise models (T1-pp-318-327), inverse filtering (T1-pp-356-357), wiener filtering (T1-pp-358-362), Constrained Least Squares Filtering (T1-pp-363-366), Geometric Mean Filter (T1-pp-367)

SKILLS:

- ✓ *Apply knowledge of science and engineering principles to image related problems.*
- ✓ *Undertake image problem identification and formulate solutions.*
- ✓ *Implement algorithms for enhancement, restoration, compression etc.*

Image Compression: Fundamentals (T1-pp-540), Image data redundancies (T1-541-548), Image Compression Model (T1-pp-549-550), Error Free Compression, Lossy Compression, Huffman coding (T1-pp-553-556), Golomb coding (T1-pp-557-559), Arithmetic Coding (T1-pp-561-563), LZW coding (T1-pp-564-566), Run-length coding (T1-pp-566-568), Bit Plane Coding (T1-pp-575-576), JPEG compression standard – DCT based image compression (T1-pp-588-593).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

MORPHOLOGICAL IMAGE PROCESSING

Erosion (T1-pp-639-640), Dilation (T1-pp-641-643), Opening, Closing (T1-pp-644-648), The hit-or-miss transformation (T1-pp-644-652); Basic morphological algorithms - boundary extraction, hole filling, extraction of connected components, thinning, thickening, skeletons, pruning (T1-pp-652-666), Some basic morphological algorithms, Gray scale Morphology (T1-pp-674-679).

UNIT-3 **6L+6T+6P+6SL=24 Hours**

SEGMENTATION AND FEATURE EXTRACTION

Image Segmentation: Fundamentals (T1-pp-700), Point, Line and edge detection (T1-pp-701-723), Thresholding (T1-pp-742-763), Segmentation by Region growing and region splitting Region-based segmentation (T1-pp-764-769)

Feature Extraction: Background (T1-pp-812-813), Boundary preprocessing (T1-pp-814-819), Boundary Feature Descriptors (T1-pp-831-839), Region Feature Descriptors (T1-pp-840-843), Principle Components as feature descriptors (T1-pp-859-867).

PRACTICES:

- Develop a module to perform add & removal of salt and pepper noise. Compute PSNR & MSE and check the impact before and after removal of noise (T2-pp-143-165)
- Develop a module to remove noise using average filter and median filter. Compute PSNR &
- MSE before and after removal of noise (T2-pp-143-165)
- Develop a module for Lossy image compression and decompression (T2-pp-282-332)
- Develop a module for DCT based image compression and decompression. (T2-pp-282-332)
- Develop a module for morphological image operations - erosion, dilation, opening & closing (T2-pp-338-349)
- Develop a module for morphological image operations - hit-or-miss transformation. (T2-pp-350-358)
- Develop a module for morphological image operations - thinning, thickening. (T2-pp-359-360)
- Develop a module for detecting lines & edges in an image. (T2-pp-379-390)
- Develop a module for segmenting region of interest (T2-pp-407-416)
- Develop a module for extracting boundary features of an image. (T2-pp-436-441).

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Learn different techniques employed for the enhancement of images.	Apply	1,2,3,5,
2	Learn different causes for image degradation and overview of image restoration techniques.	Analyze	1,2,3,5
3	Apply various compression techniques to reduce image size and morphological operations to extract features.	Evaluate	1,2,3,5
4	Employ image segmentation and representation techniques to extract region of interest.	Apply	1,2,3,5
5	Learn different feature extraction techniques for image analysis and recognition.	Analyze	1,2,3,5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



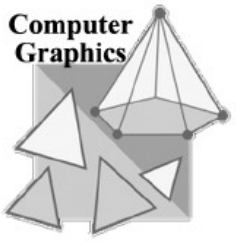


TEXT BOOKS:

1. Rafeal C Gonzalez and Richard E.Woods, “Digital Image Processing”, 4th edition, Pearson Education/ PHI, 2018.
2. Rafeal C Gonzalez and Richard E.Woods, “Digital Image Processing using MATLAB”, 4th edition, Pearson Education/ PHI, 2020.

REFERENCE BOOKS:

1. Milan Sonka, Vaclav Hlavac and Roger Boyle, “Image Processing, Analysis and Machine Vision”, 4th Edition, Cengage, 2015.
2. Alasdair McAndrew, “Introduction to Digital Image Processing with Matlab”, Thomson Course Technology, 2004 Course Technology Press, Boston, MA, United States, 2004.
3. William K. Prat, “Digital Image Processing”, 4th Edition, Wiley-Interscience, A John Wiley & Sons, Inc., Publication, 2007.



Source: https://www.javatpoint.com/computer-graphics-tutorial

25CS806 COMPUTER GRAPHICS

Hours Per Week :

L	T	P	SL	C
2	0	2	2	3

PREREQUISITE KNOWLEDGE: Basic Logical Thinking and Problem Solving Ability.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a comprehensive introduction to computer graphics leading to understanding the contemporary terminology and algorithms of computer graphics. To make the students learn the basic principles of visualization. To give an introduction to 2D and 3D modelling and animation.

MODULE-1

UNIT-1 **6L+0T+6P+6SL=18 Hours**

OVERVIEW OF GRAPHICS SYSTEM

A Survey of Computer Graphics, Computer-Aided Design, Presentation Graphics, Computer Art, Entertainment, Education & Training, Visualization, Image Processing, Graphical User Interfaces. (T1 Page:12-44)

Overview of Graphics Systems, Video Display Devices, Raster-Scan Systems, Random-Scan Systems, Graphics Monitors & Workstations, Input Devices, Hard-Copy Devices, Graphics Software. (T1 Page:45-85)

Output Primitives, Points & Lines, Line-Drawing Algorithms, Line Function, Circle-Generating Algorithms, Ellipse-Generating Algorithms, Parallel Curve Algorithms, Curve Functions, Pixel Addressing & Object Geometry, Filled-Area Primitives, Character Generation. (T1 Page:93-141).

UNIT-2 **6L+0T+6P+6SL=18 Hours**

2D GEOMETRIC TRANSFORMATIONS

Basic Transformations, Matrix representations, and Homogeneous Coordinates, Composite Transformations, Other Transformations, Transformations between Coordinate Systems, Affine Transformations, Transformation Functions, Raster Methods for Transformations. (T1 Page:193-220).

PRACTICES:

- **Basic Drawing and Algorithms**
  - Line Drawing Algorithms:
    - o Implement the DDA (Digital Differential Analyzer) algorithm for line drawing.
    - o Implement the Bresenham's Line Algorithm for rasterizing lines.
  - Circle Drawing Algorithms
    - o Implement the Midpoint Circle Algorithm to draw a circle.
    - o Implement the Bresenham's Circle Algorithm for more efficient circle drawing.
  - Rectangle and Polygon Drawing:
    - o Write a program to draw rectangles and polygons using basic line drawing.
  - Flood Fill Algorithm:
    - o Implement the Flood Fill algorithm to fill an enclosed region with color.
    - o Implement Boundary Fill algorithm for filling regions.
- **2. 2D Transformations**

- Translation:
  - o Write a program to apply translation to a 2D object (such as a square or triangle).
- Scaling:
  - o Implement a program that scales a 2D object based on a scale factor.
- Rotation:
  - o Implement 2D rotation using the rotation matrix.
- Reflection:
  - o Write a program that reflects a 2D object about an axis (X, Y, or a custom line).
- Shearing:
  - o Write a program to apply shearing transformation in 2D.

- **3. 2D Clipping**
  - Cohen-Sutherland Line Clipping Algorithm:
    - o Implement the Cohen-Sutherland algorithm to clip a line segment within a rectangular viewport.
  - Liang-Barsky Line Clipping:
    - o Implement the Liang-Barsky line clipping algorithm.
  - Polygon Clipping:
    - o Write a program to clip a polygon using the Sutherland-Hodgman algorithm.
  - Point-in-Polygon Test:
    - o Write a program to check if a point lies inside or outside a polygon.

SKILLS:

- ✓ *Learn various clipping algorithms.*
- ✓ *Compare 2-D and 3_D transformations.*
- ✓ *Learn various surface detection methods*

MODULE-2

UNIT-1 **6L+0T+6P+6SL=18 Hours**

2D VIEWING BASIC TRANSFORMATION

2-D Viewing: The viewing pipeline, window to viewport coordinate transformation, Two-Dimensional Viewing Functions, Clipping Operations, Point Clipping, Line Clipping, Polygon Clipping, Curve Clipping, Text Clipping. (T1 Page:226-254).

UNIT-2 **6L+0T+6P+6SL=18 Hours**

3D GEOMETRIC TRANSFORMATIONS

Three Dimensional Concepts, Three-Dimensional Display Methods, Three-Dimensional Object Representations, Polygon Surfaces, Curved Lines & Surfaces, Quadric Surfaces, Superquadrics, Blobby Objects. (T1 Page:306-324)

3-D Geometric & Modeling transformations, Translation, rotation, scaling, reflection and shear transformations, composite transformations. (T1 Page:417-436)

3D viewing: Viewing pipeline, viewing coordinates, Projections, View Volumes & General Projection Transformations, Clipping, Hardware Implementations, Three-Dimensional Viewing Functions. (T1 Page:441-474).

UNIT-3 **6L+0T+6P+6SL=18 Hours**

VISIBLE SURFACE DETECTION METHODS

Back-face detection, depth buffer, A-buffer, Z-buffer, scan-line, Depth-Sorting Method, Area Subdivision method, BSP tree, Visible-Surface Ray-Casting. (T1 Page:479-497).

PRACTICES:

- 3D Graphics and Transformations
  - 3D Rotation:
    - Implement 3D rotation using the rotation matrices (for each axis).
  - 3D Scaling and Translation:
    - Write a program to apply scaling and translation to 3D objects.
  - Perspective and Orthographic Projections:
    - Implement both perspective and orthographic projections for 3D objects.
  - 3D Reflection:
    - Implement 3D reflection about a plane (XY, XZ, or YZ planes).
- Ray Tracing
  - Basic Ray Tracing:
    - Implement a simple ray tracing algorithm for rendering a scene with spheres and planes.
  - Reflections and Refractions:
    - Add reflections and refractions to the ray tracing algorithm.
  - Shadows:
    - Implement shadow casting in ray tracing.
  - Anti-Aliasing:
    - Implement basic anti-aliasing techniques in ray tracing (e.g., supersampling).
- Rendering Techniques
  - Rasterization of 3D Objects:
    - Implement a basic 3D rasterization pipeline for drawing 3D objects.
  - Z-Buffering:
    - Write a program that implements Z-buffering to handle hidden surface removal.
  - Texture Mapping:
    - Implement texture mapping on a 3D object (e.g., a cube with a texture).
  - Environment Mapping:
    - Create a program that demonstrates environment mapping (e.g., reflection or refraction mapping).
- OpenGL/DirectX/WebGL
  - OpenGL Basic Setup:
    - Write a simple OpenGL program to create a window and display basic shapes.
  - 3D Object Rotation in OpenGL:
    - Create a program to apply continuous rotation to a 3D object in OpenGL.
  - Shader Programming:
    - Write simple vertex and fragment shaders in GLSL for rendering.
  - WebGL: Basic 3D Rendering:
    - Create a basic 3D rendering scene using WebGL (e.g., drawing a cube and rotating it).
  - OpenGL Lighting Effects:
    - Implement lighting and shading effects (e.g., point light, directional light) in OpenGL.

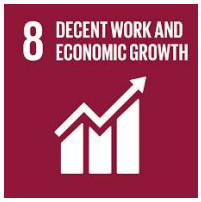
COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Implement of various graphics systems along with the algorithms used for these devices for drawing.	Apply	1, 2
2	Demonstrate 2D graphics and algorithms including line drawing, polygon fi lling, clipping, and transformations.	Apply	1, 2
3	Demonstrate Concepts and techniques used in 3D computer graphics and basic about animation.	Apply	1, 2, 3
4	Implement Surface detection methods to get a realistic screen image.	Apply	1, 2,3

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):







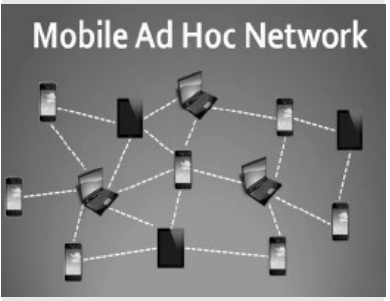


TEXT BOOKS:

1. Donald Hearn and M. Pauline Baker, “Computer Graphics C version”, Pearson education, Second Edition, 2014.
2. Zhigang Xiang, Roy Plastock, “Computer Graphics”, Schaum’s outlines, Second Edition, Tata McGraw Hill Edition, 2000.

REFERENCE BOOKS:

1. Foley, VanDam, Feiner and Hughes, “Computer Graphics Principles &Practice in C”, Pearson Education. 2nd Edition, 1996.
2. David F Rogers, “Procedural elements for Computer Graphics”, Tata McGraw Hill, 2nd Edition., 1988.
3. Neuman and Sproul, “Principles of Interactive Computer Graphics, Tata McGraw Hill”, 2nd Edition., 1978.
4. Shalini, Govil-Pai, “Principles of Computer Graphics”, Springer. 1st Edition, 2006.
5. Steven Harrington, “Computer Graphics”, TMH, 2nd Edition, 1987.



Source: https://www.educba.com/mobile-ad-hoc-network/

25CS807 MOBILE AD-HOC NETWORKS

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Computer networks**.**

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on major aspects of ad hoc networking, from design through performance issues to application requirements. It starts with the design issues and challenges associated with implementations of ad hoc network applications. This includes mobility, disconnections, and battery power consumption. The course provides a detailed treatment of proactive, reactive, and hybrid routing protocols in mobile wireless networks. It also covers the IEEE 802.11 Wireless LAN and discusses their characteristics and operations. Through activities, the course gives students hands-on experience in designing a mobile ad hoc network using the NS2 simulator.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

INTRODUCTION

Introduction to ad-hoc networks-definition (T2-Pp:1-23), characteristics, features, applications; Characteristics of the wireless channel (T1-Pp:6-10); Ad-hoc mobility models-indoor and outdoor models. (T2-Pp:1-23).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

MEDIUM ACCESS PROTOCOLS

MAC protocols- design issues, goals and classification; (T1-Pp:227-233) Contention-based protocols – with reservation (T1-Pp:248-264), without reservation(T1-Pp:233-246); Scheduling algorithms (T1-Pp:267-272); Protocols using directional antennas; IEEE standards - 802.11a, 802.11b, 802.11g, 802.15; (T1-Pp:805-808), HIPERLAN. (T1-Pp:144-146).

PRACTICES:

- Installation of NS-2 and basics of TCL scripting.
- TCL script for
  - Computing the arithmetic operations on two operands.
  - Finding the given number is prime or not using functions.
  - Finding the factorial value of a given number.
- Set the node property and routing protocol in the same MANET scenario.
- Analyse the performance of the MANET.
- Develop MAC Protocol using any suitable Network Simulator for MANETs to send the packet without any contention through wireless link using the following MAC protocols (CSMA/CA (802.11)). Analyse its performance with increasing node density and mobility.
- Simulate MANET environment using suitable Network Simulator and test with various mobility model such as Random walk, Random waypoint and Group mobility. Analyze throughput, PDR and delay with respect to different mobility models.

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

NETWORK PROTOCOLS

Routing protocols - design issues, goals, and classification (T1-Pp:299-307); Proactive vs. reactive routing (R2-Ch.7.3); Unicast routing algorithms; Multicast routing algorithms; Hybrid routing algorithm (T1-Pp:336-342); Energy-aware routing algorithm(T1-Pp:429-432); Hierarchical routing(T1-Pp:352-355); QoS aware routing(T1-Pp:432-434).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

END-END DELIVERY AND SECURITY

Transport layer - issues in designing, transport layer classification, ad-hoc transport protocols; Security issues in ad-hoc networks - issues and challenges, network security attacks; Secure routing protocols. (T1-Pp: 447-495).

UNIT-3 **6L+6T+6P+6SL=24 Hours**

CROSS LAYER DESIGN

Cross layer design - need for cross layer design, cross layer optimization; Parameter optimization techniques; Cross layer cautionary perspective; Integration of adhoc with mobile IP networks. (R5-Pp.:720–724).

PRACTICES:

- Create CBR traffic over UDP and TCP.
- Write an awk script that takes data from trace fi le and give the report for performance metrics such as packet delivery ratio, and throughput.
- Implement Transport Control Protocol in Sensor Network.
- Design and Implementation of Security algorithm for Wireless networks Implementation of security protocol for mobile network.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Summarize the protocols used at the MAC layer and scheduling mechanisms to express the mathematical properties.	Apply	1
2	Apply proactive and reactive routing algorithms to find optimal paths.	Apply	1, 2, 5
3	Analyze types of routing protocols used for unicast and multicast routing.	Analyze	1, 2, 5
4	Compare the performance of various routing protocols in ad-hoc networks.	Analyze	1, 2,
5	Develop the network security solution and routing mechanism.	Evaluate	1, 2

SKILLS:

- ✓ *Evaluate various routing protocols.*
- ✓ *Analyse the performance of MAC protocols for Ad-hoc networks.*
- ✓ *Analyse the performance of Network protocols for Ad-hoc networks.*

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





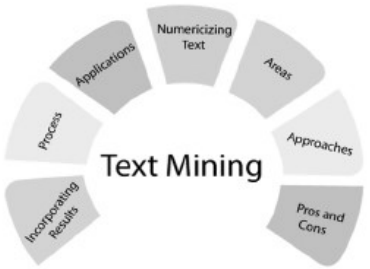
TEXT BOOKS:

1. C. Siva Ram Murthy and B. S. Manoj, “Ad hoc Wireless Networks Architecture and Protocols”, 2nd Edition, Pearson Edition, 2007.
2. Charles E. Perkins, “Ad hoc Networking”, 1st Edition, Addison – Wesley, 2000.

REFERENCE BOOKS:

1. Stefano Basagni, Marco Conti, Silvia Giordano and Ivan stojmenovic, “Mobile ad-hoc networking”, Wiley-IEEE press, 1st Edition, 2004.
2. Mohammad Ilyas, “The Handbook of Adhoc Wireless Networks”, 1st Edition, CRC press, 2002.
3. T. Camp, J. Boleng and V. Davies “A Survey of Mobility Models for Ad Hoc Network Research” Wireless Commun. and Mobile Comp., Special Issue on Mobile Ad Hoc Net working Research, Trends and Applications, vol. 2, no. 5, 2002, pp. 483–502.
4. A survey of integrating IP mobility protocols and Mobile Ad hoc networks, Fekri M. Abduljalil and Shrikant K. Bodhe, IEEE communication Survey and tutorials, v no.1, 2007.
5. V. T. Raisinhani and S.Iyer “Cross Layer Design Optimization in Wireless Protocol Stacks” Comp. Communication, Vol 27 no. 8, 2004.

25CS808 TEXT MINING



Source: https://www.javatpoint.com/text-data-mining

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Data Analysis, Databases, Basic Python, Machine Learning & Deep Learning Foundations.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a comprehensive introduction to Text Mining, covering both traditional NLP techniques and modern Large Language Model (LLM) applications. Students will learn linguistic preprocessing, POS tagging, NER, document parsing, and vector-based feature extraction. The course integrates hands-on experience with GUI and API development for real-time text analysis. Emphasis is placed on both classical models and recent advancements enabling students to build intelligent, domain specific text applications.

MODULE-1

UNIT-1 **6L+6P+6P+6SL=24 Hours**

TEXT MINING FUNDAMENTALS

Introduction to Text Mining, Applications of NLP, Corpus Design and Text Cleaning Strategies, Text Normalization, Spell Correction and Edit Distance, POS Tagging, Named Entity Recognition (NER), Document Parsing (HTML, JSON) and Chunking, Text Vectorization, Similarity Metrics, Word Cloud, Frequency Distribution, Keyword Ranking.

UNIT-2 **6L+6P+6P+6SL=24 Hours**

FUNDAMENTALS OF GUI IN TEXT SYSTEMS

Tkinter/Flask Architecture, Input Interface Design, GUI Workflow, Toggle-Based Control Panels, Inline Visualization and Preview Interface, Real-Time Frequency Analysis and Key Term Visualization, API Trigger Integration, Designing Modular and Reusable GUI Pipelines, Error Handling and Input Validation, GUI Security..

PRACTICES:

- Design a GUI that accepts raw text input and performs tokenization, stopword removal, lemmatization, and punctuation filtering with toggle buttons for each step.
- Create a tool that auto-corrects misspelled words in each paragraph using Levenshtein Distance and displays original vs corrected text side by side.
- Design a resume parsing GUI to extract fields like name, email, skills, and education using NER and POS tagging.
- Build a survey text processing tool that tokenizes, removes stopwords, and visualizes top terms via word cloud.
- Create a tool that accepts any document and visually tags parts of speech using spaCy and renders results in GUI.

SKILLS:

- ✓ *Exploratory Data analysis.*
- ✓ *Opinion Mining/ Sentiment Analysis using various deep Learning models.*
- ✓ *Text Summarization and Categorization etc.*

MODULE-2

UNIT-1 **6L+6P+6P+6SL=24 Hours**

CONTEXTUAL INTELLIGENCE

Topic Modeling, Intent Detection, Semantic Search Techniques, Text Indexing and Query Rewriting, Retrieval-Augmented Generation (RAG), Behavioral Pattern Mining from Text Logs, Explainability in NLP, Decision Tracing, Responsible AI, Hallucination Mitigation, GUI/API Integration.

UNIT-2 **6L+6P+6P+6SL=24 Hours**

TRANSFORMER MODELS

Transformer Architecture, BERT, GPT, Attention Mechanisms, Fine-Tuning Strategies, Sequence Classification, Entity Classification using Transformers, Text Summarization, Emotion and Sentiment Scoring, Evaluation Metrics for Transformer Outputs.A.

UNIT-3 **6L+6P+6P+6SL=24 Hours**

PROMPT ENGINEERING

Fundamentals of Prompt Engineering, Zero-Shot Prompting, Few-Shot Prompting, Chain-of-Thought Prompting, Prompt-based Text Classification (Emotion, Sentiment, Intent), Prompt Tuning vs Manual Prompting, Designing Prompts for Multi-Intent Tasks, Prompt Evaluation, LangChain Primer, Prompt Templates & Chaining.

PRACTICES:

- Build a topic modeling interface that analyzes customer feedback and displays key themes using LDA.
- Develop a Flask-based GPT interface to summarize long support tickets and classify urgency.
- Implement a GUI knowledge base search using LangChain and RAG pipeline for document-level Q&A.
- Build a GUI to identify customer churn risk using GPT with zero-shot classification and dashboard visualization.
- Use OpenAI API to evaluate resumes for job fitment and generate a scorecard in the frontend.
- Build a prompt GUI module that accepts long text input and auto-generate and assign tasks using summarization prompts.
- Implement a LangChain-based interface that builds workflows from goal text using prompt chaining.
- Develop an interface that allow users to compare multiple GPT prompts on the same input and view outputs with BLEU/ROUGE scores.
- Build a module to accept feedback entries and convert them into key insights using GPT + bar chart visualization.
- Build a GUI that wraps GPT output with filters to redact personal or offensive data before display.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply Pre-processing techniques, and annotation methods.	Apply	1
2	Apply classical NLP and parsing techniques through GUI-based text systems for structured analysis.	Apply	1
3	Analyze text classification, topic modeling, summarization, and emotion detection using Transformers.	Analyze	3
4	Evaluate the effectiveness of prompt-based LLMs in real-time tasks.	Evaluate	4
5	Design and develop full-stack LLM-integrated GUI applications for enterprise-level customer intelligence.	Create	5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



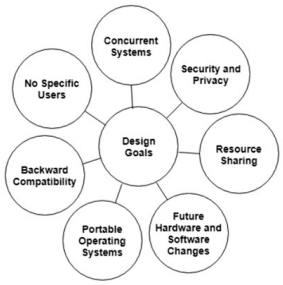


TEXT BOOK:

1. Daniel Jurafsky and James H. Martin; “Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition”, 3rd edition (Draft), Pearson Education, 2020.
2. Steven Bird, Ewan Klein, and Edward Loper; “Natural Language Processing with Python – Analyzing Text with the Natural Language Toolkit”, 1st Edition, O’Reilly Media, 2009.
3. Delip Rao and Brian McMahan; “Natural Language Processing with PyTorch: Build Intelligent Language Applications Using Deep Learning”, 1st Edition, O’Reilly Media, 2019.

REFERENCE BOOKS:

1. Karthiek Reddy Bokka; “Mastering Transformers: Build state-of-the-art models from scratch with advanced natural language processing concepts”, Packt Publishing, 2021.
2. Dennis Rothman; “Transformers for Natural Language Processing: Build and train state-of-the-art NLP models using LLMs, BERT, RoBERTa, GPT, and more”, 2nd Edition, Packt Publishing, 2022.
3. Palash Goyal, Sumit Pandey, and Karan Jain; “Deep Learning for Natural Language Processing”, Apress, 2018.



Source: https://www.tutorialspoint.com.cach3.com/operating-system-design-goals.html

25CS809 OPERATING SYSTEM DESIGN

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Knowledge of computers fundamentals, Computer organization and Digital logic design.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims at concepts and principles of Operating Systems, its overall responsibility in acting as an interface between the system's hardware components and the user. Further, it also helps students to understand the different scheduling policies, process synchronization mechanisms, deadlock handling mechanisms, memory management techniques and operating system security.

MODULE-1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

INTRODUCTION & PROCESSES

Introduction to Operating System: Introduction, Operating System Concepts, System Calls, Operating System Structure. (T1-PG:1-51)

Processes: Introduction, Inter Process Communication, Classical IPC Problems, Scheduling. (T1-PG:55-112)

UNIT-2 **6L+6T+0P+6SL=18 Hours**

INPUT/OUTPUT

Input/Output: Principles of Hardware and Software, Deadlock- Principles, Ostrich Algorithm, Detection and Recovery, Deadlock Prevention and Avoidance. (T1-PG:221-252).

PRACTICES:

- Implementation of new process creation and its communications.
- Implement of thread creation and deletion.
- Implementation of FCFS scheduling.
- Implementation of SJF and RR Scheduling.
- Implementation of producer consumer problem.
- Implementation of Banker's algorithm for Deadlock avoidance.
- Show how counting semaphores (i.e., semaphores that can hold an arbitrarily large value) can be implemented using only binary semaphores and ordinary machine instructions.
- Five batch jobs A through E, arrive at a computer center at almost the same time. They have estimated running times of 10, 6, 2, 4, and 8 minutes. Their (externally determined) priorities are 3, 5, 2, 1, and 4, respectively, with 5 being the highest priority. For each of the following scheduling algorithms, determine the mean process turnaround time. Ignore process switching overhead.

  a) Round robin.

  b) Priority scheduling.

  c) First-come, first-served (run in order 10, 6, 2, 4, 8).

  d) Shortest job first.

  For (a), assume that the system is multi programmed, and that each job gets its fair share of

the CPU. For (b) through (d) assume that only one job at a time runs, until it finishes. All jobs are completely CPU bound.

- Solve the dining philosopher's problem using monitors instead of semaphores.

MODULE-2

UNIT-1 **6L+6T+0P+6SL=18 Hours**

MEMORY MANAGEMENT

Memory Management: Basic concept of memory management, Swapping, Virtual Memory, Page Replacement Algorithms, Design issues for Paging systems, Segmentation. (T1-PG:373-420).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

FILE SYSTEMS

File Systems: Files, Directories, File System Implementation- layout, Implementing files, Disk Space Management, File System Reliability, File system Performance, Security. (T1-PG:481-537).

UNIT-3 **6L+6T+0P+6SL=18 Hours**

Operating system security- Secure operating systems, Security goals, Trust model, Threat model, Access Control Fundamentals – Protection system – Lampson's Access Matrix, Mandatory protection systems, Reference monitor, Security in Ordinary OS – Unix, Windows(T2-PG:1-19)

PRACTICES:

- Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 KB, 4 KB, 20 KB, 18 KB, 7 KB, 9 KB, 12 KB, and 15 KB. Which hole is taken for successive segment requests of (a) 12 KB (b) 10 KB (c) 9 KB for first fit? Now repeat the question for best fit, worst fit, and next fit.
- A machine has a 32-bit address space and an 8 KB page. The page table is entirely in hardware, with one 32-bit word per entry. When a process starts, the page table is copied to the hardware from memory, at one word every 100 nanoseconds. If each process runs for 100 milliseconds (including the time to load the page table), what fraction of the CPU time is devoted to loading the page tables?
  A computer with a 32-bit address uses a two-level page table. Virtual addresses are split into a 9-bit top-level page table field, an 11-bit second-level page table field, and an offset. How large are the pages and how many are there in the address space?
- Suppose that a 32-bit virtual address is broken up into four fields, a, b, c, and d. The first three are used for a three-level page table system. The fourth field, d, is the offset.
  1)Does the number of pages depend on the sizes of all four fields?
  2) If not, which ones matter and which ones do not?
- A computer has four page frames. The time of loading, time of last access, and the R and M bits for each page are as shown below (the times are in clock ticks):

Page	Loaded	Last Ref	R	M
0	126	279	0	0
1	230	260	1	0
2	120	272	1	1
3	160	280	1	1

  Which page will NRU replace?
  Which page will FIFO replace?
  Which page will LRU replace?
  Which page will second chance replace?
- A small computer has four page frames. At the first clock tick, the R bits are 0111 (page 0 is

SKILLS:

- ✓ *Know the concepts of Processes scheduling and File Systems.*
- ✓ *Identification of different disk scheduling methodologies.*
- ✓ *Interpret UNIX Commands, Shell basics, and shell environments*

0, the rest are 1). At subsequent clock ticks, the values are 1011, 1010, 1101, 0010, 1010,1100, and 0001. If the aging algorithm is used with an 8-bit counter, give the values of the four counters after the last tick.

- Free disk space can be kept track of using a free list or a bitmap. Disk addresses require D bits. For a disk with B blocks, F of which are free, state the condition under which the free list uses less space than the bitmap. For D having the value 16 bits, express your answer as a percentage of the disk space that must be free.
- A disk has 4000 cylinders, each with 8 tracks of 512 blocks. A seek takes 1 msec per cylinder moved. If no attempt is made to put the blocks of a file close to each other, two blocks that are logically consecutive (i.e., follow one another in the file) will require an average seek, which takes 5 msec. If, however, the operating system makes an attempt to cluster related blocks, the mean interblock distance can be reduced to 2 cylinders and the seek time reduced to 100 microsec. How long does it take to read a 100 block file in both cases, if the rotational latency is 10 msec and the transfer time is 20 microseconds per block?

  Assume that you have a page-reference string for a process with m frames (initially all empty). The page-reference string has length p, and n distinct page numbers occur in it.

  a) What is a lower bound on the number of page faults?

  b) What is an upper bound on the number of page faults?
- Consider the following page-replacement algorithms. Rank these algorithms on a five- point scale from “bad” to “perfect” according to their page-fault rate. Separate those algorithms that suffer from Belady’s anomaly from those that do not.

  a) LRU replacement.

  b) FIFO replacement.

  c) Optimal replacement.

  d) Second-chance replacement.
- Implement Lampson’s Access Matrix Model
- Write a program to simulate a Reference Monitor, which controls access between users (subjects) and files (objects) based on predefined security rules. The program should take user input for username, resource, and operation (read/write), and grant or deny access accordingly.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Classify the basic concepts of operating system and demonstrate knowledge of process management and inter-process communication.	Analyze	1, 2, 3
2	Synthesize the concepts of input/output management and analyze problems related to deadlock in operating systems.	Analyze	1, 2, 3
3	Design the various memory management schemes for a given scenario.	Create	1, 3, 4
4	Apply the concepts of file system interface and implementation for efficient data management and security.	Apply	1, 5, 6,
5	Create secure operating system models by applying fundamental security concepts and protection mechanisms.	Create	1, 6, 8

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





TEXT BOOK:

1. Andrew S. Tanenbaum, Albert S.Woodhull. The MINIX Book Operating Systems Design and Implementation, Pearson Publicastion, 3rd Edition, June 2006.
2. Operating System Security, Trent Jaeger, Morgan & Claypool Publishers, 2008.

REFERENCE BOOKS:

1. Richard. Stevens and Stephen A Rago, “Advanced Programming in the Unix Environment”, 3rd edition, Addison-Wesley, 2013.
2. William Stallings, “Operating Systems-Internals and Design principles” PHI, 7th Edition, 2012.
3. Gary J. Nutt. Addison-Wesley , “Operating Systems: A Modern Perspective”, Aug 2001, 2nd Edition.
4. B.A. Forouzan& R.F. Giberg, —Unix and shell Programming‖, Thomson, First Edition, New Delhi, 2003.
5. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, “Operating System Concepts”, 9th edition, John Wiley & Sons Inc, 2013. cVy



Source: https://tse4.mm.bing.net/th/id/OIP.vN86lZCAdr3RDSXc0cuHcAHaE8?pid=ImgDet&rs=1

25CS810 CLOUD COMPUTING

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE:: Operating systems and Computer Networks.

COURSE DESCRIPTION AND OBJECTIVES:

This course deals with the computing model, which enables information, software, and shared resources to be provisioned over the network as services in an on-demand manner. The main objective of this course is to enable the student to understand the evolution of cloud computing through its supporting technologies virtualization and the architectures of top cloud platforms.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

INTRODUCTION

Definition, Historical developments, Computing platforms, and technologies**.**

PRINCIPLES OF PARALLEL AND DISTRIBUTED COMPUTING: IParallel versus distributed computing, Elements of parallel computing, Elements of distributed computing, Technologies for distributed computing.

UNIT-2 **6L+6T+6P+6SL=24 Hours**

VIRTUALIZATION

Introduction, Characteristics, Virtualization techniques, Virtualization and cloud computing, Pros. and cons. of virtualization, Technology examples**.**

CLOUD COMPUTING ARCHITECTURE: Introduction, Cloud reference model, Types of clouds, Economics of clouds, Open challenges.

PRACTICES:

- Probability and statistical analysis of performing hardware virtualization using VMware workstation.
- Launch Amazon Linux EC2 Instance and connect the windows client to it.
- Launch Windows EC2 instance in AWS and connect windows client to it.
- Configure Web Server on Amazon Linux instance with Elastic IP.
- Manage Elastic Block Storage (EBS).
- Configure Amazon Simple Storage Service (Amazon s3).
- Configure Amazon S3 Glacier.
- Configure Amazon EFS.

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

CLOUD PROGRAMMING AND SOFTWARE ENVIRONMENTS

Programming on Amazon AWS and Microsoft Azure – Programming support of Google App Engine – Docker Architecture and Components –Docker Interfaces – Docker Orchestration - Emerging Cloud Software Environment.

UNIT-2 **6L+6T+6P+9SL=24 Hours**

CLOUD RESOURCE MANAGEMENT AND SCHEDULING

Policies and mechanisms for resource management, Applications of control theory to task scheduling on a cloud, Stability of a two-level resource allocation architecture, Feedback control based on dynamic thresholds, Coordination of specialized autonomic performance managers, A utility-based model for cloud-based web services, Resource bundling, combinatorial auctions for cloud resources, Scheduling algorithms for computing clouds, fair queuing, Start time fair queuing, Cloud scheduling subject to deadlines.

UNIT-3 **6L+6T+6P+9SL=24 Hours**

CLOUD APPLICATIONS & GLOBAL AND LOCAL IMPACT OF CLOUD COMPUTING ON SOCIETY

Scientific applications in healthcare, biology, geo science; Business applications in– CRM and ERP, productivity, social networking, media applications, multiplayer online gaming.

ETHICAL CONSIDERATIONS & SECURITY FOR CLOUD COMPUTING

Cloud Access: authentication, authorization and accounting - Cloud Provenance and meta-data - Cloud Reliability and fault-tolerance - Cloud Security, privacy, policy and compliance- Cloud federation, interoperability and standard.

PRACTICES:

- Configure Amazon Virtual Private Cloud (VPC).
  - a) Create your own VPC.
  - b) Create a public subnet.
  - c) Create a private subnet.
  - d) Create an Internet gateway and attach to your VPC.
  - e) Create Pubic Routing Table, associate subnet and add routing rules.
  - f) Create Private Routing Table, associate subnet and add routing Rules.
  - g) To launch Windows instance in Public subnet.
- Configure Amazon Elastic Load Balancer.
- Configure Relational Database Service (RDS).

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply the fundamental concepts and historical developments of parallel and distributed computing,.	Apply	1, 2, 10
2	Explain the principles, techniques, and characteristics of virtualization, and analyze its relationship with cloud computing including the pros and cons.	Analyze	1, 2, 4, 10
3	Demonstrate the ability to program using major cloud platforms like Amazon AWS, Microsoft Azure, and Google App Engine, and understand Docker architecture and orchestration.	Apply	1, 3, 5, 8, 10
4	Evaluate resource management and scheduling techniques in cloud computing, including policies, control theory applications, and scheduling algorithms.	Evaluate	1, 2, 3, 4, 5, 10
5	Analyze cloud applications across various sectors and discuss ethical, security, and privacy considerations involved in cloud computing.	Analyze / Evaluate	1, 2, 4, 6, 7, 10

SKILLS:

- ✓ *Gain knowledge of different types of Cloud Service Providers.*
- ✓ *Explore basic design issues of Cloud Applications.*
- ✓ *Compare & evaluate the optimum costs in the data transmissions.*

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





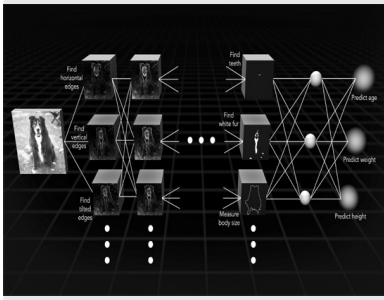
TEXT BOOK:

1. Raj Kumar Buyya, C Vecchiola and S TSelvi , “Mastering Cloud Computing”, 1st edition, Tata McGraw Hill Education (India), 2013.
2. RajKumarBuyya, Broberg J and GoscinskiA, “Cloud Computing - Principles and Paradigms”, 1st edition, Wiley, 2011.
3. Dan C Marinescu, Cloud Computing, Theory and Practice, MK Elsevier Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Cloud Computing: Principles and Paradigms, Wiley
4. Kai Hwang, Geoffrey C. Fox and Jack J. Dongarra, “Distributed and cloud computing from Parallel Processing to the Internet of Things”, Morgan Kaufmann, Elsevier – 2012.

REFERENCE BOOKS:

1. David S. Linthicum , Cloud Computing and SOA Convergence in Your Enterprise A Step-by-Step Guide, Pearson 2010.
2. Dr. Kumar Saurabh, Cloud Computing, 2nd Edition, Wiley India 2012.
3. Rittinghouse J W, and Ransome J F, “Cloud Computing - Implementation, Management, and Security”, 1st edition, CRC Press, 2009.
4. Michael Wittig and Andreas Wittig, “Amazon Web Services in Action”, 2nd edition, Manning Publications, 2015.
5. Tim Mather, Subra Kumaraswamy, Shahed Latif, “Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance”, O’ReillyMediaInc, 2009.
5. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, “Operating System Concepts”, 9th edition, John Wiley & Sons Inc, 2013.

25CS811 DEEP LEARNING



Source: https://www.symmetrymagazine.org/sites/default/files/images/standard/neural_network_visual_final.jpg

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Machine Learning.

COURSE DESCRIPTION AND OBJECTIVES:

This course offers sufficient details required to understand the basic building blocks of various deep learning based models. Especially, focuses on different types of neural network models like feed forward neural networks, convolutional neural networks, recurrent neural networks, and deep auto encoders. During this course the students build, train, and evaluate deep neural network models for various applications in image, text, and speech domains. In addition, throughout this course student will be able to understand hyper parameter tuning and other best-practices to be followed while training deep neural network models.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

EVOLUTION OF DEEP NEURAL NETWORKS

Deep Learning Intuition: History of Deep Learning, what is Deep Learning? Applications of Deep Learning.

Neural Network Basics: McCulloch–Pitts neuron, Perceptron learning rule, Perceptron convergence theorem, Sigmoidal neuron, Multi-layer feed forward neural network, back propagation algorithm, Gradient descent method, Stochastic gradient descent method. Shallow Neural Networks and Deep Neural Networks.

UNIT-2 **6L+6T+6P+6SL=24 Hours**

REGULARIZATION AND OPTIMIZATION FOR TRAINING DEEP MODELS

Optimization methods-Adagrad, Adadelta, RMSProp, Adam; Regularization Methods-Dropout, Drop connect, Batch normalization; Activation functions - Linear, sigmoid, sigmoid, ReLU and variations of ReLU; Loss Function, Improving the training process – Dataset Augmentation, Noise Robustness, Weight Initialization methods, Early stopping, Parameter sharing and tying.

INSTRUCTIONS FOR PRACTICES:

- Practice Assignments can be implemented using the Keras / Tensorflow APIs of Python
- Relevant data sets can be downloaded from standard repositories such as Kaggle/UCI or can be developed by the students.

PRACTICES:

- Using Deep pre-trained CNN model for feature extraction:
  - Extract features from the FC1 of VGG network.
  - Train any traditional ML model like SVM for classification.
  - Repeat the above by considering FC2 of VGG for feature extraction.
- Implement Shallow Neural Network model:
  - Implement a binary classification neural network with a single and multiple hidden layers.
  - Implement a Multi-class classification neural network with a single and multiple hidden layers.
  - Vary the number of neurons at suitable layers.

SKILLS:

- ✓ *Developing Vision and text based applications*
- ✓ *Hyperparameter Tuning of a deep Neural network model.*
- ✓ *Tensor Flow/ Keras tool usage for neural network implementation.*

- Hyper parameter Tuning of a Neural Network model implemented for hand-written digit classification:
  - Vary the type of activation functions.
  - Choose suitable Loss functions.
  - Vary the number of neurons at suitable layers.
  - Vary Weight Initialization methods.
  - Save the Best Model and load the saved model.
- Building a Deep Neural Network:
  - Implement a multi-class classification neural network with number of layers of your choice.
  - Include Batch Normalization layers.
  - Vary Optimization methods.
  - Add drop out layers.
- Convolutional Neural Network Models.
  - Design a Convolutional neural network with the layers of your choice
  - Compare the performance by changing the
  - Kernel size
  - Number of feature maps at each convolutional layer
  - Stride.
  - Padding.
  - Number of fully connected layers.
- Visualization of CNN Models.
  - Design a Convolutional Neural Network Model for image classification.
  - Plot Model Architecture.
  - Visualize feature maps after training of CNN.
  - Visualize class activation maps.
- Convolutional Neural Network Models.
  - Design a Convolutional neural network with the layers of your choice
  - Compare the performance by changing the
  - Kernel size
  - Number of feature maps at each convolutional layer
  - Stride.
  - Padding.
  - Number of fully connected layers.
- Visualization of CNN Models.
  - Design a Convolutional Neural Network Model for image classification.
  - Plot Model Architecture.
  - Visualize feature maps after training of CNN.
  - Visualize class activation maps.

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

CONVOLUTIONAL NEURAL NETWORKS

Convolutional Neural Networks (CNNs): Foundations of Convolutional Neural Network,

Popular Deep CNN Models: LeNet, AlexNet, VGGNet, ResNet, Google Net. Use deep Convolutional architectures for feature extraction and fine-tuning tasks.

UNIT-2 **6L+6T+6P+6SL=24 Hours**

DEEP UNSUPERVISED LEARNING

Deep Unsupervised Learning: Autoencoder- Under complete Autoencoder, regularized auto encoders, Representation power, layer size and depth, stochastic encoders and decoders, Denoising auto-encoders, Sparse auto encoder, Contractive auto-encoders, Variational Autoencoders-Nueral Network Perspective.

Generative Adversarial Networks

Generative Adversarial Networks, Applications and Architecture.

UNIT-3 **6L+6T+6P+6SL=24 Hours**

RECURRENT NEURAL NETWORKS

Architecture of an RNN, unfolding of an RNN, Backpropagation through time, Long short term memory (LSTM), Gated Recurrent Units, Applications- Text Classification, Sentiment Analysis.

PRACTICES:

- Using Deep pre-trained CNN model for feature extraction:
  - Extract features from the FC1 of VGG network.
  - Train any traditional ML model like SVM for classification.
  - Repeat the above by considering FC2 of VGG for feature extraction.
- Fine-tuning Deep pre-trained CNN for Classification:
  - Fine-tune VGG network for the task under consideration.
  - Check the performance by making.
  - all the layers trainable.
  - freezing the initial layers.
  - freezing the entire network except the final layer.
- Design MLFFNN with 3-level stacked autoencoder based pre-training for Black and white image data, display features extracted by different levels of stacked autoencoder at the end of pre-training.
- Sentiment Analysis
  - Pre-process the text.
  - Convert the text into word embedding's.
  - Implement the classification network using LSTMs/ GRUs.
  - Pre-process the text.
  - Convert the text into word embedding's.
  - Implement the classification network using LSTMs/ GRUs.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Implementation of Deep learning models to solve various real-time problems.	Apply	1, 3
2	Analyse performance of a deep network and tune its capacity and hyper parameters.	Analyze	2
3	Leveraging tools to Build deep networks and apply them for real word tasks.	Apply	1,5
4	Developing core components for deep learning algorithms.	Apply	3

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



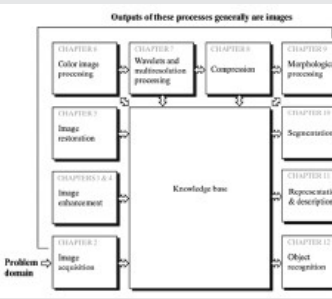
TEXT BOOKS:

1. Ian Good Fellow and Yoshua Bengio and Aaron, “Deep Learning”, 1st Edition, MIT Press, 2016.
2. Charu C Aggarwal “Neural Networks and Deep learning”, Springer International Publishing, 2018.

REFERENCE BOOKS:

1. Francois Chollet, “Deep learning with python”, 1st edition, Manning Publications, 2017.
2. S. Haykin, “Neural Networks and Learning Machines”, 3rd edition, Prentice Hall of India, 2011.
3. Josh Patterson and Adam Gibson, “Deep Learning: A Practitioner’s Approach”, 1st Edition, O’Reilly, 2017.
4. Satish Kumar, “Neural Networks, A Classroom Approach”, Tata McGraw-Hill, 2007.

25CS812 CLOUD SECURITY



Source: https://www.researchgate.net/figure/Fundamental-steps-in-digital-image-processing_fig10_333856607

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Cloud Computing and Network Security.

COURSE DESCRIPTION AND OBJECTIVES:

This Course focuses on the principles, practices, and technologies for securing cloud environments against evolving threats. The course introduces cloud fundamentals, security challenges, and the shared responsibility model, followed by identity and access management, data protection, network, and application security. It blends theory with practice, focusing on encryption, virtualization security, IAM frameworks, compliance, and zero-trust models. Students will gain the skills to assess risks, apply security controls, and ensure privacy, integrity, and resilience in cloud computing.

MODULE-1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

FUNDAMNETALS OF CLOUD SECURITY

Introduction to Cloud and Security Basics: Cloud computing overview: service models (IaaS, PaaS, SaaS),Deployment models (Public, Private, Hybrid, Community),Cloud architecture and components,Shared responsibility model in cloud,Cloud security requirements – confidentiality, integrity, availability, accountability, and compliance,Security challenges in cloud vs. traditional IT systems.

UNIT-2 **6L+6T+0P+6SL=18 Hours**

CLOUD SECURITY THREATS AND VULNERABILITIES

Symmetric Ciphers: Threat landscape in cloud environments, Common Vulnerabilities and Security Risks in Cloud Environments,Data breaches, denial of service (DoS/DDoS) in cloud, Cloud security risk management,Security standards and frameworks (ISO 27017, NIST, CSA guidelines).

PRACTICES:

- Exploring below Cloud Platforms
- Create and configure a free-tier account in AWS / Azure / GCP.
- Create Deploy a simple VM and understand the shared responsibility model.
- Vulnerability Assessment in Cloud.
- Use tools like OWASP ZAP or Nmap to scan a cloud-hosted VM or web app.
- Identify insecure APIs / open ports.

MODULE-2

UNIT-1 **6L+6T+0P+6SL=18 Hours**

CLOUD SECURITY MECHANISMS AND CONTROLS

Identity and Access Management (IAM) in Cloud: Authentication, authorization, and accounting (AAA),

Federated identity management (SAML, OAuth, OpenID Connect),Role-based and attribute-based access control in cloud,Single Sign-On (SSO) and Multi-Factor Authentication (MFA).

SKILLS:

- ✓ *Apply knowledge of science and engineering principles to image related problems.*
- ✓ *Undertake image problem identification and formulate solutions.*
- ✓ *Implement algorithms for enhancement, restoration, compression etc.*

UNIT-2 **6L+6T+0P+6SL=18 Hours**

DATA SECURITY IN CLOUD

Data protection techniques: encryption at rest, in transit, and in us,Key management in cloud (KMS, HSM),Homomorphic encryption, searchable encryption basics,Data Protection and Resilience Strategies in Cloud,Privacy preservation and compliance (GDPR, HIPAA, PCI DSS).

UNIT-3 **6L+6T+0P+6SL=18 Hours**

NETWORK AND APPLICATION SECURITY IN CLOUD

Cloud network security: Firewalls, VPNs, IDS/IPS,Secure APIs and microservices security,Virtualization and container security,Application-level security: secure coding, patch management, WAF,Zero Trust Security model in cloud environments.

PRACTICES:

- Identity and Access Management (IAM) Basics
  Create IAM users, groups, and roles in AWS/Azure, apply least-privilege policies with MFA, and verify access restrictions.
- Role-Based Access Control (RBAC)
  Define custom roles, assign them to users, and demonstrate restricted access based on role permissions.
- Data Encryption in Cloud Storage
  Upload files to cloud storage, enable encryption at rest using KMS, and verify encryption in transit with HTTPS.
- Data Backup and Recovery
  Configure automated database backups, simulate data deletion, and perform recovery to restore data.
- Network and Application Security
  Configure firewall rules and Web Application Firewall (WAF) in the cloud, then test against attacks like SQL Injection and XSS.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply cloud computing security requirements to shared responsibility model.	Apply	1,2
2	Analyze cloud-specific threats, vulnerabilities, and risks in multi-tenant environments.	Analyze	1,2,6
3	Apply identity and access management (IAM) mechanisms such as RBAC and MFA to secure cloud resources.	Apply	1,3,5
4	Implement encryption, backup, and recovery strategies for data security in cloud platforms.	Apply	1,2,6
5	Evaluate network and application-level security mechanisms, including firewalls, WAF, and zero-trust models.	Evaluate	1,2,4,5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):

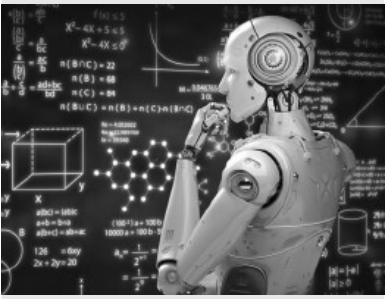


TEXT BOOKS:

1. Tim Mather, Subra Kumaraswamy, and Shahed Latif, Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance, 1st Edition, O'Reilly Media, 2009.

REFERENCE BOOKS:

1. Ronald L. Krutz and Russell Dean Vines, Cloud Security: A Comprehensive Guide to Secure Cloud Computing, Wiley, 2010
2. Vic (J.R.) Winkler, Securing the Cloud: Cloud Computer Security Techniques and Tactics, Syngress, 2011.
3. Kevin L. Jackson and Scott Goessling, Architecting Cloud Computing Solutions: Build Cloud Strategies that Align Technology with Business, Packt Publishing, 2019.
4. John Rhoton, Cloud Computing Explained: Implementation Handbook for Enterprises, 2nd Edition, Recursive Press, 2013..
5. Ben Potter, Scott Ward, AWS Security Best Practices, Amazon Web Services Whitepapers, 2019.



Source: https://www.forbes.com/sites/kalevleetaru/2019/01/15/why-machine-learning-needs-semantics-not-just-statistics/

25CS813 SOFTWARE PROJECT MANAGEMENT

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Software Engineering, Web technologies, Data Base Management Systems.

COURSE DESCRIPTION AND OBJECTIVES:

Students will be introduced to the following aspects of project management related to managing small software development and to describe activities of SPM highlights and train in the planning and implementation of project management. It brings a specific project to complete on time and on budget.

MODULE-1

UNIT-1 **9L+6T+0P+6SL=21 Hours**

SOFTWARE MANAGEMENT & SOFTWARE ECONOMICS

The waterfall model, conventional software Management performance. Evolution of Software Economics: Software Economics, pragmatic software cost estimation.

Improving Software Economics, Reducing Software product size, improving software processes, improving team effectiveness, improving automation, Achieving required quality, peer inspections.

UNIT-2 **9L+6T+0P+6SL=21 Hours**

THE OLD WAY AND THE NEW & LIFE CYCLE PHASES

The principles of conventional software Engineering, principles of modern software management, transitioning to an iterative process.

Life cycle phases: Engineering and production stages, inception, Elaboration, construction, transition phases.

PRACTICES:

- Consider a small project idea (e.g., Student Attendance System) and
- prepare a 2-page comparative report on how it would be executed under the Waterfall Model and how it would differ in an Iterative/Modern Approach.
- Key Points: Phases, feedback loops, cost impact, time to market, and risk handling.
- Estimate the size of a small software project (Ex: Library management system) using Function Point Analysis.
- Prepare a schedule for a small project of your interest using MS Project, Excel, or online tools (like ClickUp, GanttProject, or Notion), and make a Gantt chart for a 3-month mini project.
- Choose a small existing project that was developed using a traditional model (e.g., Library Management System). Design a transition plan to move it toward an Iterative/Agile process.

MODULE-2

UNIT-1 **9L+6T+0P+6SL=21 Hours**

ARTIFACTS OF THE PROCESS

The artifact sets, Management artifacts, Engineering artifacts, programmatic artifacts. Model based software architectures: A Management perspective and technical perspective.

UNIT-2 **9L+6T+0P+6SL=21 Hours**

PROJECT ORGANIZATION AND PLANNING

Work breakdown structures, planning guidelines, cost and schedule estimating, Iteration planning process, Pragmatic planning.

Project Organizations and Responsibilities, Line-of-Business Organizations, Project Organizations, evolution of Organizations.

Process Automation: Automation Building blocks, The Project Environment.

UNIT-3 **9L+6T+0P+6SL=21 Hours**

PROJECT CONTROL AND PROCESS INSTRUMENTATION

The seven core Metrics, Management indicators, quality indicators, life cycle expectations, pragmatic Software Metrics, Metrics automation.

Future Software Project Management, Modern Project Profiles, Next generation Software economics, modern process transitions.

PRACTICES:

- For any given project (e.g., Online Shopping Portal), prepare a Risk Register with following parameters.
  - Risk ID
  - Risk Description
  - Probability
  - Impact
  - Mitigation Plan
  - Contingency Plan
- Select the project of own interest and estimate the effort and development time using COCOMO (Basic) model for a project of size X KLOC (e.g., 10 KLOC).
- Pick a real-world failed software project (e.g., FBI Virtual Case File, or NHS IT Project). Prepare a 5 to 10 pages' report on reasons for the failure and particularly address the following:
  - Why it failed
  - Management mistakes
  - Lessons learned
- Develop a comprehensive Software Project Plan (SPP) for a medium-sized application. Sections to include:
  - Project Overview
  - Scope Management
  - Work Breakdown Structure (WBS)
  - Schedule (Gantt chart)
  - Cost & Resource Plan
  - Risk Management
  - Communication & Quality Plans

SKILLS:

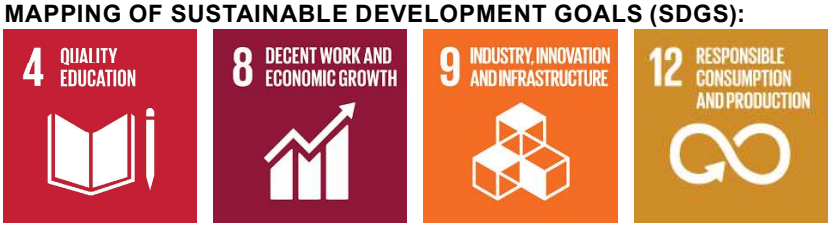
- ✓ *Develop detailed Work Breakdown Structures (WBS)*
- ✓ *Perform cost and schedule estimation using pragmatic models*
- ✓ *Plan iterations and manage lifecycle phases (Inception to Transition)*
- ✓ *Apply planning techniques aligned with both conventional and modern methodologies.*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply cost estimation techniques to improve software economics.	Apply	1,2,11
2	Analyze and differentiate between traditional and modern software management principles and describe the iterative development life cycle phases.	Analyze	1,2, 3
3	Apply various software artifacts used in project development, including their roles in management and engineering.	Apply	1, 2
4	Develop and evaluate project plans using WBS, estimation techniques, and understand organizational structures in software project environments.	Apply, Evaluate	1,2,5,11
5	Apply process instrumentation and control mechanisms using software metrics, and assess the impact of modern trends in project management.	Apply, Evaluate	1,2,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





TEXT BOOKS:

1. Walker Royce ,"Software Project Management", 1st ed., Pearson Education, 2005.

REFERENCE BOOKS:

1. Bob Hughes and Mike Cotterell, "Software ProjectManagement", 3rd ed.,Tata McGraw - Hill Edition, 2005.
2. Joel Henry, "Software Project Management", 1st ed., Pearson Education, 2006.
3. Pankaj Jalote, "Software Project Management in practice", 1st ed., Pearson Education, 2005.

25CS814 COMPILER DESIGN



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Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Data Structures and Algorithms, Theory of Computation.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the fundamental concepts of compiler design, including lexical analysis, syntax analysis, semantic analysis, intermediate code generation, optimization, and code generation. It emphasizes both theoretical aspects and implementation techniques, preparing students to design and develop efficient compilers for modern programming languages.

MODULE-1

UNIT-1 **9L+6T+0P+6SL=21 Hours**

INTRODUCTION TO COMPILERS

Compiler Structure: Phases of a Compiler, Lexical, Syntax, Semantic, Intermediate Code Generation, Optimization, and Code Generation.

Lexical Analysis: Role of Lexical Analyzer, Input Buffering, Regular Expressions, Finite Automata, Lexical Analyzer Generator (LEX).

UNIT-2 **9L+6T+0P+6SL=21 Hours**

SYNTAX ANALYSIS

Parsing Techniques: Top-down parsing (Recursive Descent, Predictive Parsing), Bottom-up parsing (Shift-Reduce, Operator Precedence, LR Parsers).

Error Handling in Parsing.

PRACTICES:

- Implement various phases of compiler in detail. Write down the output of each phase for expression Total = (b + c) + (b + c) * 50.
- Construction of DFA for regular expressions.
- Implementation of lexical analyzer using LEX.
- Design of predictive parsing table.
- Implementation of shift-reduce parser.

MODULE-2

UNIT-1 **9L+6T+0P+6SL=21 Hours**

SEMANTIC ANALYSIS AND INTERMEDIATE CODE GENERATION

Syntax-directed definitions, Dependency Graphs, Attribute Grammars, Type Checking, Intermediate Representations: Syntax Trees, Three-Address Code, Quadruples, Triples.

UNIT-2 **9L+6T+0P+6SL=21 Hours**

CODE OPTIMIZATION

Principal Sources of Optimization, Loop Optimization, Peephole Optimization, Machine-Independent Optimizations.

SKILLS:

- ✓ *Ability to design and implement compiler components.*
- ✓ *Skills in lexical analysis, parsing, and semantic analysis.*
- ✓ *Knowledge of code optimization and generation techniques.*
- ✓ *Hands-on practice with LEX and YACC tools.*

UNIT-3 **9L+6T+0P+6SL=21 Hours**

CODE GENERATION

Issues in Code Generation, Register Allocation, Instruction Selection, Code Generation Algorithms, Introduction to Code Generators (YACC).

PRACTICES:

- Implementation of syntax-directed translation.
- Generation of intermediate code for arithmetic expressions.
- Type checking examples.
- Code optimization on given intermediate code.
- Code generation using YACC.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply different phases of compiler design.	Apply	1, 2, 10
2	Apply lexical and syntax analysis techniques to programming languages.	Apply	1, 2, 3, 9, 10
3	Construct intermediate code for programming constructs.	Apply	1, 2, 3, 9, 10
4	Analyze and implement code optimization techniques.	Analyse	1, 2, 3, 9, 10
5	Design and implement a simple compiler using tools like LEX and YACC.	Create	1, 2, 3, 5, 9, 10

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



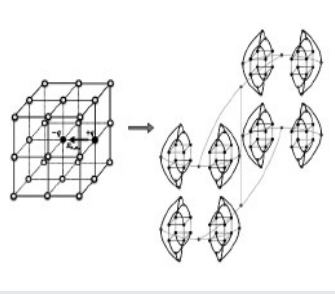
TEXT BOOKS:

1. Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, Compilers: Principles, Techniques, and Tools, 2nd Edition, Pearson, 2007.
2. Kenneth C. Louden, Compiler Construction: Principles and Practice, Cengage, 2007.

REFERENCE BOOKS:

1. Dick Grune, Henri E. Bal, Modern Compiler Design, Springer, 2007.
2. Keith D. Cooper, Linda Torczon, Engineering a Compiler, Morgan Kaufmann, 2nd Edition, 2011.
3. Steven Muchnick, Advanced Compiler Design and Implementation, Morgan Kaufmann, 1997.

25CS815 NUMERICAL ALGORITHMS



Source: https://pasquans.eu/new-numerical-algorithms-for-old-problems/

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Basics of integration, Differentiation and Polynomials.

COURSE DESCRIPTION AND OBJECTIVES:

The goal of this course is to build a solid foundation in Numerical Methods, which are essential algorithmic tools used to support and enhance computer-based predictions in modern systems science and engineering. Numerical methods serve as the backbone for solving complex real-world problems where analytical solutions are difficult or impossible to obtain. These methods encompass a wide range of techniques including, but not limited to, simple optimisation for improving performance and efficiency, interpolation methods that estimate unknown values from known data points, and linear algebra techniques used to solve large systems of equations effectively. The course also covers the use of ordinary differential equations to model dynamic systems over time and simulate their behaviour. In addition, students will explore stochastic simulation techniques that account for randomness and uncertainty, which are crucial in modeling real-life systems subject to unpredictable external influences.

MODULE–1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

ROOT FINDING METHODS, SYSTEM OF LINEAR EQUATIONS AND INTERPOLATION

Absolute error,(T1-Pp:8-19) order of Convergence (T2-Pp-30-39), Geometrical Description, method of successive approximation, Bisection method,(T1-Pp:23-28), Regula - Falsi method (T1-Pp:28-31), Newton's Method (T1-Pp:37-44 and 64 to 68). Gauss Seidal method,(T1-Pp:90-94), Crouts method, Triangularization method, Relaxation method (T2-Pp:76-100).

Interpolation – Finite differences, interpolation, Newton's forward and backward difference formulae, Newton's divided difference and Lagrange's formulae (with proof), errors in interpolation formula. ,(T1-Pp:73-125).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

APPLICATIONS

Finding positive, negative and real root of algebraic and transcendental equation(T2-Pp:2-13), Solution of simultaneous linear algebraic equation, nth difference of a polynomial, finding missing terms in a sequence, sum of n terms in a series, finding polynomial using a given set of data, estimated values of a function inside and outside the given intervals of data.

PRACTICES:

- Finding positive, negative and real root of algebraic and transcendental equation using any of your known programming knowledge.
- Solution of simultaneous linear algebraic equation using discrete methods and implement using C.
- By python program estimate the values of a function inside and outside the given intervals of data.
- Implement interpolation techniques (e.g., Newton's or Lagrange's) in Python or C to estimate missing values in a dataset.
- Write a program to calculate the nth difference of a polynomial and use it to predict future terms in a numerical sequence.

SKILLS:

- ✓ *Analyze the types and occurrence of roots.*
- ✓ *Interpolate the unknown values of function.*
- ✓ *Develop a difference equation.*
- ✓ *Gain the knowledge to solve an ODE numerically.*
- ✓ *Gain the knowledge to solve an ODE numerically.*

MODULE–2

UNIT-1 **6L+6T+0P+6SL=18 Hours**

NUMERICAL DIFFERENTIATION, INTEGRATION AND DIFFERENTIAL EQUATION

Numerical Differentiation: Newton's forward and Backward formulas to compute up to second order differentiation of a function. Numerical Integration: Trapezoidal and Simpson's 1/3 and 3/8 rules.ODE: Picard's approximation, Milne's Predictor Corrector formulas. PDE: Liebman's Iteration Process, Bender Schmidt. (T2-Pp:212-296).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

APPLICATIONS

Finding maxima and minima of a function(R4-Pp:76-98), population growth, acceleration, area bounded by the curve, Solution of ODE, Solution of Elliptic, Parabolic and Hyperbolic PDE. (R4-Pp: 1026–1041).

UNIT-3 **6L+6T+0P+6SL=18 Hours**

NUMERICAL METHODS FOR SYSTEMS MODELING AND OPTIMIZATION

Numerical Solutions to Nonlinear Systems: Iterative techniques such as Newton-Raphson for multivariable systems. Curve Fitting and Regression Analysis: Least squares method, linear and nonlinear regression. (R2-Pp:787-945).

PRACTICES:

- Implement the Newton-Raphson method for a system of nonlinear equations using any programming language. Analyze convergence for different initial guesses.
- Develop a Python or C program to compute the Jacobian matrix numerically and use it in a multivariable Newton-Raphson routine.
- Fit a linear regression model to a given dataset using the least squares method and visualize the result using plotting libraries (e.g., matplotlib in Python).
- Write a program to perform polynomial regression or exponential curve fitting on real-world data (e.g., population, temperature).
- Compare the fitting accuracy of linear vs nonlinear regression techniques using metrics like R^2 or mean squared error.
- Apply curve fitting to experimental/lab data and interpret the model parameters for practical understanding.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply numerical methods to find roots.	Analyze	1,2,4,5,9,10
2	Evaluate the unknown values using interpolation.	Evaluate	1,2,5,9,10
3	Develop a finite difference scheme.	Apply	1,2,3,5,9,10
4	Apply numerical methods to solve ODE and PDE and analyse graphically.	Create	1,2,5,9,10
5	Apply numerical methods to solve systems of equations and perform numerical integration.	Apply	1,2,4,5,10

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):



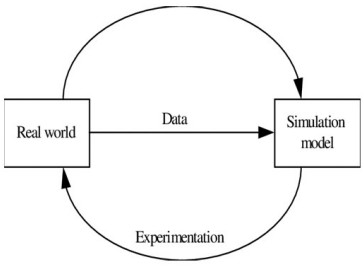


TEXT BOOKS:

1. S. S. Sastry, “Introductory Methods of Numerical Analysis”, 5th ed., PHI Learning, 2012.
2. M. K. Jain, “Numerical Methods for Scientific and Engineering Computation”, New Age International, 8th Ed., 2022.

REFERENCE BOOKS:

1. P. Kandasamy, “Numerical Methods”, S. Chand, 2nd ed., 2015.
2. C. Ray Wylie, Louis C. Barrett, “Advanced Engineering Mathematics”, 6th Edition, McGraw-Hill Book Co., New York, 1995.
3. Gupta C.B., Singh S.R., and Mukesh Kumar, “Engineering Mathematics for Semester I & II”, McGraw Hill Education (India) Pvt. Ltd., 2015.
4. H. K. Das, “Advanced Engineering Mathematics”, S. CHAND & COMPANY PVT. LTD, 2008.



Source: https://www.researchgate.net/figure/The-Simulation-Modelling-Process-Simple-Outline_fig1_3834396

25CS816 SIMULATION AND MODELLING

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Understanding of probability distributions, Statistical data analysis, Random variables, Discrete mathematics, Calculus (helpful for understanding dynamic models), Understanding of systems modeling, Queuing theory and Optimization concepts.

COURSE DESCRIPTION AND OBJECTIVES:

The objective of this course is to teach students methods for modeling of systems using discrete event simulation. Emphasis of the course will be on modeling and on the use of simulation software. The students are expected to understand the importance of simulation in IT sector, manufacturing, telecommunication, and service industries etc. By the end of the course students will be able to formulate simulation model for a given problem, implement the model in software and perform simulation analysis of the system.

MODULE-1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

INTRODUCTION TO SIMULATION

System, model and simulation(T1-Pp:12-14), model of a system,(T1-Pp:15), Static physical models, dynamic physical models, static mathematical models,(T1-Pp:18-19), dynamic mathematical models,(T1-Pp:18-19), principles used in modeling. System studies,(T2-Pp:6), a corporate model: Environment segment, production segment, management segment,(T2-Pp:6). Types of system study,(T2-Pp:7).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

MATHEMATICAL AND STATISTICAL MODELS

Probability concepts,(T1-Pp:27-64), Queuing Models, (T2-Pp:20), Properties of random numbers,(T1-Pp:33)., Generation of Pseudo-Random numbers,(T2-Pp:221)., Methods for generating random variables,(T2-Pp:222). and Validation of random numbers,(T2-Pp:222).

PRACTICES:

- Identify and classify types of models (physical vs mathematical; static vs dynamic).
- Develop a basic system model (e.g., a corporate model with environment, production, and management segments).
- Create a block diagram of a real-life system (e.g., bank, hospital, traffic system).
- Simulate simple static systems using spreadsheets or basic simulation tools.
- Generate and test pseudo-random numbers using different techniques (Linear Congruential Generator, etc.).
- Create probability distributions (e.g., uniform, exponential, normal) using programming languages or tools.
- Validate random numbers using statistical tests (Chi-square test, Kolmogorov-Smirnov test).

MODULE-2

UNIT-1 **6L+6T+0P+6SL=18 Hours**

LANGUAGE FOR SIMULATION

Input modeling: Data collection(T2-Pp:269), identifying the distribution with data(T2-Pp:270), parameter estimation(T2-Pp:280)., goodness of fit test(T2-Pp:294), fitting a non-stationary Poisson process(T2-Pp:294)., selecting input models without data, multivariate and time series input models(T2-Pp:296) Verification and validation of simulation models(T2-Pp:310), model building(T2-Pp:311), verification and validation(T2-Pp:311), verification of simulation models(T2-Pp:311), calibration and validation of models(T2-Pp:316).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

EXPERIMENTS

Experiments-Simulation of different systems (T2-Pp:327, Analysis, calibration, validation and verification of input and output simulated data(T2-Pp:327), study of alternate techniques(T2-Pp:379).

UNIT-3 **6L+6T+0P+6SL=18 Hours**

CASE STUDY

Manufacturing and material handling simulation(T2-Pp:426), goals and performance measures (T2-Pp:429), issues in manufacturing and material handling simulations(T2-Pp:430), case studies of the simulation of health care systems and smart cities. Manufacturing example(T2-Pp:435), job shop scheduling using Simpy, a job shop analysis, simulation of computer systems(T2-Pp:450): simulation tools(T2-Pp:452), model input, high level computer system simulation, memory simulation(T2-Pp:456):

PRACTICES:

- Input modeling: Collect and analyze real or simulated data (e.g., inter arrival times).
- Fit data to statistical distributions using tools (e.g., Arena, MATLAB, Python).
- Parameter estimation and performing a goodness-of-fit test.
- Model verification and validation of a small simulation project (e.g., coffee shop service line).
- Simulate and analyze input/output data of a multi-server queuing system.
- Perform calibration of input data and analyze the output of a simulation run.
- Compare alternative designs or policies using simulation (e.g., staffing strategies in a call center).
- Conduct sensitivity analysis using simulation parameters.
- Manufacturing System Simulation: Simulate a job shop environment using any simulation software.
- Material Handling System: Model and simulate a conveyor system or warehouse picking system.
- Computer System Simulation: Simulate CPU scheduling (e.g., Round Robin, FCFS).
- Memory Simulation: Model paging or memory allocation algorithms in a simplified computer system.

SKILLS:

- ✓ *Interpret the model and apply the results to resolve critical issues in a real world environment.*
- ✓ *Analyse the Simulation models using input analyzer, and output analyzer.*
- ✓ *Verify and Validate of simulation model.*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Apply dynamic physical models and static mathematical models.	Apply	1, 2
2	Generate random variables and random numbers.	Apply	1, 2, 4
3	Build the simulation models.	Apply	1, 3, 5
4	Verify and validate simulation models.	Analyze	1,2,4,5,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):





TEXT BOOK:

1. Geoffrey Gordon, “System Simulation”, Second edition, Prentice Hall, India, 2002[unit I]
2. Jerry Banks and John S.Carson, Barry L.Nelson, David M.Nicol, “Discrete Event System Simulation”, fourth edition, Prentice Hall, India, 2002[unit II,III,IV,V].

REFERENCE BOOKS:

1. Robert E.Shannon, “System Simulation The art and science”, Prentice Hall, New Jersey, 1995.
2. D.S. Hira, “System Simulation”, S.Chand and company Ltd, New Delhi, 2000.